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The impact of transitional care interventions on hospital readmissions in older medical patients: a systematic review

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TITLE PAGE

The impact of transitional care interventions on hospital readmissions in older medical patients: a systematic review

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ABSTRACT

Objectives

To identify and synthesise available evidence on the impact of transitional care interventions with both pre-and post-discharge elements on readmission rates in older medical patients.

Design

A systematic review.

Method

PubMed, The Cochrane Library, Embase, Cinahl, and Web of Science were searched from January 2008 to August 2019 using a systematic search strategy complimented by an extensive manual search. The study selection at title level was undertaken by one author and the remaining selection process as well as data extraction and methodological quality assessment were undertaken by two authors independently. Inclusion criteria were: either older medical patients >65 years or mean age in study population of 75+ years; exposure was transitional care interventions between hospital and home with both pre- and post-discharge components; outcome was hospital readmissions. A narrative synthesis was performed and effect sizes were estimated.

Result

A total of 1951 records were identified. 11 studies were included: five randomised controlled trials, four controlled clinical trials and two pre-post cohort studies. The 11 studies represent 15 different interventions and 29 outcome results measuring readmission rates within 7-182 days after discharge. 22 of the 29 outcome results showed a reduction in readmission rates in the intervention groups compared to the control groups. The most significant impact is seen when interventions: are high intensity, last at least one month and target patient at risk. The methodological quality of included studies was generally poor.

Conclusion

Transitional care interventions have a positive impact as they reduce readmission rates among older medical patients although the impact varied at different times of outcome assessment. There was insufficient evidence to recommend a specific intervention. High quality studies that examine the impact of interventions are needed, preferably complimented by a process evaluation in order to refine and improve future interventions.

PROSPERO registration number CRD42019121795.

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STRENGTHS AND LIMITATIONS

- Focus on trans-sectorial interventions with both pre- and post-discharge elements
- High internal validity as PRISMA and AMSTAR guidelines have been applied
- Results may underrepresent negative study findings as negative impact is often not published
- Assessing intervention impact on unplanned readmissions as the only outcome

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INTRODUCTION

Transitional care interventions (TCI) may be essential in the transition of older medical patients from hospital to home as it may prevent adverse events and unplanned hospital readmissions. These events can have detrimental consequences for the individual patient.

The global demography is radically changing. In the European Union, elderly above 65 years are estimated to account for 29.1% of the total population in 2080 compared to 19.2% in 2016. Additionally, the prevalence of the population above 80 years will double between 2016 and 2080[1]. As a result, the utility of healthcare services and costs will dramatically increase[2]. As these changes will bring substantial challenges to the healthcare systems[3 4], the potential need for TCIs will also increase. Older people in need of healthcare are often medical patients that have several concurrent diseases, reduced physical or mental functionalities, limited ability to provide self-care, are living alone and need care from primary or secondary healthcare services[5].

Older people with complex co-morbid conditions are at high risk of adverse events and safety incidents immediately after hospital discharge[6].

Unplanned readmission seems to be related to insufficient discharge planning, inadequate patient transition from hospital to home and unintended events such as medication errors and poor communication between hospital and primary care professionals[7-10]. By contrast, optimised, customised and patient-centred discharge planning and transitions may reduce length of hospital stay (LOS), risk of readmission, medication discrepancies and mortality as well as improve the patients' activity of daily living (ADL) and reduce the healthcare costs[11].

One approach to address these challenges is to examine the impact of interventions that aim to reduce readmissions.

Previous systematic reviews have mainly evaluated the impact of hospital- and/or home-based interventions on readmissions[12 13] and included populations with specific conditions or both medical and surgical patients. To our knowledge, no systematic review examines the impact of transitional care interventions that take place in both hospital and home on readmissions among older medical patients based on recent data.

The purpose of this systematic review is to evaluate the impact of transitional care interventions, with both pre-and- post-discharge elements, on readmission among older medical patients.

METHODS

This review was registered in the PROSPERO database prior to data collection (CRD42019121795)[14]. It is presented in accordance with the Preferred Reporting Items for Systematic Reviews (PRISMA)[15].

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Eligibility criteria

We identified studies that aimed to reduce readmission rates through TCI's among older medical patients. The PICO process for framing the research question was applied[16] and defined as:

i. ***P****opulation*: Older medical patients discharged from a general medical ward or emergency department (ED);
ii. ***I****ntervention*: In the transitional phase between hospital and home which examined the impact of the intervention on readmission rates. The interventions had to include both pre- and post-discharge components;
iii. ***C****omparison*: Usual care defined as standard care and treatment;
iv. ***O****utcome*: Unplanned readmission to hospital.

Selecting tool is displayed in Supplementary 1.

Studies were excluded if:

i. the population was aged under 65 years or the mean age was less than 75 years;
ii. they included other patient groups than medical patients (e.g. surgical patients);
iii. they included participants with only one medical diagnosis (Internal Classification of Disease -10th edition);
iv. studies only included participants with psychiatric disorders;
v. they compared interventions with anything other than usual care;
vi. readmission was not an outcome;
vii. intervention only included either pre- or post-discharge components; they were reviews, case-reports or case-studies without comparison groups.

Information sources

To identify eligible studies, we searched the following bibliographic databases: PubMed, The Cochrane Library, Embase, Cinahl, and Web of Science from January 2008 to August 2019. An extensive snowball search was performed where the reference lists from relevant studies, systematic reviews and included studies were examined. Additionally, publication lists from prominent researchers within the field were examined. Grey literature was searched in all relevant resources listed by Paez[17] from January 2008 to August 2019. Authors from relevant study protocols and grey literature were contacted by e-mail in order to examine whether the studies were published or study results were available. The Negative Results Scientific Journal was searched.

Search strategy

The search string (Supplementary 2) was developed in collaboration between the authors and a dedicated university research librarian. Key terms, free-text-words, subject headings, index terms and appertaining synonyms were

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identified through relevant theory, research and indexed keywords from relevant studies. The searches were limited by only including studies published in English or Scandinavian languages. The bibliographic searches were conducted on December 13 and 14, 2018 and the searches were regularly updated until August 31, 2019.

Study selection
First, titles were screened for their potential relevance according to population and outcome by the first author (LFR). Second, two authors independently screened titles and abstracts for intervention eligibility (LFR and MG). Third, an assessment of the full text was performed by two authors independently (always including LFR). In case of disagreement, a third author, who was chosen a priori, was consulted.

Data extraction
The Cochrane Data Extraction Form was modified to fit the present patient group and intervention type[18]. Data from all the included studies were extracted by two researchers independently (always including LFR). Extracted data included study characteristics and results such as author, year of publication, country, study design, setting, participants, study size, outcomes, follow-up time and impact of intervention in numbers and/or percent. Only data on the outcome "readmission" were extracted and analysed.

Quality assessment
"The Quality Assessment Tool for Quantitative Studies" (EPHPP) (Supplementary 3) was applied to assess bias in each study included in the review. The validated tool is recommended by The Cochrane Collaboration[16 19] and provides a standardised means to assess study quality. EPHPP assess six methodological dimensions: selection bias, study design, confounders, blinding, data collection as well as withdrawals and drop-outs. It distributes to the overall methodological rating into a strong, moderate or weak measure of internal validity[20]. Quality assessment will be conducted at study level and results will be analysed in the synthesis and taken into account in the conclusion. Risk of bias across studies will be analysed and the impact hereof will be discussed.

Data synthesis
Inspired by Pigott and Shepperd[21], following aspects that can lead to heterogeneity will be assessed: 1) context, 2) target population, 3) intervention, 4) methodological features and 5) researcher characteristics and reporting context. This assessment will either result in a descriptive synthesis or a meta-analysis of data. Risk ratios (RR) and their 95% confidence intervals (CI) are calculated when possible and presented in forest plots stratified by subgroups.

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RESULTS

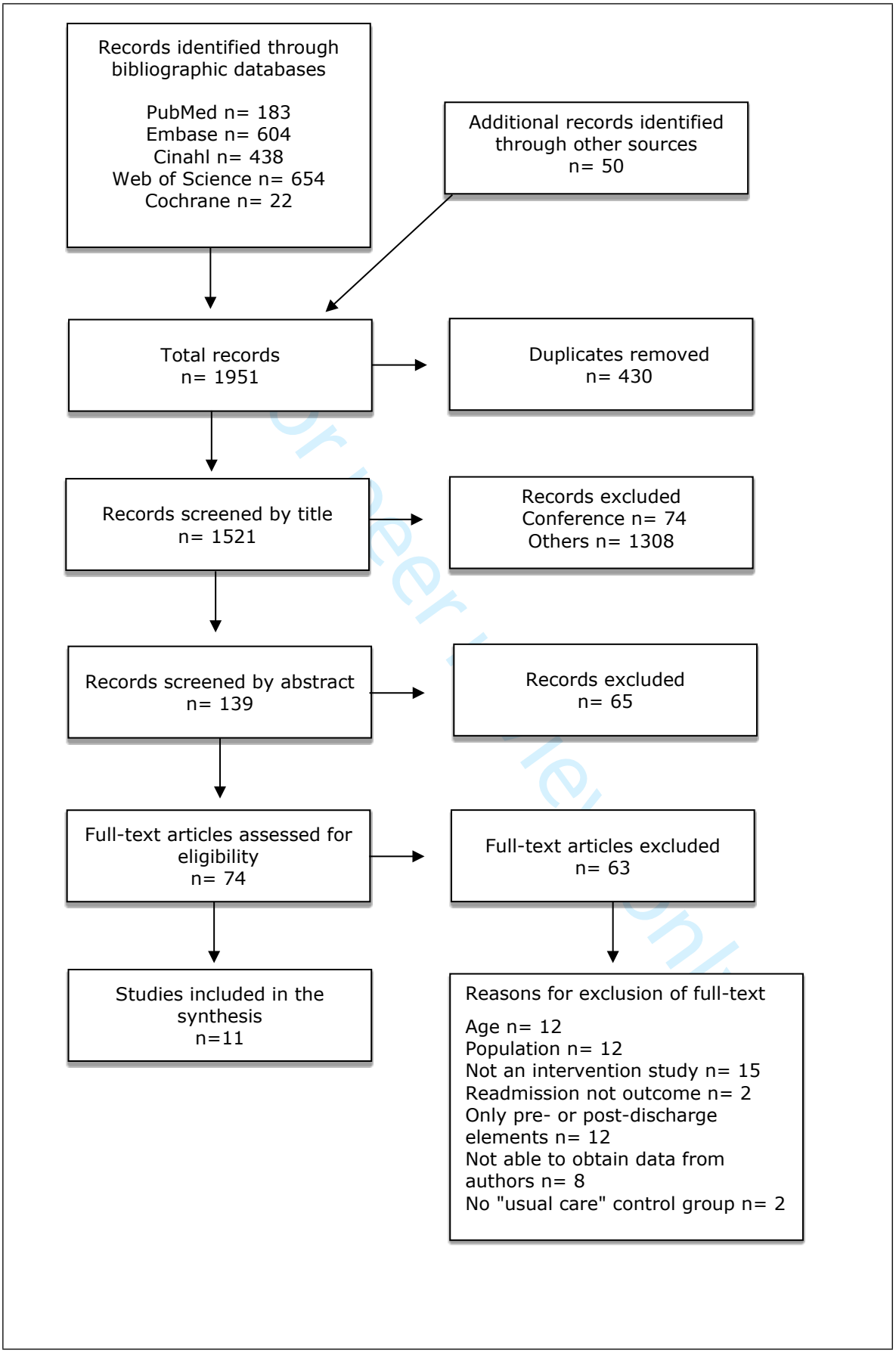
Study selection

In total, 1951 records were identified. Of those, 1901 records were identified through bibliographic databases and 50 records were found through other sources. After removing duplicates, 1521 records were screened by title and abstract. 74 records were considered for full-text review of which 11 met the eligibility criteria (Figure 1). The 63 remaining records were excluded due to participants' age, population, study design, readmission not listed as an outcome, only pre- or post-discharge elements in the intervention, not able to obtain data from authors and no "usual care" control group (Supplementary 4).

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Figure 1 Flow chart



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Study characteristics

The 11 included studies represented 15 different interventions and 29 outcome assessments measuring readmission rates at different time-points. Two studies were multi-arm studies[22 23]. Study characteristics are shown in Table 1. Five of the included studies were randomised controlled trials (RCT's)[23-27], four controlled clinical trials (CCT's)[22 28-30] and two pre-post cohort studies[31 32]. Sample sizes of individual studies ranged from 41 to 19,157. Allocation to intervention- or control group was performed on an individual level. The majority of studies reported that readmissions were unplanned or acute[23-28 32]. Outcome assessments were conducted 7-182 days after discharge. In addition to readmission, the 11 studies assessed outcomes on physical functioning[24 27], cognitive functioning[24], quality of life[22 25 27], time to either hospital readmission or discharge from nursing homes[24 28], self-efficacy[22], self-rated health[22], visits to emergency departments, general practitioners or allied health professional[25 26 28 31 32], length of stay[27 31], co-morbidity[27], cost-effectiveness[27] and mortality[24 28].Three studies took place in USA[26 29 30] , two in Australia[23 25], two in Hong Kong[22 31], one in the Netherlands [24], one in the United Kingdom[27], one in Denmark[28] and one in New Zealand[32]. The studies were published between 2009 and 2018.

Study population

In total, approximately 24,500 patients were evaluated in our review; 8,800 in the intervention groups and 15,700 in the control groups. The mean age of the participants in the intervention groups was approximately 78 (range 74.9–83.6), whilst the mean age among the control groups was approximately 79 (range 75.2-84.5). Approximately 59% (range 3.6-85) of the participants in the intervention groups were females and 52% (range 2.2-67) of the control group were females.

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Table 1 Study characteristic

Author Year	Country Setting	Design	Intervention	Study population	Mean age, years (SD) or median (range)	Female n, (%)	Outcomes	OA	Readmission (event/total)	RR
Buurman[24] 2016	Netherland Multi-centre study	RCT	Transitional Care Bridge Program Intervention	T: 674 I: 337 C:337 Unselected patients	I: 79.7 (7.3) C: 80.0 (7.8)	I: 195 (57.9) C: 195 (57.9)	-Readmission - Katz Index of ADL - Mortality - Cognitive functioning - Time to hospital readmission - Time to discharge from a nursing home	182	I: 106/316 C: 88/303	1.15
Chow[22] 2014	Hong Kong Single-centre study	Controlled clinical trial	The nurse case management intervention	T: 312 Ia: 96 Ib: 108 C: 108 Unselected patients	Ia: 75.00 (60–92) Ib: 75.50 (60–89) C: 77.00 (60–89)	Ia: 46 (52.9) Ib: 52 (54.2) C: 49 (50.0)	- Readmission - Quality of life - Self-efficacy - Self-rated health	28 84	Ia:14 /91 Ib: 16/100 C: 24/105 Ia: 32/97 Ib: 30/106 C: 59/130	Ia: 0.67 Ib: 0.70 Ia: 0.73* Ib: 0.62*
Courtney[25] 2009	Australia Single-centre study	RCT	Older Hospitalised Patients' Discharge Planning and In-home Follow-up Protocol (OHP-DP)	T: 128 I: 64 C: 64 Patients at risk	I: 78.1 (6.3) C: 79.4 (7.3)	I: 36 (62.1) C: 40 (62.5)	- Readmission - Visits to emergency department - Visits to general practitioner - Visits to allied health professional - Health-related quality of life	28 84 168	I: 2/49 C: 9/58 I: 10/49 C: 16/58 I: 11/49 C: 27/58	0.26 0.74 0.48*
Finlayson[23] 2018	Australia Multi-centre study	RCT	NR	T: 222 Ic: 57 Id:56 Ie:54 C:55 Patients at risk	Ic: 77.1 (7.64) Id: 77.6 (6.50) Ie: 77.8 (6.23) C: 77.9 (6.20)	Ic: 46 (80.7) Id: 42 (75.0) Ie: 37 (68.5) C: 37 (67.3)	- Readmission	28 84 168	Ic: 4/53 Id: 7/49 Ie: 5/49 C: 13/53 Ic: 11/53 Id: 16/44 Ie: 9/48 C: 19/50 Ic: 18/52 Id: 18/42 Ie: 16/47 C: 23/50	Ic: 0.31* Id: 0.58 Ie: 0.42 Ic: 0.55* Id: 0.96 Ie: 0.49* Ic: 0.75 Id: 0.93 Ie: 0.74
Koehler[26]	USA	RCT (pilot)	Elderly Care	T: 41	I: 77.2 (5.3)	I: 17 (85)	- Readmission	30	I: 2/20	0.26*

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2009	Single-centre study		Bundle	I: 20 C:21 Patients at risk	C: 79.8 (5.6)	C: 13 (62)	- Emergency department visits	60	C: 8/21 I: 6/20 C: 9/21	0.70
Lin[31] 2015	Hong Kong, Multi-centre study	Cohort analytic (two groups pre - post study)	Integrated Care and Discharge Support for elderly patients (ICDS)	T: NR I: 1,090 C: NR Patients at risk	Total: 80.4 ± 7.6 I: NR C: NR	T: 557 (51.1)	- Readmission -AED attendance - LOS	182 pre 182 post	NR Reports a 47% reduction in readmissions	NR*
Nielsen[28] 2018	Denmark Single-centre study	Controlled clinical trial	Elderly Activity Performance Intervention	T: 375 I:144 C: 231 Unselected patients	I: 81 (7.9) C: 78 (8.6)	I: 79 (55) C: 122 (53)	- Readmission - All-cause mortality - Number of contacts to general practitioners and emergency departments without admission - Time to first readmission	30 182	I: 25/139 C: 55/231 I: 64/144 C: 99/231	0.76 1.04
Robinson[32] 2015	New Zealand Multi-centre study	Cohort analytic (two groups pre - post study)	Integrated Transition of Care (ITC)	T: 19157 I: 5172 C:13,985 Patients at risk	I: 78.2 (9.2) C: 77.6 (9.1)	I: 2486 (48.1) C: 6765 (48.4)	- Readmission - Emergency attendances	7 28 90	I: 500/5,172 C: 1,239/13,985 I: 1370/5,172 C: 3,588/13,985 I: 2,281/5,172 C: 6,079/13,985	1.09 1.03 1.01
Rottman-Sagebiel[29] 2018	USA Single-centre study	Controlled clinical trial	The Geriatrics Medication Education at Discharge project (GMED)	T: 1,624 I: 435 C: 1,189 Patients at risk	I: 74.9 (7.6) C: 75.2 (8.35)	I: 14 (3.6) C: 26 (2.2)	- Readmission	30	NR	NR
Sahota[27] 2017	UK Single-centre study	RCT	The Community In-reach Rehabilitation and Care Transition (CIRACT)	T: 250 I: 125 C: 125 Unselected patients	I: 83.6 (6.6) C: 84.5 (5.9)	I: 82 (66) C: 79 (63)	- Readmission - LOS - Day 91-super spell bed days -Functional ability - Co-morbidity - Health-related quality of life - Cost-effectiveness analysis	28 91	I: 18/106 C: 14/106 I: 45/106 C: 39/106	1.29 1.15
Voss[30] 2011	USA Multi-centre study	Controlled clinical trial	Care Transitions Intervention (CTI)	T: 1888 I: 1042 C: 846 Unselected patients	I: NR C: NR	I: NR C: NR	- Readmission	30	NR	NR

T: total, OA: Readmission outcome assessment in days after hospital discharge, RR: relative risk, *: statistically significant, NR: not reported or author could not access the raw data. CGA: comprehensive geriatric assessment, LOS: length of hospital stay

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I: intervention, Ia: intervention "Home visit", Ib: intervention "Call", C: control, Ic: EN-HaT (Exercise and Nurse Home Visit and Telephone follow up), Id: Exercise, Ie: N-HaT (Nurse Home visit and Telephone follow-up)

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Transitional care interventions

All studies included pre- and post-discharge components and some included bridging components. The most pronounced difference between the interventions was in the pre-discharge phase, whereas elements in the bridging and post-discharge phase were somewhat similar. Table 2 outlines the intervention components and Supplementary 5 contains detailed descriptions of the interventions.

Pre-discharge components

Across studies, pre-discharge components consisted of; patient assessments, health record or plan, discharge planning, involvement of caregivers, concerns or barriers, patient education, physical exercise, medication reconciliation, counselling, nutrition screening or pre-discharge home visit.

Bridging components

Bridging components consisted of; written hand-over between hospital and primary sector, telephone hand-over or in-person hand-over.

Post-discharge components

Post-discharge components comprised of; home visits, telephone follow-up, additional healthcare services, nurse availability from hospital discharge until seven days after discharge or rapid and intensive community support.

Synthesis of results

All five aspects that can lead to heterogeneity were analysed and found to differ significantly between the included studies. Therefore, it was not possible to conduct a meta-analysis due to heterogeneity.

Considering the impact on outcome level, 22 of the 29 outcome results (76%) showed positive impact on readmission in the intervention groups compared to the control groups. Three studies did not report readmission rates but reported a positive impact on readmission rates and it was not possible to obtain these numbers from the authors[29-31]. It was not possible to calculate RR and its 95% CI for those studies, and they are not presented in the forest plots.

The impact size (RR) was calculated using the reported readmission rates from eight studies. RR ranged from 0.26 to 1.29. Nineteen outcome results were <1 of which one was close to 1 (RR=0.96). Seven estimates were >1 of which three were close to 1 (RR= 1.01, 1.03, 1.04). Of the 29 outcome results, eight were statistically significant[22 23 25 26 31].

Considering the impact on study level, seven of the eleven studies showed an entirely positive impact on readmissions[22 23 25 26 29-31], whilst one study presented both a positive impact and no impact[28] and three studies showed no impact at all[24 27].

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Subgroup analysis

Study population

The RRs for participants categorised as "unselected patients" ranged between 0.67 and1.29 whilst RRs for "patients at risk" ranged between 0.26 and 1.09. Supplementary 6 presents estimates stratified according to the included study population.

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Table 2 Intervention components

Study	Pre-discharge components												Bridging components					Post-discharge components					
	Patient assessment	Personal health record and/or plan	Education	Medication reconciliation	Physical exercise	Address concerns and/or barriers	Counselling	Discharge planning	Nutrition screening	Caregiver involvement	Home visit to assess future needs	Multidisciplinary care	In-person hand-over	Written hand-over	Telephone hand-over	Community-based nurse visits hospital	Communications path unknown	Number of home visits	Number of telephone follow-up	Nurse availability	Referral to additional services	Intensive community support	Intervention intensity
Buurman et al.[24]	+	+										+	+			+		5[g]					9
Chow et al.[22] (1)	+	+				+												2[g]	2				6
Chow et al.[22] (2)	+	+				+													4				6
Courtney et al.[25]	+	+			+	+		+				+		+				1[a,g]	9				12
Finlayson et al.[23] (3)	+	+			+					+		+						6[b]					10
Finlayson et al.[23] (4)	+	+						+		+		+						1[a,g]	10	+			12
Finlayson et al.[23] (5)	+	+			+			+		+		+						1[a,g] 6[b]	10	+			12
Koehler et al.[26]		+	+	+		+	+	+		+		+		+				2[c]					7
Lin et al.[31] (6)	+							+				+						?[h]	?				6
Lin et al.[31] (7)	+							+				+									+[d]	+	5
Nielsen et al.[28]	+	+[d]			+												+	1[g]					4
Robinson et al.[32]	+		+[f]	+[f]					+			+		+	+[d]				2[e]				7
Rottman-Sagebiel et al.[29]			+[f]	+[f]						+		+		+	+[d]				1		+[d]		6
Sahota et al.[27]	+	+								+	+	+						1[a,h]			+[d]		3
Voss et al.[30]		+													+			1[g]	3				7

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1; post-discharge home visit group, 2; post-discharge call group , 3; Exercise intervention, 4; N-Hat (Nurse Home visit and Telephone follow-up) intervention, 5; Ex-Hat (Exercise and Nurse Home Visit and Telephone follow up) intervention, 6; ICM case management group, 7; HST service group a; additional home visits available if needed, b; by physiotherapist, c; one call by nurse and one call by pharmacist, d; if deemed necessary, e; to participants who return to independent living, f; by pharmacist, g; first visit within 3 days after discharge, h; time for first home visit unknown, ?; number not reported

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Intervention intensity
Calculation of intervention intensity is inspired by Verhaegh et al.[33] Ten interventions were categorised as low intensity[22 26-32] and five were categorised as highintensity[23-25].
Supplementary 7 illustrates a clear difference in the impact between the low and high intensity interventions. RRs for low intensity interventions ranged between 0.26-1.29 whilst RRs ranged between 0.26-1.15 among high intensity interventions.

Length of support
Duration of the interventions ranged from enrolment until one day to six months after hospital discharge. Four studies lasted between one and seven days[26 28 29 32], two studies lasted 28 and 30 days[22 30], one study 84 days[31], two 168 days[23 24], one 182 days[25] and one study did not report the duration of the intervention[27]. Supplementary 8 displays the impact on readmission rates stratified on the duration of the interventions. A duration of one month or more seemed to have a larger impact than interventions with a shorter length of support.

Country of origin
Studies conducted outside the European countries seemed to have a greater impact on readmissions compared to studies conducted within the European Union (Supplementary 9).

Outcome assessment
The impact on readmissions was largest within 30 days after hospital discharge. The impact decreased hereafter and was similar between 1 month and 6 months after discharge (Supplementary 10).

Quality assessment within studies
Of the 11 studies, two studies were assessed to have a strong methodological quality[22 25], three had a moderate[24 27 32] and six had a weak methodological quality[23 26 28-31]. Of the RCT's, one had a strong[25], two had a moderate[24 27] and two had a weak quality[23 26]. Of the CCT's, one study had a strong[22] and three had a weak quality[28-30]. One of the pre-post cohort studies had a moderate[32] and one had a weak quality[31] (Table 3).

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Table 3 Quality assessment

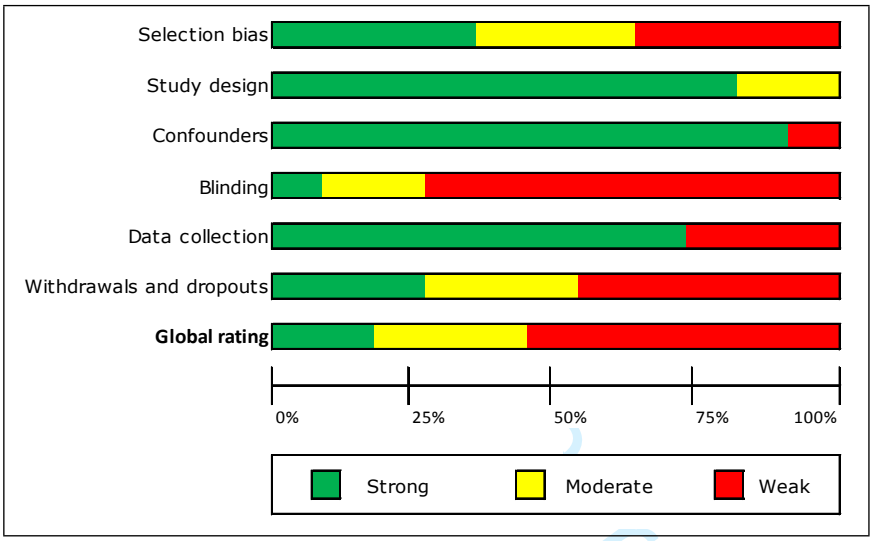
Author	Selection bias	Study design	Confounders	Blinding	Data collection methods	Withdrawals and drop-outs	Global rating
Buurman et al.[24]	Moderate	Strong	Strong	Strong	Strong	Weak	**Moderate**
Chow et al.[22]	Moderate	Strong	Strong	Moderate	Strong	Strong	**Strong**
Courtney et al.[25]	Moderate	Strong	Strong	Moderate	Strong	Moderate	**Strong**
Finlayson et al.[23]	Weak	Strong	Strong	Weak	Weak	Moderate	**Weak**
Koehler et al.[26]	Weak	Strong	Strong	Weak	Strong	Weak	**Weak**
Lin et al.[31]	Strong	Moderate	Weak	Weak	Weak	Strong	**Weak**
Nielsen et al.[28]	Strong	Strong	Strong	Weak	Strong	Weak	**Weak**
Robinson et al.[32]	Strong	Moderate	Strong	Weak	Strong	Strong	**Moderate**
Rottman-Sagebiel et al.[29]	Weak	Strong	Strong	Weak	Weak	Weak	**Weak**
Sahota et al.[27]	Strong	Strong	Strong	Weak	Strong	Moderate	**Moderate**
Voss et al.[30]	Weak	Strong	Strong	Weak	Strong	Weak	**Weak**

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Risk of bias across studies

The majority of the studies did not meet the criteria in the components *selection bias* and *blinding* and were thus assessed moderate or weak. The vast majority of studies met the criteria of *study design*, *confounders*, and *data collection* and were therefore rated strong (Figure 2).

Figure 2 Risk of bias across studies



DISCUSSION

We found that the majority of interventions in the transitional phase between hospital and home appear to reduce readmission rates among older patients discharged from a medical ward.

However, some studies reported both positive impact and no impact on the readmission rate on similar care interventions. These divergent results may have several plausible explanations as discussed below.

Explanations related to study characteristic

Country of origin

Studies conducted in European countries have less impact compared to studies from non-European countries. The impact of complex interventions is, among others, altered by the context of the implementation[34 35]. Therefore, differences in impact between countries may be explained by diversity in the social, political, economic, clinical and geographical setting. The accessibility, type, character and quality of healthcare services of basic healthcare services may also affect the study impacts[36]. USA and Australia have a long history of discharge planning and transitional care. Therefore, those countries have a high quantity of research as well as refined strategies and guidelines[37-39].

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Study population
The impact on readmissions is greater among "patients at risk" compared to "unselected older patients". This is to be expected as readmission rates are higher among "patients at risk" than among "unselected patients". Furthermore, the "patients at risk" are more frail and may have a higher degree of morbidity which could have affected the risk of readmission[40].

Readmission rates prior to the study
Pre-intervention readmission rates may mirror the differences in impact between studies. It may be assumed that hospitals with low pre-intervention readmission rates may not experience a reduction of post-intervention readmission rates as the remaining readmissions may not be preventable. In contrast hereto, it may be assumed that hospitals with high pre-intervention readmission rates are expected to attain a reduction in post-intervention readmission rates.

Sample size
Several of the studies have small sample sizes. As seen in Table 1, studies with small sample sizes had a stronger impact compared to studies with larger sample sizes. This is in line with Dechartres et al. who also showed that larger effect sizes are seen in small to moderate sized trials compared to larger trials[41] One reason may be that small studies are more prone to publication bias compared to larger studies[42].

Explanations related to review findings
Interventions
Our findings suggest that the intensity of interventions influences the impact on readmission rates as high intensity interventions generally have a more pronounced impact compared to low intensity. It is in line with previous findings demonstrated by Verhaegh et al.[33]. The higher quantity of elements means that more aspects in a complex patient cohort and complex settings can be addressed.
Intervention components that at first glance appear to be similar across studies are comprised of different features. The diverging impact on readmissions may be affected by differences in intervention content across studies. It is not possible to make an intervention component analysis across studies that evaluate which components that positively affect the readmissions. However, some trends are seen in Table 2. Interventions with positive impact on readmission comprise the following components; patient assessment, personal health record or plan, addresses concerns and barriers, discharge planning, caregiver involvement, home visits and telephone follow-up.

Intervention fidelity

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The impact on readmission rates may be affected by the fidelity of interventions[43]. Intervention fidelity is not described in several studies. This result is likely to be related to inconsistency in monitoring the implementation or lack of transparency in reporting the study - thus hindering the assessment of whether lack of positive impact is caused by poorly implemented intervention or if the intervention did not work.

Outcome assessment
Time of the outcome assessment has an important impact on the possible preventive readmission rate[44]. The findings of this review suggest that interventions are most impactful within the first 30 days after discharge. A potential impact is likely to have obliterated if outcome is assessed later than 30 days after discharge.

Residual confounding
Several of the included studies did not adjust for potential confounders such as length of stay, fall within the last 12 months, living conditions, prior admissions, poor overall health condition and functional disability[45-47]. This could have affected the internal validity and the impact on readmission rate. The lack of analysis that adjusts for potential confounders is highlighted in previous studies[48], underlining the problematic trend in this research area.

Risk of Bias
Differences in the quality assessment and thus risk of bias may explain the variation in the impact of the included studies[16]. Supplementary 11 shows that studies assessed to have weak and strong methodological quality have positive impact on readmission rates whereas studies with moderate quality have no impact.

Statistical analysis
Several authors fail to report whether the statistical analysis are performed based on first readmission. One patient may therefore represent several readmissions within a specific follow-up period. If this is the case, results from those studies may report a higher readmission rate and thus greater impact compared to studies using unique observations in their statistical analysis.

Quality assessment
The methodological quality of the majority of included studies was low, thus indicating a high risk of bias. Low global rating can both reflect the methodological quality or insufficient reporting of the methodology. The latter makes it difficult to accurately assess the true quality and thus risk of bias. If the low rating is caused by insufficient methodological reporting, it may not affect the impact. However, if the low rating reflects methodological problems, it probably underestimates the true impact of the interventions.
The component "blinding" represents the weakest rating. Blinding of participants was not always possible due to the nature of the interventions.

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Blinding is crucial in pharmaceutical trials and in other clinical studies. However, little is known about the benefits or disadvantages of blinding participants in complex interventions and how bias due to blinding may affect the results. Blinding outcome assessors may not be crucial when outcome is collected through digital records.

COMPARISON WITH PRIVIOUS RESEARCH

Prior systematic reviews included other study populations such as surgical patients or patients with only one medical condition. This review includes a broader population, namely older medical patients. Direct comparison to previous findings is therefore difficult. This review suggests that transitional care services reduce the risk of readmission among older medical patients These findings are in line with similar reviews that found positive effects of TCIs on hospital readmission[49-51]. These reviews also found very low to moderate methodological quality of the included studies.

LIMITATION OF INCLUDED EVIDENCE BASE

There is a lack of transparency across studies despite the fact that the CONSORT statement recommends reporting a sufficient description of interventions[52]. A call for more transparency in clinical trials and adherence to appropriate guidelines is also reported elsewhere[53]. If the template for the Intervention Description and Replication (TIDieR) is followed, the replicability may improve thus making it possible to build on prior research findings[54]. Evaluating TCIs provides insight to whether interventions reduce readmissions among intervention groups compared to control groups. However, we gain no knowledge on causality between exposure and outcome. In complex interventions, such as TCIs, it is difficult to evaluate which mechanisms or components result in change[55]. A process evaluation may have captured the interventions' fidelity and, thus provided insight on which mechanisms and components actually work[56]. Process evaluation has previously been requested[57] stressing the unmet need to identify essential components in TCIs. We have requested missing intervention details and other relevant data from authors; some requests were met while others were not.

STRENGTHS AND LIMITATIONS OF THIS REVIEW

The study has several strengths. This review has adhered to PRISMA[15] and AMSTAR[58] which ensures that all important methodological considerations are made. By reducing the risk of bias in this review, systematic errors are minimised during all stages of the process.

Despite these strengths, we note some limitations. Only studies published in English, Danish, Norwegian or Swedish were eligible for inclusion in this review. Therefore, studies published in other languages were omitted, potentially excluding useful evidence.

This review only focuses on hospital readmission as an outcome, although included studies evaluate the intervention impact on multiple outcomes. It is

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possible that some studies report a positive impact in terms of other outcomes than readmission.
Positive findings are more likely to be published in English language journals whereas negative findings are more often published in local-language journals[59]. Publication bias is most likely present in this research field. During this review process, we have contacted several researchers who had registered TCIs on ClinicalTrials.gov. All researchers who found no or negative impact either struggled to publish or decided not to publish due to lack of positive findings. Abovementioned issues may result in an underrepresentation of negative results in the evidence base and thus this present review.

IMPLICATIONS AND FUTURE RESEARCH

Future research projects may benefit from the knowledge gained from this review when designing and developing new studies. Knowing that interventions with a minimum duration of one month that are high intensity and targeting high-risk patients may result in more effective interventions.
This review highlights an unmet need for high methodological quality studies that evaluate the impact of TCIs among older medical patients. Therefore, future research can benefit from a higher level of adherence to relevant guidelines and detailed description of 1) interventions and comparison, 2) the implementation process, 3), and actions taken to minimise bias and confounding. Further research should be undertaken to investigate the process evaluations of complex interventions to identify how and why interventions either work or do not work. To develop a broader picture of TCIs, additional studies need to focus on psychological outcomes. The societal cost of such interventions versus individual and societal benefits needs to be further evaluated. High-risk patients such as physically disabled, chronically ill patients regardless of age and co-morbidity may benefit from these kinds of interventions[60].

CONCLUSION

The majority of transitional care interventions have a positive impact on readmission rates among older medical patients, although the most significant impact was seen within 30 days after hospital discharge. Therefore, we believe there is evidence for recommending transitional care that includes both pre- and post-discharge components. However, there is no evidence for recommending a specific intervention. The key finding shows that there is an apparent pronounced positive impact among patients categorised as "patients at risk" in "high intensity" interventions and in interventions with duration of one month or more.

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Author Contribution

Lisa Fønss Rasmussen: first author writing the manuscript, screening records for inclusion, data extraction of included studies, quality assessment of included studies,

Louise Grode: screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Jeppe Lange: screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Ishay Barat: screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Merete Gregersen: screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

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Supplementary 1, selecting tool

Review question	"What is the impact of transitional care interventions between hospital and home on readmissions among older medical patients?"	
Reviewer Name:		Date:
Author name/study ID:		Year:
Title:		Journal:
	Include	**Exclude**
Patient population	- Elderly patients discharged from a general medical ward or emergency departments - Patients ≥ 65 years or mean age ≥ 75 years	- Patients ≤ 65 and mean age ≤ 75 - Not only medical patients - Discharging ward not specified - Only one patient group/diagnosis such as COPD, diabetes - Patients with psychiatric disorders or patients who are cognitively impaired - Other
Interventions	- Interventions in the transitional phase between hospital and home which examine the effect of the intervention on readmission rates between intervention group and control group	- No comparison/control group - Not an intervention study - Other
Comparators	- Usual or standard procedure	- Comparison between two interventions - Comparison with anything else than usual procedure - Other
Outcomes	- Readmission	- Readmission is not an outcome
Setting	- Intervention includes both pre- and post-discharge components	- Only pre-discharge components - Only post-discharge components - Other
Study design	- Quantitative studies	- Systematic reviews, meta-analysis, literature reviews - Case-reports or case-studies - Meeting abstract, study protocol - Other
Overall decision	- **Included**	- **Excluded**
Notes: Reason for exclusion:		

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Supplementary 2, search string

Search string with free-text words used in PubMed

1	elder* (title)
2	aged (title)
3	old* (title)
4	geriatric* (title)
5	1 OR 2 OR 3 OR 4
6	"discharge intervention" (title)
7	transition* (title)
8	transition* care (title)
9	6 OR 7 OR 8
10	readmission* (text word)
11	re-admission* (text word)
12	rehospitalization* (text word)
13	re-hospitalization* (text word)
14	rehospitalisation* (text word)
15	re-hospitalisation* (text word)
16	10 OR 11 OR 12 OR 13 OR 14 OR 15
17	5 AND 9 AND 16
18	Filters: published in the last 10 years; Danish; English; Norwegian;

Search string with subject headings used in PubMed

1	Patient Discharge/organization and administration (MESH)
2	Transitional Care/organization and administration (MESH)
3	Transitional care/trends (MESH)
4	1 OR 2 OR 3
5	Readmission/organization and administration (MESH)
6	Patient Readmission/trends (MESH)
7	5 OR 6
8	Aged (MESH)
9	Aged, 80 and over (MESH)
10	4 AND 7 AND 9
12	Filters: published in the last 10 years; Danish; English; Norwegian;

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Quality Assessment Tool for Quantitative Studies Dictionary



The purpose of this dictionary is to describe items in the tool thereby assisting raters to score study quality. Due to under-reporting or lack of clarity in the primary study, raters will need to make judgements about the extent that bias may be present. When making judgements about each component, raters should form their opinion based upon information contained in the study rather than making inferences about what the authors intended. Mixed methods studies can be quality assessed using this tool with the quantitative component of the study.

A) SELECTION BIAS

(Q1) Participants are more likely to be representative of the target population if they are randomly selected from a comprehensive list of individuals in the target population (score very likely). They may not be representative if they are referred from a source (e.g. clinic) in a systematic manner (score somewhat likely) or self-referred (score not likely).

(Q2) Refers to the % of subjects in the control and intervention groups that agreed to participate in the study before they were assigned to intervention or control groups.

B) STUDY DESIGN

In this section, raters assess the likelihood of bias due to the allocation process in an experimental study. For observational studies, raters assess the extent that assessments of exposure and outcome are likely to be independent. Generally, the type of design is a good indicator of the extent of bias. In stronger designs, an equivalent control group is present and the allocation process is such that the investigators are unable to predict the sequence.

Randomized Controlled Trial (RCT)

An experimental design where investigators randomly allocate eligible people to an intervention or control group. A rater should describe a study as an RCT if the randomization sequence allows each study participant to have the same chance of receiving each intervention and the investigators could not predict which intervention was next. If the investigators do not describe the allocation process and only use the words 'random' or 'randomly', the study is described as a controlled clinical trial.

See below for more details.

Was the study described as randomized?

Score YES, if the authors used words such as random allocation, randomly assigned, and random assignment.

Score NO, if no mention of randomization is made.

Was the method of randomization described?

Score YES, if the authors describe any method used to generate a random allocation sequence.

Score NO, if the authors do not describe the allocation method or describe methods of allocation such as alternation, case record numbers, dates of birth, day of the week, and any allocation procedure that is entirely transparent before assignment, such as an open list of random numbers of assignments.
If NO is scored, then the study is a controlled clinical trial.

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Was the method appropriate?

Score YES, if the randomization sequence allowed each study participant to have the same chance of receiving each intervention and the investigators could not predict which intervention was next. Examples of appropriate approaches include assignment of subjects by a central office unaware of subject characteristics, or sequentially numbered, sealed, opaque envelopes.

Score NO, if the randomization sequence is open to the individuals responsible for recruiting and allocating participants or providing the intervention, since those individuals can influence the allocation process, either knowingly or unknowingly.

If NO is scored, then the study is a controlled clinical trial.

Controlled Clinical Trial (CCT)
An experimental study design where the method of allocating study subjects to intervention or control groups is open to individuals responsible for recruiting subjects or providing the intervention. The method of allocation is transparent before assignment, e.g. an open list of random numbers or allocation by date of birth, etc.

Cohort analytic (two group pre and post)
An observational study design where groups are assembled according to whether or not exposure to the intervention has occurred. Exposure to the intervention is not under the control of the investigators. Study groups might be non-equivalent or not comparable on some feature that affects outcome.

Case control study
A retrospective study design where the investigators gather 'cases' of people who already have the outcome of interest and 'controls' who do not. Both groups are then questioned or their records examined about whether they received the intervention exposure of interest.

Cohort (one group pre + post (before and after)
The same group is pretested, given an intervention, and tested immediately after the intervention. The intervention group, by means of the pretest, act as their own control group.

Interrupted time series
A study that uses observations at multiple time points before and after an intervention (the 'interruption'). The design attempts to detect whether the intervention has had an effect significantly greater than any underlying trend over time. Exclusion: Studies that do not have a clearly defined point in time when the intervention occurred and at least three data points before and three after the intervention

Other:
One time surveys or interviews

C) CONFOUNDERS

By definition, a confounder is a variable that is associated with the intervention or exposure and causally related to the outcome of interest. Even in a robust study design, groups may not be balanced with respect to important variables prior to the intervention. The authors should indicate if confounders were controlled in the design (by stratification or matching) or in the analysis. If the allocation to intervention and control groups is randomized, the authors must report that the groups were balanced at baseline with respect to confounders (either in the text or a table).

D) BLINDING

(Q1) Assessors should be described as blinded to which participants were in the control and intervention groups. The purpose of blinding the outcome assessors (who might also be the care providers) is to protect against detection bias.

(Q2) Study participants should not be aware of (i.e. blinded to) the research question. The purpose of blinding the participants is to protect against reporting bias.

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E) DATA COLLECTION METHODS

Tools for primary outcome measures must be described as reliable and valid. If 'face' validity or 'content' validity has been demonstrated, this is acceptable. Some sources from which data may be collected are described below:

Self reported data includes data that is collected from participants in the study (e.g. completing a questionnaire, survey, answering questions during an interview, etc.).

Assessment/Screening includes objective data that is retrieved by the researchers. (e.g. observations by investigators).

Medical Records/Vital Statistics refers to the types of formal records used for the extraction of the data.

Reliability and validity can be reported in the study or in a separate study. For example, some standard assessment tools have known reliability and validity.

F) WITHDRAWALS AND DROP-OUTS

Score YES if the authors describe BOTH the numbers and reasons for withdrawals and drop-outs.

Score NO if either the numbers or reasons for withdrawals and drop-outs are not reported.

Score NOT APPLICABLE if the study was a one-time interview or survey where there was not follow-up data reported.

The percentage of participants completing the study refers to the % of subjects remaining in the study at the final data collection period in all groups (i.e. control and intervention groups).

G) INTERVENTION INTEGRITY

The number of participants receiving the intended intervention should be noted (consider both frequency and intensity). For example, the authors may have reported that at least 80 percent of the participants received the complete intervention. The authors should describe a method of measuring if the intervention was provided to all participants the same way. As well, the authors should indicate if subjects received an unintended intervention that may have influenced the outcomes. For example, co-intervention occurs when the study group receives an additional intervention (other than that intended). In this case, it is possible that the effect of the intervention may be over-estimated. Contamination refers to situations where the control group accidentally receives the study intervention. This could result in an under-estimation of the impact of the intervention.

H) ANALYSIS APPROPRIATE TO QUESTION

Was the quantitative analysis appropriate to the research question being asked?

An intention-to-treat analysis is one in which all the participants in a trial are analyzed according to the intervention to which they were allocated, whether they received it or not. Intention-to-treat analyses are favoured in assessments of effectiveness as they mirror the noncompliance and treatment changes that are likely to occur when the intervention is used in practice, and because of the risk of attrition bias when participants are excluded from the analysis.

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Component Ratings of Study:

For each of the six components A – F, use the following descriptions as a roadmap.

A) SELECTION BIAS

Good: The selected individuals are very likely to be representative of the target population (Q1 is 1) and there is greater than 80% participation (Q2 is 1).

Fair: The selected individuals are at least somewhat likely to be representative of the target population (Q1 is 1 or 2); and there is 60 - 79% participation (Q2 is 2). 'Moderate' may also be assigned if Q1 is 1 or 2 and Q2 is 5 (can't tell).

Poor: The selected individuals are not likely to be representative of the target population (Q1 is 3); or there is less than 60% participation (Q2 is 3) or selection is not described (Q1 is 4); and the level of participation is not described (Q2 is 5).

B) DESIGN

Good: will be assigned to those articles that described RCTs and CCTs.

Fair: will be assigned to those that described a cohort analytic study, a case control study, a cohort design, or an interrupted time series.

Weak: will be assigned to those that used any other method or did not state the method used.

C) CONFOUNDERS

Good: will be assigned to those articles that controlled for at least 80% of relevant confounders (Q1 is 2); or (Q2 is 1).

Fair: will be given to those studies that controlled for 60 – 79% of relevant confounders (Q1 is 1) and (Q2 is 2).

Poor: will be assigned when less than 60% of relevant confounders were controlled (Q1 is 1) and (Q2 is 3) or control of confounders was not described (Q1 is 3) and (Q2 is 4).

D) BLINDING

Good: The outcome assessor is not aware of the intervention status of participants (Q1 is 2); and the study participants are not aware of the research question (Q2 is 2).

Fair: The outcome assessor is not aware of the intervention status of participants (Q1 is 2); or the study participants are not aware of the research question (Q2 is 2).

Poor: The outcome assessor is aware of the intervention status of participants (Q1 is 1); and the study participants are aware of the research question (Q2 is 1); or blinding is not described (Q1 is 3 and Q2 is 3).

E) DATA COLLECTION METHODS

Good: The data collection tools have been shown to be valid (Q1 is 1); and the data collection tools have been shown to be reliable (Q2 is 1).

Fair: The data collection tools have been shown to be valid (Q1 is 1); and the data collection tools have not been shown to be reliable (Q2 is 2) or reliability is not described (Q2 is 3).

Poor: The data collection tools have not been shown to be valid (Q1 is 2) or both reliability and validity are not described (Q1 is 3 and Q2 is 3).

F) WITHDRAWALS AND DROP-OUTS - a rating of:

Good: will be assigned when the follow-up rate is 80% or greater (Q1 is 1 and Q2 is 1).

Fair: will be assigned when the follow-up rate is 60 – 79% (Q2 is 2) OR Q1 is 4 or Q2 is 5.

Poor: will be assigned when a follow-up rate is less than 60% (Q2 is 3) or if the withdrawals and drop-outs were not described (Q1 is No or Q2 is 4).

Not Applicable: if Q1 is 4 or Q2 is 5.

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QUALITY ASSESSMENT TOOL FOR QUANTITATIVE STUDIES



COMPONENT RATINGS

A) SELECTION BIAS

(Q1) Are the individuals selected to participate in the study likely to be representative of the target population?

1 Very likely
2 Somewhat likely
3 Not likely
4 Can't tell

(Q2) What percentage of selected individuals agreed to participate?

1 80 - 100% agreement
2 60 – 79% agreement
3 less than 60% agreement
4 Not applicable
5 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

B) STUDY DESIGN

Indicate the study design

1 Randomized controlled trial
2 Controlled clinical trial
3 Cohort analytic (two group pre + post)
4 Case-control
5 Cohort (one group pre + post (before and after))
6 Interrupted time series
7 Other specify _____
8 Can't tell

Was the study described as randomized? If NO, go to Component C.

No Yes

If Yes, was the method of randomization described? (See dictionary)

No Yes

If Yes, was the method appropriate? (See dictionary)

No Yes

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

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C) CONFOUNDERS

(Q1) Were there important differences between groups prior to the intervention?

1 Yes
2 No
3 Can't tell

The following are examples of confounders:

1 Race
2 Sex
3 Marital status/family
4 Age
5 SES (income or class)
6 Education
7 Health status
8 Pre-intervention score on outcome measure

(Q2) If yes, indicate the percentage of relevant confounders that were controlled (either in the design (e.g. stratification, matching) or analysis)?

1 80 – 100% (most)
2 60 – 79% (some)
3 Less than 60% (few or none)
4 Can't Tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

D) BLINDING

(Q1) Was (were) the outcome assessor(s) aware of the intervention or exposure status of participants?

1 Yes
2 No
3 Can't tell

(Q2) Were the study participants aware of the research question?

1 Yes
2 No
3 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

E) DATA COLLECTION METHODS

(Q1) Were data collection tools shown to be valid?

1 Yes
2 No
3 Can't tell

(Q2) Were data collection tools shown to be reliable?

1 Yes
2 No
3 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

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F) WITHDRAWALS AND DROP-OUTS

(Q1) Were withdrawals and drop-outs reported in terms of numbers and/or reasons per group?

1 Yes
2 No
3 Can't tell
4 Not Applicable (i.e. one time surveys or interviews)

(Q2) Indicate the percentage of participants completing the study. (If the percentage differs by groups, record the lowest).

1 80 -100%
2 60 - 79%
3 less than 60%
4 Can't tell
5 Not Applicable (i.e. Retrospective case-control)

RATE THIS SECTION	STRONG	MODERATE	WEAK	
See dictionary	1	2	3	Not Applicable

G) INTERVENTION INTEGRITY

(Q1) What percentage of participants received the allocated intervention or exposure of interest?

1 80 -100%
2 60 - 79%
3 less than 60%
4 Can't tell

(Q2) Was the consistency of the intervention measured?

1 Yes
2 No
3 Can't tell

(Q3) Is it likely that subjects received an unintended intervention (contamination or co-intervention) that may influence the results?

4 Yes
5 No
6 Can't tell

H) ANALYSES

(Q1) Indicate the unit of allocation (circle one)

community organization/institution practice/office individual

(Q2) Indicate the unit of analysis (circle one)

community organization/institution practice/office individual

(Q3) Are the statistical methods appropriate for the study design?

1 Yes
2 No
3 Can't tell

(Q4) Is the analysis performed by intervention allocation status (i.e. intention to treat) rather than the actual intervention received?

1 Yes
2 No
3 Can't tell

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GLOBAL RATING

COMPONENT RATINGS

Please transcribe the information from the gray boxes on pages 1-4 onto this page. See dictionary on how to rate this section.

A	**SELECTION BIAS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
B	**STUDY DESIGN**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
C	**CONFOUNDERS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
D	**BLINDING**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
E	**DATA COLLECTION METHOD**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
F	**WITHDRAWALS AND DROPOUTS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	Not Applicable

GLOBAL RATING FOR THIS PAPER (circle one):

1	STRONG	(no WEAK ratings)
2	MODERATE	(one WEAK rating)
3	WEAK	(two or more WEAK ratings)

With both reviewers discussing the ratings:

Is there a discrepancy between the two reviewers with respect to the component (A-F) ratings?

No Yes

If yes, indicate the reason for the discrepancy

1 Oversight
2 Differences in interpretation of criteria
3 Differences in interpretation of study

Final decision of both reviewers (circle one):

1 STRONG
2 MODERATE
3 WEAK

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Supplementary 4, exclusion of full text

Altfeld S.J., Shier G.E., Rooney M., Johnson T.J., Golden R.L., Karavolos K., Avery E., Nandi V. & Perry, A.J. 2013, "Effects of an enhanced discharge planning intervention for hospitalized older adults: a randomized trial.", The Gerontologist, vol. 53, no. 3, pp. 430-440. **Reason for exclusion:** not only medical patients.

Alvarez R, Ginsburg J, Grabowski J, Post S, Rosenberg W. The Social Work Role in Reducing 30-Day Readmissions: The Effectiveness of the Bridge Model of Transitional Care. J Gerontol Soc Work 2016 Apr;59(3):222-227. **Reason for exclusion:** not an intervention study.

Arbaje A.I., Maron D.D., Yu Q., Wendel V.I., Tanner E., Boult C., Eubank K.J. & Durso, S.C. 2010, "The geriatric floating interdisciplinary transition team.", Journal of the American Geriatrics Society, vol. 58, no. 2, pp. 364-370. **Reason for exclusion:** readmission is not an outcome.

Arnold ME, Buys L, Fullas F. Impact of pharmacist intervention in conjunction with outpatient physician follow-up visits after hospital discharge on readmission rate (Reprinted from Journal of Health-System Pharmacy Residents JHPR, vol 2, issue 4, 2014). American Journal of Health-System Pharmacy 2015 JUN 1;72(11):S36-S42. **Reason for exclusion:** only post-discharge elements, not only medical patients.

Arevalo-Flechas, L.C., Noel, P., Rottman-Sagebiel, R., Cupples, N., Conde, A., Sanchez-Reilly, S., Powers, B. & Espinoza, S. 2016, "Presence of a Caregiver Associated with Increased Odds of Readmission in a Cohort of Older Adults Included in a Transitional Care Intervention", Journal of the American Geriatrics Society, vol. 64, pp. S141-S141. **Reason for exclusion:** meeting abstract.

Bellon, J.E., Bilderback, A., Ahuja-Yende, N.S., Wilson, C., Altieri Dunn, S.C., Brodine, D. & Boninger, M.L. 2018, "University of Pittsburgh Medical Center Home Transitions Multidisciplinary Care Coordination Reduces Readmissions for Older Adults", Journal of the American Geriatrics Society. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Burt S, Berry D, Quackenbush P. Implementation of Transition in Care and Relationship Based Care to Reduce Preventable Rehospitalizations. Home Healthc Now 2015 Jul-Aug;33(7):390-393. **Reason for exclusion:** not an intervention study.

Cheen MHH, Goon CP, Ong WC, Lim PS, Wan CN, Leong MY, et al. Evaluation of a care transition program with pharmacist-provided home-based medication review for elderly Singaporeans at high risk of readmissions. Int J Qual Health Care 2017 Apr 1;29(2):200-205. **Reason for exclusion:** not only medical patients, only post-discharge elements, mean age < 75 year or age < 65 years.

Damiani G, Federico B, Venditti A, Sicuro L, Rinaldi S, Cirio F, et al. Hospital discharge planning and continuity of care for aged people in an Italian local health unit: does the care-home model reduce hospital readmission and mortality rates? BMC Health Serv Res 2009 Feb 4;9:22-6963-9-22. **Reason for exclusion:** not able to obtain data from the authors.

Dedhia P., Kravet S., Bulger J., Hinson T., Sridharan A., Kolodner K., Wright S. & Howell, E. 2009, "A quality improvement intervention to facilitate the transition of older adults from three hospitals back to their homes.", Journal of the American Geriatrics Society, vol. 57, no. 9, pp. 1540-1546. **Reason for exclusion:** only pre-discharge elements.

Dizon, M.L. & Reinking, C. 2017, "Reducing Readmissions: Nurse-Driven Interventions in the Transition of Care From the Hospital", Worldviews on evidence-based nursing, vol. 14, no. 6, pp. 432-439. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Edmans J, Bradshaw L, Franklin M, Gladman J, Conroy S. Specialist geriatric medical assessment for patients discharged from hospital acute assessment units: randomised controlled trial. BMJ 2013 Oct 8;347:f5874. **Reason for exclusion:** not able to obtain data from the authors.

Farrell T.W., TomoaiaCotisel A., Scammon D.L., Brunisholz K., Kim J., Day J., Gren L.H., Wallace S., Gunning K., Tabler J. & Magill, M.K. 2015, "Impact of an integrated transition management program

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in primary care on hospital readmissions.", Journal for healthcare quality : official publication of the National Association for Healthcare Quality, vol. 37, no. 1, pp. 81-92. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Field, T.S., Ogarek, J., Garber, L., Reed, G. & Gurwitz, J.H. 2015, "Association of early post-discharge follow-up by a primary care physician and 30-day rehospitalization among older adults", Journal of general internal medicine, vol. 30, no. 5, pp. 565-571. **Reason for exclusion:** not an intervention study.

Gardner R, Li Q, Baier RR, Butterfield K, Coleman EA, Gravenstein S. Is implementation of the care transitions intervention associated with cost avoidance after hospital discharge? J Gen Intern Med 2014 Jun;29(6):878-884. **Reason for exclusion:** readmission is not an outcome.

Gurwitz, J.H., Field, T.S., Ogarek, J., Tjia, J., Cutrona, S.L., Harrold, L.R., Gagne, S.J., Preusse, P., Donovan, J.L., Kanaan, A.O., Reed, G. & Garber, L. 2014, "An Electronic Health Record-Based Intervention to Increase Follow-Up Office Visits and Decrease Rehospitalization in Older Adults", Journal of the American Geriatrics Society, vol. 62, no. 5, pp. 865-871. **Reason for exclusion:** not only medical patients.

Hamar, B., Rula, E.Y., Wells, A.R., Coberley, C., Pope, J.E. & Varga, D. 2016, "Impact of a Scalable Care Transitions Program for Readmission Avoidance", American Journal of Managed Care, vol. 22, no. 1, pp. 28-34. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Harrison, J.D., Auerbach, A.D., Quinn, K., Kynoch, E. & Mourad, M. 2014, "Assessing the impact of nurse post-discharge telephone calls on 30-day hospital readmission rates", JGIM: Journal of General Internal Medicine, vol. 29, no. 11, pp. 1519-1525. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Hendrix C., Tepfer S., Forest S., Ziegler K., Fox V., Stein J., McConnell E.S., Hastings S.N., Schmader K. & ColonEmeric, C. 2013, "Transitional Care Partners: a hospital-to-home support for older adults and their caregivers.", Journal of the American Association of Nurse Practitioners, vol. 25, no. 8, pp. 407-414. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Heppenstall C.P., Hanger H.C., Wilkinson T.J. & Dhanak, M. 2018, "Telephone discharge support for frail, vulnerable older people discharged from hospital: Impact on readmission rates - Participant and general practitioner feedback.", Australasian journal on ageing, vol. 37, no. 2, pp. 107-112. **Reason for exclusion:** only post-discharge elements.

Hitch B, Parlier AB, Reed L, Galvin SL, Fagan EB, Wilson CG. Evaluation of a Team-Based, Transition-of-Care Management Service on 30-Day Readmission Rates. N C Med J 2016 Mar-Apr;77(2):87-92. **Reason for exclusion:** not only medical patients, mean age < 75 year or age < 65 years.

Hung D. & Leidig, R.C. 2015, "Implementing a transitional care program to reduce hospital readmissions among older adults.", Journal of nursing care quality, vol. 30, no. 2, pp. 121-129. **Reason for exclusion:** not an intervention study.

Hung D., Truong Q., Yakir M. & Nicosia, F. 2018, "Hospital-Community Partnerships to Aid Transitions for Older Adults: Applying the Care Transitions Framework.", Journal of nursing care quality, vol. 33, no. 3, pp. 221-228. **Reason for exclusion:** not an intervention study.

Hwang, U., Dresden, S.M., Rosenberg, M.S., Garrido, M.M., Loo, G., Sze, J., Gravenor, S., Courtney, D.M., Kang, R., Zhu, C.W., Vargas-Torres, C., Grudzen, C.R., Richardson, L.D. & The GEDI, W.I. 2018, "Geriatric Emergency Department Innovations: Transitional Care Nurses and Hospital Use", Journal of the American Geriatrics Society, vol. 66, no. 3, pp. 459-466. **Reason for exclusion:** only pre-discharge elements.

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Expert-by-experience, and Novice Nurses, to Hospitalized Elders with Chronic Healthcare Conditions", Pacific Rim International Journal of Nursing Research, vol. 16, no. 4, pp. 343-360. **Reason for exclusion:** not an intervention study.

Jenq G.Y., Doyle M.M., Belton B.M., Herrin J. & Horwitz, L.I. 2016, "Quasi-experimental evaluation of the effectiveness of a large-scale readmission reduction program.", JAMA Internal Medicine, vol. 176, no. 5, pp. 681-690. **Reason for exclusion:** not able to obtain data from the authors.

Kane R.L., Huckfeldt P., Tappen R., Engstrom G., Rojido C., Newman D., Yang Z. & Ouslander, J.G. 2017, "Effects of an intervention to reduce hospitalizations from nursing homes: A randomized implementation trial of the INTERACT program.", JAMA Internal Medicine, vol. 177, no. 9, pp. 1257-1264. **Reason for exclusion:** only post-discharge elements.

Kirkham, H., Clark, B., Paynter, J., Lewis, G. & Duncan, I. 2014, "The effect of a collaborative pharmacist-hospital care transition program on the likelihood of 30-day readmission", American Journal of Health-System Pharmacy, vol. 71, no. 9, pp. 739-745. **Reason for exclusion:** not only medical patients, mean age < 75 year or age < 65 years.

Latour, C.H., Bosmans, J.E., van Tulder, M.W., de Vos, R., Huyse, F.J., de Jonge, P., van Gemert, L.A. & Stalman, W.A. 2007, "Cost-effectiveness of a nurse-led case management intervention in general medical outpatients compared with usual care: an economic evaluation alongside a randomized controlled trial", Journal of psychosomatic research, vol. 62, no. 3, pp. 363-370. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Legrain, S., Tubach, F., Bonnet-Zamponi, D., Lemaire, A., Aquino, J., Paillaud, E., Taillandier-Heriche, E., Thomas, C., Verny, M., Pasquet, B., Moutet, A.L., Lieberherr, D. & Lacaille, S. 2011, "A New Multimodal Geriatric Discharge-Planning Intervention to Prevent Emergency Visits and Rehospitalizations of Older Adults: The Optimization of Medication in AGEd Multicenter Randomized Controlled Trial", Journal of the American Geriatrics Society, vol. 59, no. 11, pp. 2017-2028. **Reason for exclusion:** only pre-discharge elements.

Leung D.Y.P., Lee D.T.F., Lee I.F.K., Lam L.W., Lee S.W.Y., Chan M.W.M., Lam Y.M., Leung S.H., Chiu P.C., Ho N.K.F., Ip M.F. & Hui, M.M.Y. 2015, "The effect of a virtual ward program on emergency services utilization and quality of life in frail elderly patients after discharge: A pilot study.", Clinical Interventions in Aging, vol. 10, pp. 413-420. Participant age unknown, outcome readmission **Reason for exclusion:** not able to obtain data from the authors.

Lin, C.Y., Barnato, A.E. & Degenholtz, H.B. 2011, "Physician Follow-Up Visits After Acute Care Hospitalization for Elderly Medicare Beneficiaries Discharged to Noninstitutional Settings", Journal of the American Geriatrics Society, vol. 59, no. 10, pp. 1947-1954. **Reason for exclusion:** not an intervention study.

Mabire C., Bula C., Morin D. & Goulet, C. 2015, "Nursing discharge planning for older medical inpatients in Switzerland: A cross-sectional study"., Geriatric nursing (New York, N.Y.), vol. 36, no. 6, pp. 451-457. **Reason for exclusion:** not an intervention study.

Moore A.B., Krupp J.E., Dufour A.B., Sircar M., Travison T.G., Abrams A., Farris G., Mattison M.L.P. & Lipsitz, L.A. 2017, "Improving Transitions to Postacute Care for Elderly Patients Using a Novel Video-Conferencing Program: ECHO-Care Transitions.", American Journal of Medicine, vol. 130, no. 10, pp. 1199-1204. **Reason for exclusion:** not only medical patients.

Morrison, J., Palumbo, M.V. & Rambur, B. 2016, "Reducing Preventable Hospitalizations With Two Models of Transitional Care", Journal of Nursing Scholarship, vol. 48, no. 3, pp. 322-329. **Reason for exclusion:** no "usual care" control group.

Naylor MD, Bowles KH, McCauley KM, Maccoy MC, Maislin G, Pauly MV, et al. High-value transitional care: translation of research into practice. J Eval Clin Pract 2013 Oct;19(5):727-733. **Reason for exclusion:** not able to obtain data from the authors.

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Naylor M.D., Hirschman K.B., Hanlon A.L., Bowles K.H., Bradway C., Mccauley K.M. & Pauly, M.V. 2014, "Comparison of evidence-based interventions on outcomes of hospitalized, cognitively impaired older adults.", Journal of Comparative Effectiveness Research, vol. 3, no. 3, pp. 245-257. **Reason for exclusion:** only includes cognitively impaired patients.

Nazir A., Khan B., Counsell S., Henderson M., Gao S. & Boustani, M. 2015, "Impact of an inpatient geriatric consultative service on outcomes for cognitively impaired patients.", Journal of Hospital Medicine, vol. 10, no. 5, pp. 275-280. **Reason for exclusion:** only pre-discharge elements.

Occelli, P., Touzet, S., Rabilloud, M., Ganne, C., Bourdy, S.P., Galamand, B., Debray, M., Dartiguepeyrou, A., Chuzeville, M., Comte, B., Turkie, B., Tardy, M., Luiggi, J., Jacquet-Francillon, T., Gilbert, T., Bonnefoy, M. & Poupon Bourdy, S. 2016, "Impact of a transition nurse program on the prevention of thirty-day hospital readmissions of elderly patients discharged from short-stay units: study protocol of the PROUST stepped-wedge cluster randomised trial", BMC Geriatrics, vol. 16, pp. 1-9. **Reason for exclusion:** study protocol.

Ohuabunwa U., Jordan Q., Shah S., Fost M. & Flacker, J. 2013, "Implementation of a care transitions model for low-income older adults: A high-risk, vulnerable population.", Journal of the American Geriatrics Society, vol. 61, no. 6, pp. 987-992. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Ohuabunwa, U., Johnson, E., Cruel, S., Turner, J. & Jordan, Q. 2016, "Care Transitions: an Integrated Model of Care for Elderly Patients at High Risk for Readmissions", Gerontologist, vol. 56, pp. 190-190. **Reason for exclusion:** meeting abstract.

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Supplementary 5, intervention characteristics

Study, year	Title	Description of intervention components	Providers	Context	Length of support	Treatment of comparison group
Buurman et al.[24] **2016**	Transitional Care Bridge Program Intervention	Pre-discharge: *≤ 48 hours after hospital admission*: - participants received a systematic CGA. - Care and treatment plan with prioritazion by patient and geriatric team -Multidisciplinary care provided by geriatric team Bridging: - In-person handover of CGA and care treatment plan by geriatric team to CCRN - Visit of CCRN to participant Post-discharge: *≤ 2 days after hospital discharge:* - Home visit or visit to the NH, with medication reconciliation; initiate home care; check for activities of daily living and availability of adequate help; target geriatric conditions; contact GP or NH physician *2 weeks after hospital discharge*: - Home visit or visit to the NH, evaluation of hospital-based care and treatment plan, goal setting by participant, protocol adherence for incident and prevalent geriatric conditions *6 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on the proxy and caregiver burden; further actions on geriatric conditions *12 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on care needs for incident and prevalent geriatric conditions *24 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on care needs; evaluation of geriatric conditions; handover to GP	Geriatric team A community care registered nurse (CCRN)	Three hospitals with affiliated home care organizations in the Netherlands	From ≤ 48 hours after hospital admission to 24 weeks after hospital discharge	CGA-Only Control Arm (usual care): - a systematic CGA - care and treatment plan with prioritazion by patient and geriatric team -multidisciplinary care provided by geriatric team
Chow et al.[22]	The nurse case management intervention	Two intervention arms: 1. "Home Visit" 2. "Call"	Senior year nursing students	Patients were recruited from the medical	From enrolment until 4 weeks after discharge	Social calls twice within 4 weeks.

2014		Pre-discharge in both "Home Visit" and "Call" arms: - Pre-discharge assessment: the Problem Classification Scheme, Intervention Scheme and Problem Rating Scale for Outcomes (e.g. analysis of the barriers and developed mutual goals with the patients to ensure that they were able to perform health maintenance activities, such as nutrition, monitoring of symptoms and medication adherence) Bridging: None Post-discharge - Home Visit arm: - First visit within 72 hours of discharge. - Week 2: a telephone call to evaluate the interventions and advice given during the home visit - Week 3: home visit to examine whether patients were able to continue managing their health needs after hospital discharge. - Week 4: final telephone call in the fourth week post discharge to remind patients about adherence behaviors and to motivate and support them before concluding the interventions Post-discharge – Call arm: - Week 1: first telephone call based on the patients' needs, which had been identified prior to discharge. - Week 2. call the patients: referred to the mutual goals developed by the NCM and the patients to continue with the interventions. - Week 3. call the patients referred to the mutual goals developed by the NCM and the patients to continue with the interventions. - Week 4: closing call to motivate and support the patients in maintaining self-management behaviors	Nurse Case Managers (NCM)	department of a 1700-bedded acute, general regional hospital in Hong Kong		No assessments and related interventions were conducted to the control group.
Courtney et al.[25] **2009**	Older Hospitalised Patients' Discharge Planning and In-home	Pre-discharge: *Exercise components*: - Muscle stretching, balance training, walking for endurance, and muscle strengthening using resistance exercises	A registered nurse (RN) A physio-therapist	A tertiary referral hospital in Brisbane, Australia	From enrolment until 6 months after discharge	Control group received the routine care, discharge planning, and rehabilitation advice normally provided. If

	Follow-up Protocol (OHP-DP)	*Nurse components:* - The nurse visited daily during participants' hospital stays to address concerns, facilitate the exercise program, and oversee discharge planning - Developed a transitional care plan covering the areas of functional ability and need for assistance with activities of daily living, post-discharge treatments and follow-up care, social support, chronic disease management plans and information, medication information, community services, and assistance with the exercise program. <u>Bridging</u>: - Written guidelines were provided on post-discharge management, including diagrams and specific instructions for their exercise program <u>Post-discharge</u>: - Within 48 hours of discharge, the nurse undertook a home visit to assess availability of support, address transitional concerns, provide advice and support, and ensure that the exercise program could be safely undertaken at home. - Extra home visits were provided if required - Weekly follow-up telephone calls were provided for 4 weeks, followed by monthly follow-up for a further 5 months				in-home follow-up was necessary, it was organized in the routine manner (e.g., referral to community health services)
Finlayson et al.[23] **2018**		Three intervention arms: 1. "Exercise" 2. "Nurse Home visit and Telephone follow-up" (N-HaT) 3. "Exercise and Nurse Home Visit and Telephone follow up" (ExN-HaT) <u>Exercise arm:</u> *Pre-discharge*: - Within 72 hours of admission, a physiotherapist will visit measure ADL and conduct performance tests of balance and gait. This information will be used to plan individualised exercise programs designed to improve strength, stability, coordination, endurance, mobility, and improve self confidence with respect to ADL. The exercise prescription will be developed by the patient, care givers, doctors, and ward nurses. -Physiotherapist will provide a pedometer	Advanced Practice Gerontic Nurse (APGN) A physiotherapist or exercise physiologist	Two tertiary metropolitan hospitals in Australia	From enrolment until 24 weeks after discharge	Routine hospital and follow-up care as provided by the health service. This involved a needs assessment by the hospital health staff, discharge planning, and referrals for follow up services as appropriate.

		together with simple verbal and written instructions regards usage and journaling of activity levels - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required *Bridging:* None *Post-discharge:* - An exercise physiologist conducts 6 weekly in home visits to reassess the patient's physical measures and functional capacity, evaluate progress with the exercise program, and reset program goals accordingly <u>N-HaT arm:</u> *Pre-discharge*: - Within 72 hours of admission, an APGN will undertake a health assessment, and prepare a comprehensive transitional care plan. This process includes discussing the planned discharge plan with patient's caregivers, doctor, and ward nurses to individualise the plan - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required. *Bridging:* None *Post-discharge:* - Within 48 hours after discharge, an APGN conducts a home visit. This home visit is undertaken to 1) identify sufficient caregiver support is available at home, 2) assess the home environment is safe, 3) determine the patient has the required medications and dressings (if required), 4) ensure the patient (and caregiver) fully understand their medication and treatment regimes; 5) reinforce and further explain exercise program and use of pedometer and journal in the home, and 6) provide advice and support to the caregiver				

		-Additional home visits are available by APGN if required - over the initial 4 week period after discharge, a APGN provides weekly telephone follow-up calls -APGN is available to be contacted via telephone 7 days per week - For 6 months, monthly telephone follow-up calls are conducted by the APGN. <u>ExN-HaT arm</u>: *Pre-discharge*: - Within 72 hours of admission, an APGN will undertake a health assessment, and prepare a comprehensive transitional care plan. This process includes discussing the planned discharge plan with patient's caregivers, doctor, and ward nurses to individualise the plan - Within 72 hours of admission, a physiotherapist will visit the intervention patients to determine the functional capacity of the patient using measures of ADL and performance tests of balance and gait. This information will be used to plan individualised exercise programs designed to improve strength, stability, coordination, endurance, mobility, and improve self confidence with respect to ADL. The exercise prescription will be developed by the patient, caregivers, doctors, and ward nurses. - Physiotherapist will provide a pedometer together with simple verbal and written instructions regards usage and journaling of activity levels - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required. *Bridging:* None *Post-discharge:* - Within 48 hours post discharge, an APGN provides one home visit. This home visit is undertaken to 1) identify sufficient caregiver support is available at home, 2) assess the home environment is safe, 3) determine the patient has the required medications and dressings (if				

		required), 4) ensure the patient (and caregiver) fully understand their medication and treatment regimes; 5) reinforce and further explain exercise program and use of pedometer and journal in the home, and 6) provide advice and support to the caregiver - Additional home visits are available by APGN if required - An exercise physiologist conducts 6 weekly visits to reassess the patient's physical measures and functional capacity, evaluate progress with the exercise program, and reset program goals accordingly - Over the initial 4 week period after discharge, a APGN provides weekly telephone follow-up calls - APGN is available to be contacted via telephone 7 days per week - For 6 months, monthly telephone follow-up calls are conducted by the APGN				
Koehler et al.[26] **2009**	Early Care Bundle or Supplemental Care Bundle	<u>Pre-discharge</u>: By Study Care Coordinator - Daily condition specific education - Additional time to identify and address discharge barriers - Extra discharge teaching with focus on self-management By Study Pharmacist: - Medication reconciliation at admit - Daily medication review and education - Additional medication change recommendations as felt to be indicated - Medication reconciliation at discharge - Counselling on discharge Structured documents: - personal health record (completed by the patient and family for future use) <u>Bridging</u>: Structured documents: - Supplemental discharge form given to patient and faxed to PCP <u>Post-discharge</u>: By Study Care Coordinator:	Care coordinators (CC) Clinical pharmacists (CP)	1 of 2 hospital-medicine groups at the 900-bed Baylor University Medical Center, Texas	Max 24 hours after enrolment to 1 week after discharge	Usual care: - Needs assessment - discharge planning - Medication reconciliation at admit - Daily direct patient care - Discharge medication education - Medication reconciliation at discharge - Discharge education - Medication review with new orders and as needed - Medication change recommendations - Standard BUMC discharge form - New discharge prescriptions

		- Follow up call at 5-7 days post-discharge By Study Pharmacist: - Follow up call at 5-7 days post-discharge				
Lin et al.[31] **2015**	Integrated Care and Discharge Support for elderly patients (ICDS)	Pre-discharge: - Risk stratification - Comprehensive geriatric assessment - Discharge planning are performed Bridging: None Post-discharge: Patients receive either: - ICM Case Management or - HST service *ICM Case Management* : - Each high-risk older patient is followed up by a case manager for a period of around 3 months following hospital discharge - Case managers provide home visits and telephone support and they are responsible for community service coordination and ensuring patient compliance with planned services and management. *HST service*: - Rapid and intensive community support for discharged patients, offering services such as meal delivery, household cleaning, respite care, and home assessment and modification Link nurses, ICM case managers, and the HST hold weekly multidisciplinary case conferences chaired by an ICM geriatrician. If needed, referral for rehabilitation in a geriatric day hospital, fast track clinic, or early specialist clinic follow-up can be offered to patients.	At hospital: - Link nurses - Geriatricians Community ICM Case Management: - Social workers - Physiotherapist (PT) - Occupational therapist (OT) - Advanced practice nurse (APN) Community HST service: - Nurses - PT - OT - Other allied health members	Four hospitals of Hong Kong West Cluster.	ICM case management: - From recruitment until 3 months after discharge HST service: - NR	NR
Nielsen et al.[28] **2018**	Elderly Activity Performance Intervention	Pre-discharge: *Component 1:* -Assessment of the patients' performance of daily activities - Assessment of Motor and Process Skills Based on the results in component 1, patients with limitations in performing daily activities	Occupational therapist (OT) Physiotherapist	Short-stay unit (ED) at a 1,150-bed university hospital in Denmark	From enrolment until the day after discharge	Usual practice: relevant medical treatment and care. Referral to occupational therapy and physiotherapy took place only if the medical or nursing

		received components 2 and 3 *Component 2:* - A rehabilitation plan was prescribed. The rehabilitation plan included a description of the patient's previous and current performance of daily activities and specified the need for further rehabilitation. Bridging: - Primary care was informed about the discharge, and visitation of the patient to further rehabilitation Post-discharge: *Component 3:* - Home visit by an OT the day after discharge for patients with a prescribed rehabilitation plan. The home visit aimed to enhance the patient's performance of daily activities and to start rehabilitation. - OT screened the home for safety risks and factors that potently could limit the performance of daily activities.				staff considered it necessary. If the occupational therapist was summoned to asses a patient, the occupational therapist performed a short interview and a non-standardized observation of the patient's performance of basic daily activities and the results were communicated to the municipality homecare staff. If necessary, nurses from the ED organized referral to nursing home care after discharge
Robinson et al.[32] **2015**	Integrated Transition of Care (ITC)	Pre-discharge: - Nutrition screening, and if necessary referral to a dietitian - Allied health review - Discharge medicines reconciliation and patient education by a pharmacist. Bridging: - Records were communicated to primary care providers electronically and via mail, and when required, by telephone. Post-discharge: Provided only to patients who were returning to independent community living - 1 and 3 days after discharge: a telephone assessment, education, and support. The aim was to identify problems not dealt with during the discharge process, assist with self-management, and ensure appropriate health and social supports.	Community nurses A geriatrician A pharmacist Cultural support workers	Two acute general hospitals with 536 and 250 beds in Auckland, New Zealand.	From recruitment until 3 days after discharge	NR

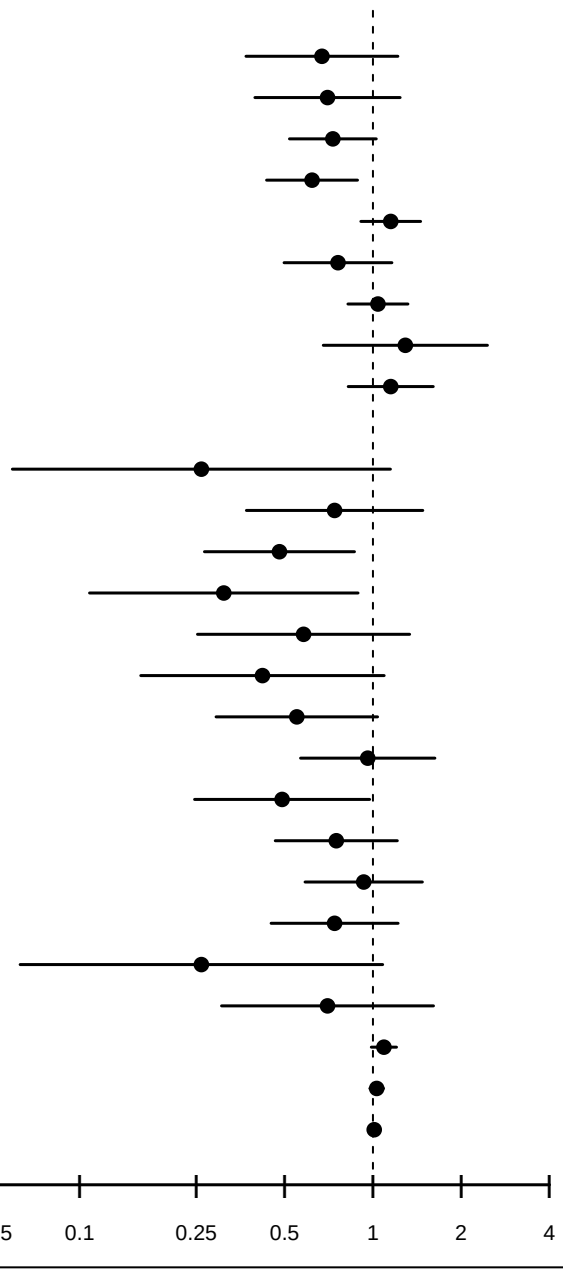
Rottman-Sagebiel et al.[29] **2018**	The Geriatrics Medication Education at Discharge project (GMED)	Pre-discharge: - An individual, face-to-face meeting between the CPS and the patient. - Medication reconciliation - Medication education regarding administration and usage of the patient's medications. - The caregiver, if any, was included in the discussion either at the bedside or by telephone following the face-to-face visit with the patient. - Discuss findings with inpatient team in person or by telephone and through documentation in the CPRS. Bridging: - All information the telephone encounter was communicated to the PCP through CPRS documentation and by telephone as needed. Post-discharge: - A telephone visit within 2 to 3 days of discharge. The purpose of the telephone visit was to perform medication reconciliation, identify and rectify medication errors, provide further patient education, and assist in facilitating appropriate follow-up by the patient's PCP, if required.	Primary care provider (PCP) A clinical pharmacist specialist (CPS) with oversight from a senior CPS with geriatric pharmacology expertise An internist or geriatrician	Veterans Memorial Hospital of the South Texas Veterans Health Care System (STVHCS)	From recruitment until 2 or 3 days after discharge	NR
Sahota et al.[27] **2017**	CIRACT	Pre-discharge: - Comprehensive assessment - Formed a rehabilitation plan - Treated daily (7 days a week, if appropriate) and the duration of rehabilitation they received depended on their needs. - The team liaised with the participant and their carer(s) to visit the participant's home to assess and provide recommendations for equipment and make adaptations and/or modifications as required Bridging: None Post-discharge: - Team visited the participant at home to assess the level of rehabilitation required	Senior occupational therapist (transition coach) Senior physiotherapist Assistant practitioner A social services practitioner	Queen's Medical Centre: a 1,800-bed hospital in Nottingham, UK.	NR	The standard THB-Rehab service was provided on weekdays only. An assessment of the participant's ability to perform certain tasks and provided recommendations for rehabilitation. Referral to community based services for provision of equipment at home, personal care and ongoing rehabilitation in community services where appropriate at

		- Further follow-up visits as deemed necessary - Appropriate referral to additional community services				discharge.
Voss et al.[30] **2011**	Care Transitions Intervention (CTI)	Pre-discharge: - Meet with the patient to establish initial rapport - Introduce the personal health record, including their main health problems, their medications, and questions for their health care providers - Arrange a home visit, ideally within 48 to 72 hours after hospital discharge Bridging: - Discuss topics related to participants with their outpatient providers Post-discharge: *Home visit within 3 days after discharge*: - Reconcile all of the patient's medication regimens - TC imparted skills for effectively communicating care needs during subsequent encounters with health care professionals. - TC also reviewed with the patient any red flags that indicated a condition was worsening and provided education about the initial steps to take to manage the red flags and when to contact the appropriate health care professional. *Phone call*: - Telephoning 3 times during a 28-day post-hospitalization discharge period. The first telephone call within 7 to 10 days generally focused on determining whether the patient had received appropriate services. In the 2 subsequent telephone calls, TC reviewed the patient's progress toward goals established during the home visit, discussed any encounters that took place with other health care professionals, reinforced the importance of maintaining and sharing the personal health record, and supported the patient's role in chronic illness self-management. The final telephone call was conducted by day 30	Transition coach (TC) who had a background in nursing or social work	6 Rhode Island acute care hospitals, including 2 community hospitals, 3 teaching hospitals, and a tertiary care center and teaching hospital, ranging from 129 beds to 719 beds	From enrolment until 30 days after discharge	From randomization to the day of discharge

NR: not reported, CGA: comprehensive geriatric assessment, NH: nursing home, GP: general practitioner, ADL: activity of daily living, PCP: primary care provider, ICM: integrated care model, HST: home support team, ED: emergency department.

Supplementary 6, forest plot grouped by study population

Author	Intervention	OA	RR (95% CI)
Unselected patients			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Patients at risk			
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)

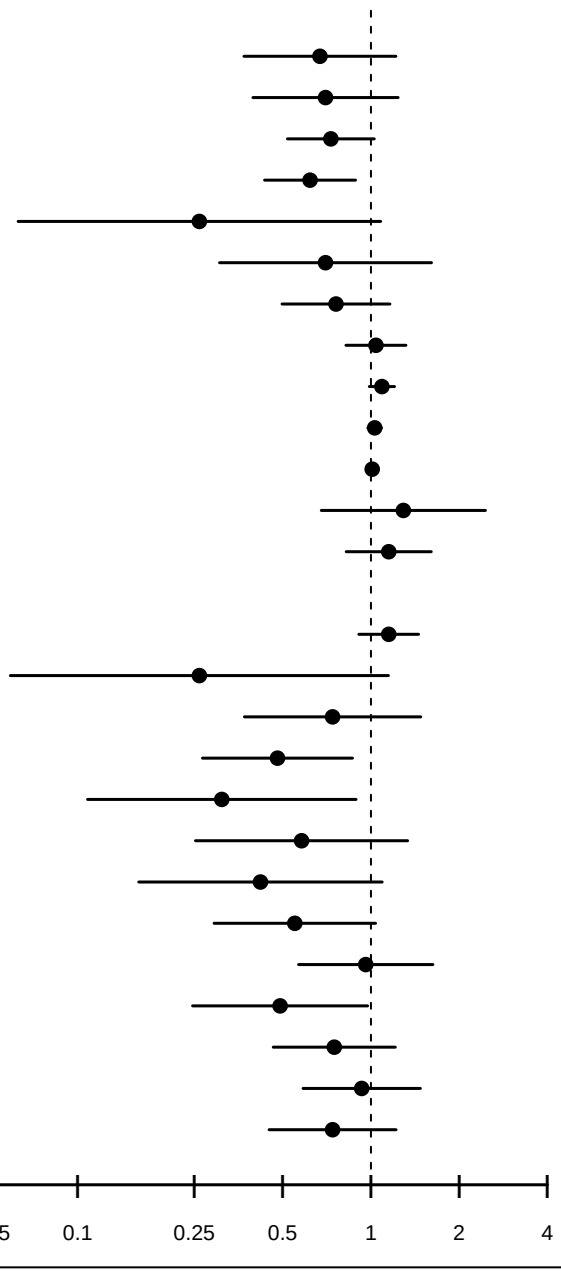


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 7, forest plot grouped by intervention intensity

Author	Intervention	OA	RR (95% CI)
Low intensity			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
High intensity			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)

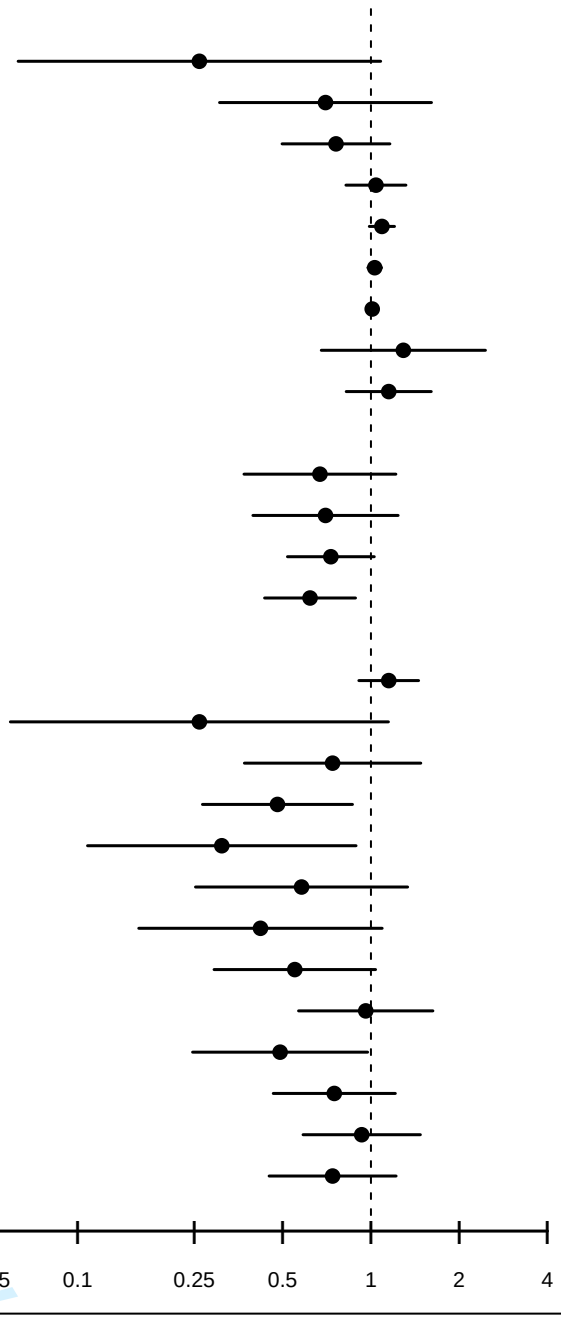


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 8, forest plot grouped by length of support

Author	Intervention	OA	RR (95% CI)
≤ **7 days**			
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
8−30 days			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
≥ **31 days**			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)

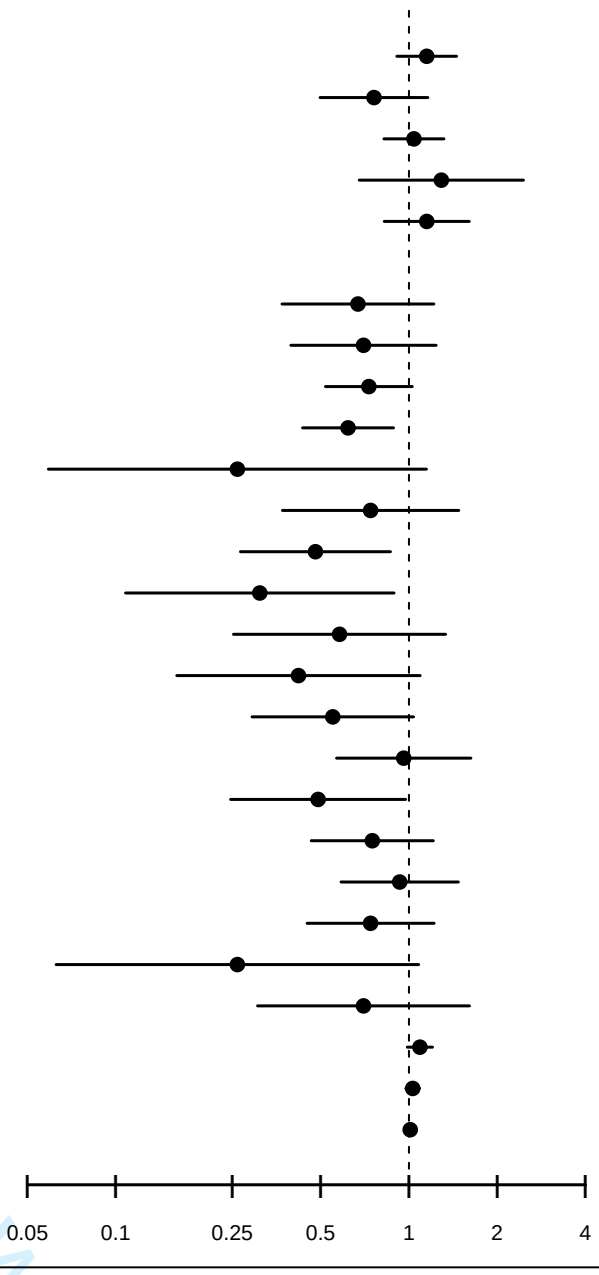


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 9, forest plot grouped by country of origin

Author	Intervention	OA	RR (95% CI)
European studies			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Non−European studies			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)

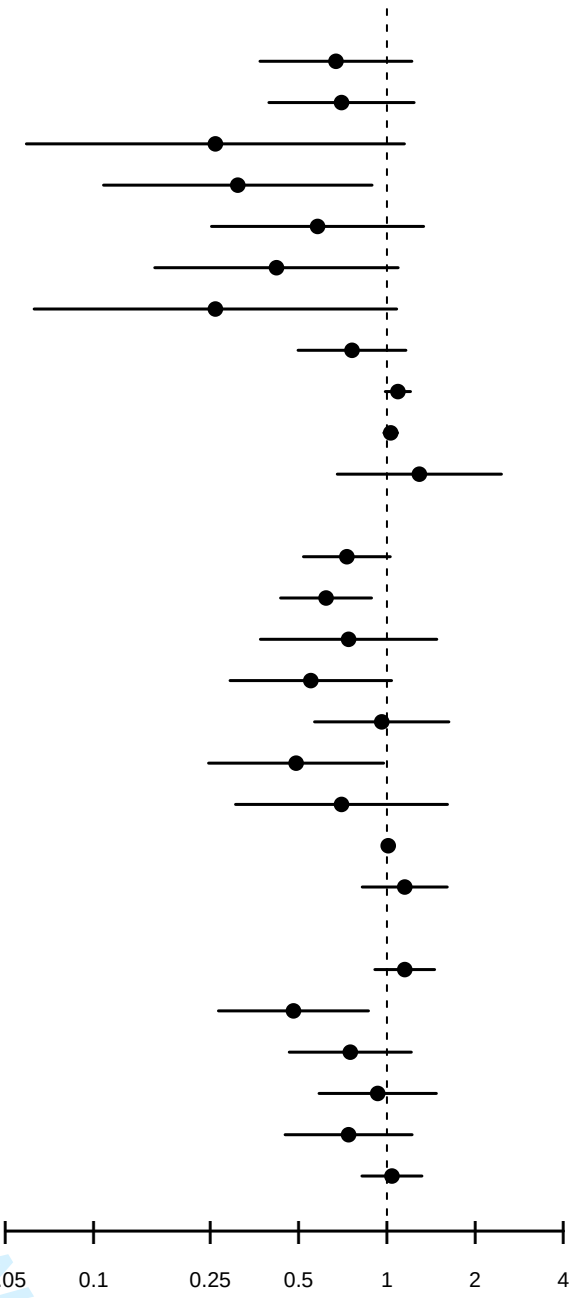


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 10, forest plot grouped by outcome assessment

Author	Intervention	RR (95% CI)
≤ **30 days**		
Chow et al.	Home visit	0.67 (0.37−1.22)
Chow et al.	Call	0.70 (0.40−1.24)
Courtney et al.		0.26 (0.06−1.15)
Finlayson et al.	EN−Hat	0.31 (0.11−0.89)
Finlayson et al.	Exercise	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	0.42 (0.16−1.09)
Koehler et al.		0.26 (0.06−1.08)
Nielsen et al.		0.76 (0.50−1.16)
Robinson et al.		1.09 (0.99−1.20)
Robinson et al.		1.03 (0.98−1.09)
Sahota et al.		1.29 (0.68−2.46)
31−91 days		
Chow et al.	Home visit	0.73 (0.52−1.03)
Chow et al.	Call	0.62 (0.43−0.89)
Courtney et al.		0.74 (0.37−1.48)
Finlayson et al.	EN−Hat	0.55 (0.29−1.04)
Finlayson et al.	Exercise	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	0.49 (0.25−0.97)
Koehler et al.		0.70 (0.30−1.61)
Robinson et al.		1.01 (0.97−1.05)
Sahota et al.		1.15 (0.82−1.61)
≥ **92 days**		
Buurman et al.		1.15 (0.91−1.45)
Courtney et al.		0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	0.75 (0.46−1.21)
Finlayson et al.	Exercise	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	0.74 (0.45−1.22)
Nielsen et al.		1.04 (0.82−1.32)

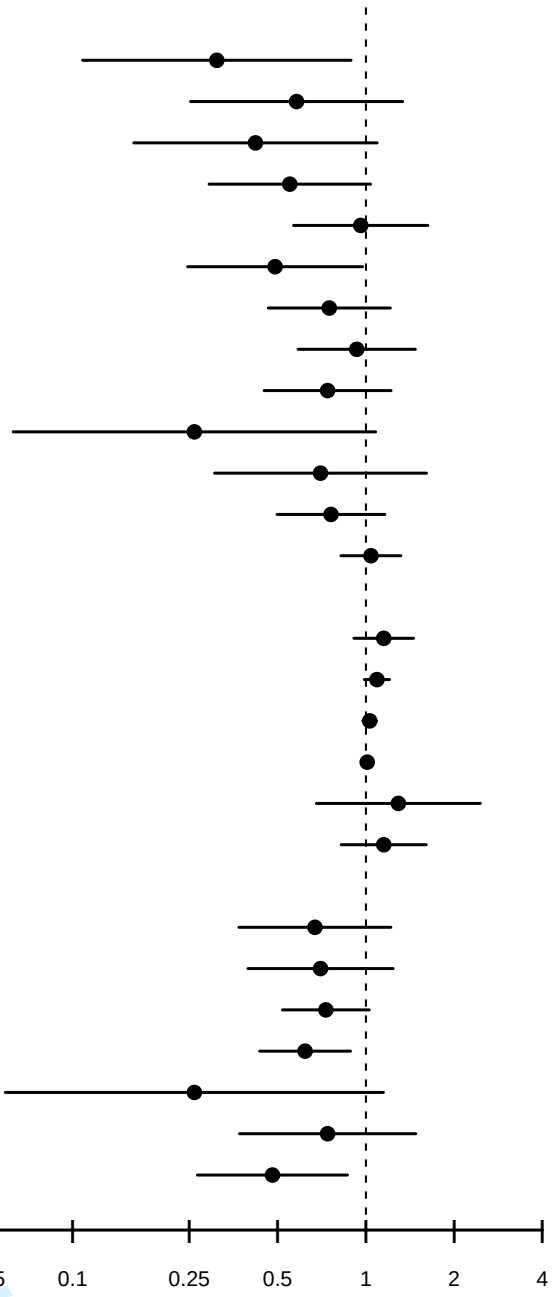


Abbreviations: RR, relative risk; CI, confidence interval

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Supplementary 11, forest plot grouped by quality assessment

Author	Intervention	OA	RR (95% CI)
Weak			
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Moderate			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Strong			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)



Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Effects of transitional interventions between hospital and home to older patients discharged from a Medical Ward aiming at reducing unplanned readmissions: a systematic review

Lisa Rasmussen, Merete Gregersen, Louise Grode, Ishay Barat, Jeppe Lange

Citation

Lisa Rasmussen, Merete Gregersen, Louise Grode, Ishay Barat, Jeppe Lange. Effects of transitional interventions between hospital and home to older patients discharged from a Medical Ward aiming at reducing unplanned readmissions: a systematic review. PROSPERO 2019 CRD42019121795 Available from: http://www.crd.york.ac.uk/PROSPERO/display_record.php?ID=CRD42019121795

Review question

"What is the relationship between discharge interventions in the transitional between hospital and home and readmissions among older patients discharged from a Medical Ward?"

Sub-questions:
- What are the contents of the interventions and how are they structured?
- How effective are the interventions?
- Which possible correlations are seen between intervention elements and aspects?
- Which elements seem to be essential in order to achieve positive effects?
- How are the methodological quality of the included studies?

P: Patients aged 65 or above discharged from a Medical ward
I: Clinical transitional phase between hospital and home
C: Not named
O: Hospital readmission
S: The interventions have to take place in both hospital and home settings

Searches

The strategy aims to find both published and grey literature. The search strategy has already been developed by the undersigned and a research librarian.
Studies published English, Danish, Swedish and Norwegian in between January 1st 2007 to December 31st 2018 are included in the review.
Search strategy is divided in 2 steps:
First, following bibliographic databases will systematically be searches: PubMed, CINAHL, Embase, Cochrane and Web of Science.
Second, a hand search will be conducted. This include: references from relevant systematic reviews, references from all included articles, references of references, personal bibliographies of authors of all included studies, "similar articles" to all included studies will be screened in PubMed, subject specific journals will be searched and finally, keywords from all relevant studies will be searched
Both text Words and subject headings will be searched
P: aged OR elder* OR old* OR geriatric* (title)
AND
I: "discharge intervention" OR transition* OR transition* care (title)
AND
O: readmission* OR re-admission* OR rehospitalization* OR re-hospitalization* OR rehospitalisation* OR re-hospitalisation* (text Word)

Types of study to be included

Inclusion: Quantitative designs
Exclusion: Systematic reviews, qualitative studies

Condition or domain being studied

In Denmark, 20% of all admitted patients above 65 are readmitted within the first 30 days after discharge.

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Readmissions negatively affect patients and their families, healthcare systems, healthcare costs and mortality. Readmissions that occur shortly after discharge are often caused by problems in the transition between hospital and home. Insufficient discharge planning is associated with increased risk of unintended events such as medication errors, poor communication between hospital and primary care professionals in the post-discharge period as well as readmission. Research on discharge interventions show various effects on readmission rates. More research is needed to expand the knowledge on effective and non-effective strategies in order to reduce readmissions and improve the quality of discharge processes. Published reviews on readmissions mainly focus on either interventions based on hospitals or in home settings. This review focus on interventions involving both hospital and home setting

Participants/population

Inclusion: patients aged 65 or above discharged from a medical ward
Exclusion: patients aged 64 or below discharged from other wards than general medicine wards. Patients with specific diagnosis such as COPD or psychiatric disorders

Intervention(s), exposure(s)

Transitional care interventions are often complex as they are multifaceted and thus the content, elements and delivery differ from intervention to intervention. Due to the heterogeneity between interventions the primary inclusion criteria is that the interventions have to take place in both hospital and home settings. This means that elements of the intervention have to be physically delivered both at the hospital and in the patients own homes or wherever the discharge destinations may be.
Interventions which only include pharmaceutical elements such as medication reconciliation are excluded. Additionally, interventions which only include rehabilitation or therapeutic elements are also excluded.

Comparator(s)/control

Studies comparing interventions to "usual care" will be included. Usual care is not pre-defined.

Context

Main outcome(s)

Unplanned hospital readmission rates.
Readmission is defined as a return hospitalization to an acute care hospital that follows a prior acute care admission within a specified time interval

Timing and effect measures

Dueto expected heterogeneity of the studies, it is assumed that it will not bepossible to conduct a meta-analysis. However, when possible reported effectmeasures will be listed and discussed.

Additional outcome(s)

None

Timing and effect measures

Data extraction (selection and coding)

PRISMA flow chart will be followed.
A research librarian will be consulted in order to improve the search strategy. Literature search will be done by one reviewer (the undersigned). Titles of studies retrieved using the search strategy and those from the hand search will be screened by one reviewer (the undersigned). Abstracts will be screened independently by two reviewers to identify studies that potentially meet inclusion criteria. The full text of those studies will be retrieved and independently assessed for eligibility by two reviewers. If any disagreements or uncertainties occur, a third reviewer will be consulted and disagreements or uncertainties will be solved through discussions.
A data extraction form has been developed to address the review question of this current review. The data extraction form is based on components from “Data Extraction Form for Experimental/Observational Studies” from Joanna Briggs Institute and “Data Extraction and Assessment Template” from The Cochrane Public Health Group. It will be pilot tested before data extraction of included studies. Data includes descriptive data as well as analytical data such as: study setting, study population, participant demographics,

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outcomes, times of measurements, details on the interventions, control conditions, results, conclusions strength and limitations will be extracted from all included studies.

Risk of bias (quality) assessment

Methodological quality of included studies will be assessed independently by two reviewers. Both will follow the same procedure. The methodological quality assessment will be conducted using The Quality Assessment Tool for Quantitative Studies (EPHPP) tool. The tool includes components of intervention integrity. Final results of the Quality Assessment Tool for Quantitative Studies lead to an overall methodological rating of strong, moderate or weak measures of the internal validity. Following is assessed in the present tool: selection bias, allocation bias, confounding, detection/performance, bias, information bias and attrition bias. Analysis and results of the extracted data as well as conclusions and external validity of the review will also be discussed in the light of methodological quality of included studies. Regardless of the results of the methodological quality assessment, all studies meeting the eligibility criteria will be included in the review.

Strategy for data synthesis

Descriptive data as well as data on interventions and outcomes will be tabulated in a table.
A meta-analysis will only be included if the included studies are sufficiently homogeneous. However, a high level of heterogeneity between included studies is expected in this review, Thus a quantitative synthesis is not planned.
Results:

- It will be a combined presentation of the results rather than a presentation of results from all individual studies
- A narrative description of the overall results from data extraction will be presented in a table
- Possible associations deriving from the data extraction are presented in order to give an overall picture of similarities and differences between the studies
- Results will be presented in the light of effects and statistically significance
- Results from methodological quality assessment will be illustrated in a table and then described

Analysis:

- Analysis will be based on a combined analysis rather than a analysis of individual studies
- Analysis of extracted data will be discussed in the light of results from the methodological assessments

Analysis of subgroups or subsets

This systematic review will most likely be qualitative/descriptive. Therefore, it is not possible to describe statistical analysis.
It will be a combined presentation of findings from all included studies, rather than a presentation of findings from each study individually. There will be a combined presentation of study population, participant demographics, outcomes, times of Measurements, description of interventions and results. Similarities and

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differences between the included studies will be presented.
This review will most likely be descriptive, and for now it is not possible to state whether subgroup analysis will be undertaken or not and Thus not possible to specify the Groups in advance.

Contact details for further information

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Organisational affiliation of the review

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www.regionshospitalet-horsens.dk

Review team members and their organisational affiliations

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Mrs Merete Gregersen. Aarhus University Hospital
Mrs Louise Grode. Horsens Regional Hospital
Mr Ishay Barat. Horsens Regional Hospital
Mr Jeppe Lange. Horsens Regional Hospital

Anticipated or actual start date

16 December 2018

Anticipated completion date

31 July 2019

Funding sources/sponsors

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Conflicts of interest

Language

Danish, English

Country

Denmark

Stage of review

Review_Ongoing

Subject index terms status

Subject indexing assigned by CRD

Subject index terms

Hospitals; Humans; Patient Discharge; Patient Readmission

Date of registration in PROSPERO

30 January 2019

Date of publication of this version

30 January 2019

Details of any existing review of the same topic by the same authors

Stage of review at time of this submission

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Stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	Yes
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Versions

30 January 2019

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	none
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	17 Table 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Only 1 outcome Table 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	13
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	19 Fig. 2
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	None
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	19-22, 23
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	22
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

The impact of transitional care interventions on hospital readmissions in older medical patients: a systematic review

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Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Health services research, Evidence based practice
Keywords:	GERIATRIC MEDICINE, PREVENTIVE MEDICINE, INTERNAL MEDICINE

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TITLE PAGE

The impact of transitional care interventions on hospital readmissions in older medical patients: a systematic review

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ABSTRACT

Objectives

To identify and synthesise available evidence on the impact of transitional care interventions with both pre-and post-discharge elements on readmission rates in older medical patients.

Design

A systematic review.

Method

Inclusion criteria were: medical patients ≥65 years or mean age in study population of ≥75 years; interventions were transitional care interventions between hospital and home with both pre- and post-discharge components; outcome was hospital readmissions. Studies were excluded if they: included other patient groups than medical patients, included patients with only one diagnosis or patients with only psychiatric disorders. PubMed, The Cochrane Library, Embase, Cinahl, and Web of Science were searched from January 2008 to August 2019. Study selection at title level was undertaken by one author; the remaining selection process, data extraction and methodological quality assessment were undertaken by two authors independently. A narrative synthesis was performed, and effect sizes were estimated.

Result

We identified 1951 records and included 11 studies: five randomised controlled trials, four controlled clinical trials and two pre-post cohort studies. The 11 studies represent 15 different interventions and 29 outcome results measuring readmission rates within 7-182 days after discharge. Twenty-two of the 29 outcome results showed a drop in readmission rates in the intervention groups compared with the control groups. The most significant impact was seen when interventions were of high intensity, lasted at least one month and targeted patient at risk. The methodological quality of the included studies was generally poor.

Conclusion

Transitional care interventions reduce readmission rates among older medical patients although the impact varied at different times of outcome assessment. High-quality studies examining the impact of interventions are needed, preferably complimented by a process evaluation to refine and improve future interventions.

PROSPERO registration number CRD42019121795.

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STRENGTHS AND LIMITATIONS

- Focus on trans-sectorial interventions with both pre- and post-discharge elements
- High internal validity as PRISMA and AMSTAR guidelines have been applied
- Results may underrepresent negative study findings as negative impact often remains unpublished
- Assessing intervention impact on unplanned readmissions as the only outcome

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INTRODUCTION

Transitional care interventions (TCIs) may be essential in older medical patients' transition from hospital to home as they may prevent adverse events and unplanned hospital readmissions. These events can have detrimental consequences for the individual patient.

The global demography is radically changing. In the European Union, elderly above 65 years are estimated to account for 29.1% of the total population in 2080 compared with 19.2% in 2016. Additionally, the fraction of the population above 80 years is expected to double between 2016 and 2080[1]. We may therefore expect a dramatic increase in healthcare service demands and costs[2]. As these changes will bring substantial challenges to healthcare systems[3 4], the potential need for TCIs will also increase. Older people needing healthcare are often medical patients with several concurrent diseases, reduced physical or mental functionalities, limited ability to provide self-care, and they are often living alone and need care from primary or secondary healthcare services[5].

Older people with complex co-morbid conditions are at high risk of adverse events and safety incidents immediately after their discharge from hospital[6]. Unplanned readmission seems to be related to insufficient discharge planning, and unintended events during discharge and transition such as medication errors and inadequate communication between hospital- and primary care professionals[7-10]. By contrast, optimised, customised and patient-centred discharge planning and transitions may reduce length of hospital stay (LOS), risk of readmission, medication discrepancies and mortality; and may as well as improve the patients' activity of daily living (ADL) and reduce healthcare costs[11].

One approach to addressing these challenges is to examine the impact of interventions aimed at reducing readmissions.

Previous systematic reviews have mainly evaluated the impact of hospital- and/or home-based interventions on readmissions[12 13] and included populations with specific conditions or both medical and surgical patients. To our knowledge, no systematic review has been conducted examining the impact of TCIs that take place in both hospital and home on older medical patients readmission rates based on recent data.

The purpose of this systematic review is to evaluate the impact of TCIs with both pre- and -post-discharge elements on readmission among older medical patients.

METHODS

This review was registered in the PROSPERO database prior to data collection (CRD42019121795)[14]. It is presented in accordance with the Preferred Reporting Items for Systematic Reviews (PRISMA)[15].

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Eligibility criteria

We identified studies that aimed to reduce readmission rates through TCIs among older medical patients. The PICO process for framing the research question was applied[16] and defined as:

i. ***P****opulation*: Older medical patients discharged from a general medical ward or emergency department (ED);
ii. ***I****ntervention*: In the transitional phase between hospital and home which examined the impact of the intervention on readmission rates. The interventions had to include both pre- and post-discharge components;
iii. ***C****omparison*: Usual care defined as standard care and treatment;
iv. ***O****utcome*: Unplanned readmission to hospital.

Selecting tool is displayed in Supplementary 1.

Studies were excluded if:

i. The population was aged under 65 years or the mean age was below 75 years
ii. They included other patient groups than medical patients (e.g. surgical patients)
iii. They included participants with only one medical diagnosis (International Classification of Disease -10th edition)
iv. Studies included participants with psychiatric disorders only
v. They compared interventions with anything other than usual care
vi. Readmission was not an outcome
vii. Intervention only included either pre- or post-discharge components; they were reviews, case reports or case studies without comparison groups.

Information sources

To identify eligible studies, we searched the following bibliographic databases: PubMed, The Cochrane Library, Embase, Cinahl and Web of Science from January 2008 to August 2019. An extensive snowball search was performed where the reference lists from relevant studies, systematic reviews and included studies were examined. Additionally, publication lists from prominent researchers within the field were examined. Grey literature was searched in all relevant resources listed by Paez[17] from January 2008 to August 2019. Authors from relevant study protocols and grey literature were contacted in order to examine whether the studies were published or study results were available. The Negative Results Scientific Journal was searched.

Search strategy

The search string (Supplementary 2) was developed in collaboration between the authors and a university research librarian. We used key terms, free text-words, subject headings, index terms and appertaining synonyms, which were

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identified through relevant theory and research. The searches were limited by only including studies published in English or Scandinavian languages. The bibliographic searches were conducted on 13 and 14 December 2018 and the searches were regularly updated until 31 August 2019.

Study selection
First, titles were screened for their potential relevance according to population and outcome by the first author (LFR). Second, two authors independently screened titles and abstracts for intervention eligibility (LFR and MG). Third, an assessment of the full text was performed by two authors independently (always including LFR). In case of disagreement, a third author, who was chosen *a priori*, was consulted.

Data extraction
The Cochrane Data Extraction Form was modified to fit the present patient group and intervention type[18]. Data from the included studies were extracted by two researchers independently (always including LFR). Extracted data included study characteristics and results such as author, year of publication, country, study design, setting, participants, study size, outcomes, follow-up time and impact of intervention in numbers and/or per cent. Only data on the outcome "readmission" were extracted and analysed.

Quality assessment
"The Quality Assessment Tool for Quantitative Studies" (EPHPP) (Supplementary 3) was applied to assess bias in each study included in the review. The validated tool is recommended by The Cochrane Collaboration[16 19] and provides a standardised means to assess study quality. The EPHPP assesses six methodological dimensions: selection bias, study design, confounders, blinding, data collection as well as withdrawals and dropouts. It distributes the overall methodological rating into a strong, moderate or weak measure of internal validity[20]. Quality assessment will be conducted at study level. All studies meeting the inclusion criteria are included in the synthesis regardless of the results of the quality assessments. Quality assessment across studies will be analysed and the impact hereof will be discussed.

Data synthesis
Inspired by Pigott and Shepperd[21], the following aspects that can entail heterogeneity will be assessed: 1) context, 2) target population, 3) intervention, 4) methodological features and 5) researcher characteristics and reporting context. This assessment will result in a descriptive synthesis or a meta-analysis of data. Risk ratios (RR) and their 95% confidence intervals (CIs) are calculated[16(section 6.4.1)] when possible and presented in forest plots stratified by subgroups. Only main results will be presented and discussed according to sub-groups based on study- and intervention characteristics and review findings. The certainty of the synthesised results will be assessed using GRADE[16(section 14-2-1)].

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Patient and public involvement

Patients or the public were not involved in this study.

RESULTS

Study selection

In total, 1,951 records were identified. Of those, 1,901 records were identified through bibliographic databases and 50 records were found through other sources. After removing duplicates, 1,521 records were screened by title and abstract. Seventy-four records were considered for full-text review of which 11 met the eligibility criteria (Figure 1). The 63 remaining records were excluded due to participants' age, population, study design, readmission not listed as an outcome, only pre- or post-discharge elements in the intervention, not able to obtain data from authors and no "usual care" control group (Supplementary 4).

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Study characteristics

The 11 included studies represented 15 different interventions and 29 outcome assessments measuring readmission rates at different time points. Two studies were multi-arm studies[22 23]. Study characteristics are shown in Table 1. Five of the included studies were randomised controlled trials (RCTs)[23-27], four controlled clinical trials (CCTs)[22 28-30] and two pre-post cohort studies[31 32]. Sample sizes of individual studies ranged from 41 to 19,157. Allocation to intervention or control group was performed on an individual level. The majority of studies reported that readmissions were unplanned or acute[23-28 32]. Outcome assessments were conducted 7-182 days after discharge. In addition to readmission, the 11 studies assessed outcomes according to physical functioning[24 27]; cognitive functioning[24]; quality of life[22 25 27]; time to either hospital readmission or discharge from nursing homes[24 28]; self-efficacy[22]; self-rated health[22]; visits to emergency departments, general practitioners or allied health professional[25 26 28 31 32]; length of stay[27 31]; co-morbidity[27]; cost-effectiveness[27] and mortality[24 28].Three studies took place in USA[26 29 30] , two in Australia[23 25], two in Hong Kong[22 31], one in the Netherlands [24], one in the United Kingdom[27], one in Denmark[28] and one in New Zealand[32]. The studies were published between 2009 and 2018.

Study population

In total, approximately 24,500 patients were evaluated in our review; 8,800 in the intervention groups and 15,700 in the control groups. The mean age of the participants in the intervention groups was approximately 78 years (range 74.9–83.6), whilst the mean age among the control groups was approximately 79 years (range 75.2-84.5). Approximately 59% (range 3.6-85) of the participants in the intervention groups were females and 52% (range 2.2-67) of the control group were females.

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Table 1 Study characteristic

Author Year	Country Setting	Design	Intervention	Study population	Mean age, years (SD) or median (range)	Female n, (%)	Outcomes	OA	Readmission (event/total)	RR (CI)
Buurman[24] 2016	Netherland Multi-centre study	RCT	Transitional Care Bridge Program Intervention	T: 674 I: 337 C:337 Unselected patients	I: 79.7 (7.3) C: 80.0 (7.8)	I: 195 (57.9) C: 195 (57.9)	-Readmission - Katz Index of ADL - Mortality - Cognitive functioning - Time to hospital readmission - Time to discharge from a nursing home	182	I: 106/316 C: 88/303	1.15 (0.91−1.45)
Chow[22] 2014	Hong Kong Single-centre study	Controlled clinical trial	The nurse case management intervention	T: 312 Ia: 96 Ib: 108 C: 108 Unselected patients	Ia: 75.00 (60–92) Ib: 75.50 (60–89) C: 77.00 (60–89)	Ia: 46 (52.9) Ib: 52 (54.2) C: 49 (50.0)	- Readmission - Quality of life - Self-efficacy - Self-rated health	28 84	Ia:14 /91 Ib: 16/100 C: 24/105 Ia: 32/97 Ib: 30/106 C: 59/130	Ia: 0.67 (0.37−1.22) Ib: 0.70 (0.40−1.24) Ia: 0.73 (0.52−1.03) Ib: 0.62 (0.43−0.89)
Courtney[25] 2009	Australia Single-centre study	RCT	Older Hospitalised Patients' Discharge Planning and In-home Follow-up Protocol (OHP-DP)	T: 128 I: 64 C: 64 Patients at risk	I: 78.1 (6.3) C: 79.4 (7.3)	I: 36 (62.1) C: 40 (62.5)	- Readmission - Visits to emergency department - Visits to general practitioner - Visits to allied health professional - Health-related quality of life	28 84 168	I: 2/49 C: 9/58 I: 10/49 C: 16/58 I: 11/49 C: 27/58	0.26 (0.06−1.15) 0.74 (0.37−1.48) 0.48 (0.27−0.87)
Finlayson[23] 2018	Australia Multi-centre study	RCT	NR	T: 222 Ic: 57 Id:56 Ie:54 C:55 Patients at risk	Ic: 77.1 (7.64) Id: 77.6 (6.50) Ie: 77.8 (6.23) C: 77.9 (6.20)	Ic: 46 (80.7) Id: 42 (75.0) Ie: 37 (68.5) C: 37 (67.3)	- Readmission	28 84 168	Ic: 4/53 Id: 7/49 Ie: 5/49 C: 13/53 Ic: 11/53 Id: 16/44 Ie: 9/48 C: 19/50	Ic: 0.31 (0.11−0.89) Id: 0.58 (0.25−1.33) Ie: 0.42 0.42 (0.16−1.09) Ic: 0.55 (0.29−1.04) Id: 0.96 (0.57−1.63)

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									Ic: 18/52 Id: 18/42 Ie: 16/47 C: 23/50	Ie: 0.49 (0.25−0.97) Ic: 0.75 (0.46−1.21) Id: 0.93 (0.59−1.47) Ie: 0.74 (0.45−1.22)
Koehler[26] 2009	USA Single-centre study	RCT (pilot)	Elderly Care Bundle	T: 41 I: 20 C:21 Patients at risk	I: 77.2 (5.3) C: 79.8 (5.6)	I: 17 (85) C: 13 (62)	- Readmission - Emergency department visits	30 60	I: 2/20 C: 8/21 I: 6/20 C: 9/21	0.26 (0.06−1.08) 0.70 (0.30−1.61)
Lin[31] 2015	Hong Kong, Multi-centre study	Cohort analytic (two groups pre - post study)	Integrated Care and Discharge Support for elderly patients (ICDS)	T: NR I: 1,090 C: NR Patients at risk	Total: 80.4 ± 7.6 I: NR C: NR	T: 557 (51.1)	- Readmission -AED attendance - LOS	182 pre 182 post	NR Reports a statistically significant reduction on 47% in readmissions	NR
Nielsen[28] 2018	Denmark Single-centre study	Controlled clinical trial	Elderly Activity Performance Intervention	T: 375 I:144 C: 231 Unselected patients	I: 81 (7.9) C: 78 (8.6)	I: 79 (55) C: 122 (53)	- Readmission - All-cause mortality - Number of contacts to general practitioners and emergency departments without admission - Time to first readmission	30 182	I: 25/139 C: 55/231 I: 64/144 C: 99/231	0.76 (0.50−1.16) 1.04 (0.82−1.32)
Robinson[32] 2015	New Zealand Multi-centre study	Cohort analytic (two groups pre - post study)	Integrated Transition of Care (ITC)	T: 19157 I: 5172 C:13,985 Patients at risk	I: 78.2 (9.2) C: 77.6 (9.1)	I: 2486 (48.1) C: 6765 (48.4)	- Readmission - Emergency attendances	7 28 90	I: 500/5,172 C: 1,239/13,985 I: 1370/5,172 C: 3,588/13,985 I: 2,281/5,172 C: 6,079/13,985	1.09 (0.99−1.20) 1.03 (0.98−1.09) 1.01 (0.97−1.05)
Rottman-Sagebiel[29] 2018	USA Single-centre study	Controlled clinical trial	The Geriatrics Medication Education at Discharge project (GMED)	T: 1,624 I: 435 C: 1,189 Patients at risk	I: 74.9 (7.6) C: 75.2 (8.35)	I: 14 (3.6) C: 26 (2.2)	- Readmission	30	NR	NR

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Sahota[27] 2017	UK Single-centre study	RCT	The Community In-reach Rehabilitation and Care Transition (CIRACT)	T: 250 I: 125 C: 125 Unselected patients	I: 83.6 (6.6) C: 84.5 (5.9)	I: 82 (66) C: 79 (63)	- Readmission - LOS - Day 91-super spell bed days -Functional ability - Co-morbidity - Health-related quality of life - Cost-effectiveness analysis	28 91	I: 18/106 C: 14/106 I: 45/106 C: 39/106	1.29 (0.68–2.46) 1.15 (0.82–1.61)
Voss[30] 2011	USA Multi-centre study	Controlled clinical trial	Care Transitions Intervention (CTI)	T: 1888 I: 1042 C: 846 Unselected patients	I: NR C: NR	I: NR C: NR	- Readmission	30	NR	NR

T: total, OA: Readmission outcome assessment in days after hospital discharge, RR: relative risk, NR: not reported or author could not access the raw data.
CGA: comprehensive geriatric assessment, LOS: length of hospital stay
I: intervention, Ia: intervention "Home visit", Ib: intervention "Call", C: control, Ic: EN-HaT (Exercise and Nurse Home Visit and Telephone follow up), Id: Exercise, Ie: N-HaT (Nurse Home visit and Telephone follow-up)

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Transitional care interventions

All studies included pre- and post-discharge components and some included bridging components. The most pronounced difference between the interventions was in the pre-discharge phase, whereas elements in the bridging and post-discharge phase were somewhat similar. Table 2 outlines the intervention components, and Supplementary 5 contains detailed descriptions of the interventions.

Pre-discharge components

Across studies, pre-discharge components consisted of health record or plan, discharge planning, involvement of caregivers, concerns or barriers, patient education, physical exercise (physical exercise (muscle strength, stretch, balance, and walking), medication reconciliation, counselling, nutritional screening, pre-discharge home visit or patient assessments (Comprehensive Geriatric Assessment, Problem Classification Scheme, Assessment of Patients' Performance of Daily Activities and Assessment of Motor and Process Skills).

Bridging components

Bridging components consisted of written hand-over between hospital and primary sector, telephone hand-over or in-person hand-over.

Post-discharge components

Post-discharge components comprised of home visits, telephone follow-up, referral to additional healthcare services (rehabilitation, early specialist follow-up, general practitioner or other community services), nurse availability from hospital discharge until seven days after discharge or rapid and intensive community support.

Synthesis of results

All five aspects that may introduce heterogeneity were analysed and a pronounced diversity between the included studies was found. Additionally, some studies had incomplete reporting of study effects. Therefore, because of the heterogeneity, a meta-analysis could not be conducted.

Considering the impact on outcome level, 22 of the 29 outcome results (76%) showed a positive impact on readmission in the intervention groups compared with the control groups. Three studies did not report readmission rates but reported a positive impact on readmission rates, and it was not possible to obtain these numbers from the authors[29-31]. It was not possible to calculate RRs and their 95% CIs for those studies, and they are therefore not presented in the forest plots.

The impact size (RR) from individual studies was calculated using the reported readmission rates from eight studies. RRs ranged from 0.26 to 1.29. Nineteen outcome results were <1 of which one was close to 1 (RR=0.96). Seven estimates were >1 of which three were close to 1 (RR= 1.01, 1.03, 1.04). Of the 29 outcome results, five were statistically significant[22 23 25 26 31].

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Considering the impact on study level, seven of the 11 studies showed an entirely positive impact on readmissions[22 23 25 26 29-31], whilst one study presented both a positive impact and no impact[28], and three studies showed no impact at all[24 27].

Subgroup analysis

Study population

Supplementary 6 exhibits impacts according to the included study population. There is a clear difference in impacts between the groups. In total, 14 of 17 (82%) interventions including "patients at risk" reported positive impacts on readmission rates. In contrast, only five out of nine (56%) interventions including "unselected patients" reported positive impacts. In addition, the positive impacts were larger among interventions including "patient at risk" than among interventions with "unselected patients". Three of the four statistically significant results are found among the "patient at risk" group. However, the 95% Cis are wider among "patients at risk" compared with the "unselected patients" indicating less precise effect estimates.

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Table 2 Intervention components

Study	Pre-discharge components												Bridging components					Post-discharge components					
	Patient assessment	Personal health record and/or plan	Education	Medication reconciliation	Physical exercise	Address concerns and/or barriers	Counselling	Discharge planning	Nutrition screening	Caregiver involvement	Home visit to assess future needs	Multidisciplinary care	In-person hand-over	Written hand-over	Telephone hand-over	Community-based nurse visits hospital	Communications path unknown	Number of home visits	Number of telephone follow-up	Nurse availability	Referral to additional services	Intensive community support	Intervention intensity
Buurman et al.[24]	+	+										+	+			+		5[g]					9
Chow et al.[22] (1)	+	+				+												2[g]	2				6
Chow et al.[22] (2)	+	+				+													4				6
Courtney et al.[25]	+	+			+	+		+				+		+				1[a,g]	9				12
Finlayson et al.[23] (3)	+	+			+					+		+						6[b]					10
Finlayson et al.[23] (4)	+	+						+		+		+						1[a,g]	10	+			12
Finlayson et al.[23] (5)	+	+			+			+		+		+						1[a,g] 6[b]	10	+			12
Koehler et al.[26]		+	+	+		+	+	+		+		+		+				2[c]					7
Lin et al.[31] (6)	+							+				+						?[h]	?				6
Lin et al.[31] (7)	+							+				+									+[d]	+	5
Nielsen et al.[28]	+	+[d]			+												+	1[g]					4
Robinson et al.[32]	+		+[f]	+[f]					+			+		+	+[d]				2[e]				7
Rottman-Sagebiel et al.[29]			+[f]	+[f]						+		+		+	+[d]				1		+[d]		6
Sahota et al.[27]	+	+								+	+	+						1[a,h]			+[d]		3
Voss et al.[30]		+													+			1[g]	3				7

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1; post-discharge home visit group, 2; post-discharge call group , 3; Exercise intervention, 4; N-Hat (Nurse Home visit and Telephone follow-up) intervention, 5; Ex-Hat (Exercise and Nurse Home Visit and Telephone follow up) intervention, 6; ICM case management group, 7; HST service group a; additional home visits available if needed, b; by physiotherapist, c; one call by nurse and one call by pharmacist, d; if deemed necessary, e; to participants who return to independent living, f; by pharmacist, g; first visit within 3 days after discharge, h; time for first home visit unknown, ?; number not reported

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Intervention intensity
Calculation of intervention intensity was inspired by Verhaegh et al.[33] Ten interventions were categorised as low intensity[22 26-32] and five were categorised as high intensity[23-25].
Supplementary 7 illustrates a clear difference. The vast majority of interventions with high intensity reported a positive impact on readmission rates whereas only half of the interventions with a low intensity showed a positive impact. The impacts were larger and statistically significant among high-intensive interventions.

Length of support
The interventions lasted from enrolment until 1 day to 6 months after hospital discharge. Four studies lasted between 1 and 7 days[26 28 29 32], two studies lasted 28 and 30 days[22 30], one study 84 days[31], two 168 days[23 24], one 182 days[25] and one study did not report the duration of the intervention[27].
Supplementary 8 displays the impact on readmission rates stratified according to the duration of the interventions. A short length of support is associated with less or no impact on readmission rates. A length of support of one month or more is associated with positive, larger and statistically significant impacts.

Country of origin
Studies conducted outside the European countries seemed to have a greater impact on readmissions than studies conducted within the European Union and all statistically significant results are found in studies conducted in Non-European countries (Supplementary 9).

Outcome assessment
The impact on readmissions was largest within 30 days after hospital discharge. The impact decreased hereafter and was similar between 1 month and 6 months after discharge. Most statistically significant results are found when outcomes are assessed between 1 and 3 months (Supplementary 10).

Quality assessment within studies
Of the 11 studies, two studies were assessed to have a strong methodological quality[22 25], three had a moderate[24 27 32] and six had a weak methodological quality[23 26 28-31]. Of the RCTs, one had a strong[25], two had a moderate[24 27] and two had a weak quality[23 26]. Of the CCTs, one study had a strong[22] and three had a weak quality[28-30]. One of the pre-post cohort studies had a moderate[32] and one had a weak quality[31] (Table 3).

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Table 3 Quality assessment

Author	Selection bias	Study design	Confounders	Blinding	Data collection methods	Withdrawals and drop-outs	Global rating
Buurman et al.[24]	Moderate	Strong	Strong	Strong	Strong	Weak	**Moderate**
Chow et al.[22]	Moderate	Strong	Strong	Moderate	Strong	Strong	**Strong**
Courtney et al.[25]	Moderate	Strong	Strong	Moderate	Strong	Moderate	**Strong**
Finlayson et al.[23]	Weak	Strong	Strong	Weak	Weak	Moderate	**Weak**
Koehler et al.[26]	Weak	Strong	Strong	Weak	Strong	Weak	**Weak**
Lin et al.[31]	Strong	Moderate	Weak	Weak	Weak	Strong	**Weak**
Nielsen et al.[28]	Strong	Strong	Strong	Weak	Strong	Weak	**Weak**
Robinson et al.[32]	Strong	Moderate	Strong	Weak	Strong	Strong	**Moderate**
Rottman-Sagebiel et al.[29]	Weak	Strong	Strong	Weak	Weak	Weak	**Weak**
Sahota et al.[27]	Strong	Strong	Strong	Weak	Strong	Moderate	**Moderate**
Voss et al.[30]	Weak	Strong	Strong	Weak	Strong	Weak	**Weak**

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Quality assessment across studies
The majority of the studies did not meet the criteria in the components *selection bias* and *blinding* and were thus rated moderate or weak. The vast majority of studies met the criteria of *study design*, *confounders* and *data collection* and were therefore rated strong (Figure 2).

DISCUSSION

We found that the majority of interventions in the transitional phase between hospital and home appear to reduce readmission rates among older patients discharged from a medical ward.
However, some studies reported both a positive impact and no impact on the readmission rate following similar care interventions. These divergent results may have several plausible explanations as discussed below.

Explanations related to study characteristic

Country of origin
Studies conducted in European countries have less impact than studies conducted in non-European countries. The impact of complex interventions is, among others, altered by the context of the implementation[34 35], and differences in impact between countries may therefore be explained by diversity in the social, political, economic, clinical and geographical setting. The accessibility, type, character, quality and overall comprehensiveness of healthcare services provided may also play a role[36]. The USA and Australia have a long history of discharge planning and transitional care, and these countries therefore have a high quantity of research as well as refined strategies and guidelines[37-39].

Study population
The impact on readmissions is greater among "patients at risk" than among "unselected older patients". This is to be expected as readmission rates are higher among "patients at risk" than among "unselected patients". Furthermore, "patients at risk" are more frail and may have a higher degree of morbidity, which could have affected the risk of readmission[40].

Readmission rates prior to the study
Pre-intervention readmission rates may mirror differences in impact between studies. It may be assumed that hospitals with low pre-intervention readmission rates may experience no reduction of post-intervention readmission rates as the remaining readmissions may not be preventable. In contrast, it may also be assumed that hospitals with high pre-intervention readmission rates will achieve a reduction in post-intervention readmission rates.

Sample size

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Several studies had small sample sizes. As seen in Table 1, studies with small sample sizes reported a stronger impact than studies with larger sample sizes. This is in line with Dechartres et al. who also reported larger effect sizes in small-to-moderate-sized trials than in larger trials[41] One reason may be that small studies are more prone to publication bias than larger studies[42].

Explanations related to review findings

Interventions

Our findings suggest that the intensity of interventions influences the impact on readmission rates as high-intensity interventions generally have a stronger impact than low-intensity interventions. This is in line with previous findings presented by Verhaegh et al.[33]. The higher quantity of elements means that more aspects in a complex patient cohort and complex settings can be addressed.

Intervention components that at first glance appear to be similar across studies may comprise different features. The diverging impact on readmissions may be affected by differences in intervention contents across studies. It is not possible to make an intervention component analysis across studies that evaluates which components positively affect readmissions. However, some trends are seen in Table 2. Interventions with a positive impact on readmission comprise the following components; patient assessment, personal health record or plan, concerns and barriers, discharge planning, caregiver involvement, home visits and telephone follow-up.

Intervention fidelity

The impact on readmission rates may be affected by the fidelity of the interventions[43]. However, several studies do not describe intervention fidelity, which may be due to inconsistency in monitoring of the implementation or lack of transparency in reporting the study. These shortcomings thus hinder assessment of whether lack of positive impact is caused by poor implementation of the intervention or if the intervention did not work.

Outcome assessment

The timing of the outcome assessment has an important bearing on the possible preventive readmission rate[44]. The findings of this review suggest that interventions have the largest impact within the first 30 days after discharge. A potential impact is likely to have obliterated if outcome is assessed later than 30 days after discharge.

Residual confounding

Several of the included studies did not adjust for potential confounders such as length of stay, fall within the past 12 months, living conditions, prior admissions, poor overall health condition and functional disability[45-47]. This could have affected the internal validity and the impact on readmission rates.

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The lack of analysis adjusting for potential confounders is highlighted in previous studies[48], underlining the problematic trend in this research area.

Risk of bias
Differences in the quality assessment and thus risk of bias may explain the variation in the impact of the included studies[16]. Supplementary 11 shows that studies assessed to have weak and strong methodological quality have a positive impact on readmission rates whereas studies with moderate quality have no impact.

Statistical analysis
Several authors fail to report whether the statistical analysis is performed based on the first readmission. One patient may therefore represent several readmissions within a specific follow-up period. If this is the case, results from those studies may report a higher readmission rate and thus greater impact than studies using unique observations in their statistical analysis.

Quality assessment
The methodological quality of the majority of the included studies was low, thus indicating a high risk of bias. Low global rating can reflect both the methodological quality and insufficient reporting of the methodology. The latter makes it difficult to assess accurately the true quality and thus the risk of bias. If the low rating is caused by insufficient methodological reporting, it may have no effect on the impact. However, if the low rating reflects methodological problems, it probably underestimates the true impact of the interventions.
The component "blinding" represents the weakest rating. Blinding of participants was not always possible due to the nature of the interventions. Blinding is crucial in pharmaceutical trials and in other clinical studies. However, little is known about the benefits or disadvantages of blinding participants in complex interventions and how bias due to blinding may affect the results. Blinding outcome assessors may not be crucial when outcome is collected through digital records.

COMPARISON WITH PREVIOUS RESEARCH
Prior systematic reviews included other study populations such as surgical patients or patients with only one medical condition. The present review includes a broader population, namely older medical patients. Direct comparison of the present findings with previous findings is therefore difficult. The present review suggests that transitional care interventions reduce the risk of readmission among older medical patients.
These findings are in line with findings in similar reviews that found positive effects of TCIs on hospital readmission[49-51]. These reviews also found that the included studies had a very-low-to-moderate methodological quality.

LIMITATION OF INCLUDED EVIDENCE BASE

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Transparency across studies is lacking even if the CONSORT statement recommends reporting a sufficient description of interventions[52]. A call for more transparency in clinical trials and adherence to appropriate guidelines is also reported elsewhere[53]. If the template for the Intervention Description and Replication (TIDieR) is followed, the replicability may improve, thus making it possible to build on prior research findings[54].
Evaluating TCIs provides insight into whether interventions reduce readmissions among intervention groups compared with control groups. However, we gain no knowledge about causality between exposure and outcome. In complex interventions, such as TCIs, it is difficult to evaluate which mechanisms or components result in change[55]. A process evaluation may have captured the fidelity of the interventions and may thus have provided insight into which mechanisms and components actually work[56]. Process evaluation has previously been requested[57], stressing the unmet need to identify essential components in TCIs. We have requested missing intervention details and other relevant data from authors; some requests were met while others were not. Lastly, all GRADE domains except one (indirectness of evidence) were assessed and found to downgrade the evidence and hence, lower the certainty of the evidence of this review.

STRENGTHS AND LIMITATIONS OF THIS REVIEW

The study has several strengths. This review adhered to PRISMA[15], SWiM[58] and AMSTAR[59] which ensures that all important methodological considerations were made. By reducing the risk of bias in this review, systematic errors were minimised during all stages of the process.
Despite these strengths, we note some limitations. The eligibility criteria were somewhat narrow, which limited the number of included studies. This, however, was deemed necessary to meet the need for evidence focusing on clinical settings where a large number of hospital wards are general medical wards.
Only studies published in English, Danish, Norwegian or Swedish were eligible for inclusion in this review. Therefore, studies published in other languages were omitted, potentially excluding useful evidence.
This review only focuses on hospital readmission as an outcome although the included studies evaluated the intervention impact on multiple outcomes. It is therefore possible that some studies report a positive impact in terms of other outcomes than readmission.
Positive findings are more likely to be published in English language journals, whereas negative findings are more often published in local-language journals[60]. Publication bias is therefore likely present. During this review process, we have contacted several researchers who had registered TCIs on ClinicalTrials.gov. All researchers who found no or negative impact either struggled to publish or decided not to publish. The above-mentioned issues may result in an underrepresentation of negative results in the evidence base and thus in this present review.

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IMPLICATIONS AND FUTURE RESEARCH

Future research projects may benefit from the knowledge gained from the present review when designing and developing new studies. Knowing that interventions with a minimum duration of 1 month that have a high intensity and target high-risk patients may result in more effective interventions.
This review highlights an unmet need for studies of high methodological quality that evaluate the impact of TCIs among older medical patients. Therefore, future research can benefit from a higher level of adherence to relevant guidelines and more detailed descriptions of 1) interventions and comparison groups, 2) the implementation process and 3) actions taken to minimise bias and confounding. Further research should be undertaken to investigate the process evaluations of complex interventions to identify how and why interventions either work or do not work. To develop a broader picture of TCIs, additional studies need to focus on psychological outcomes. The societal cost of such interventions versus individual and societal benefits needs to be further evaluated. High-risk patients such as physically disabled, chronically ill patients may benefit from these kinds of interventions regardless of age and co-morbidity [61].

CONCLUSION

The majority of TCIs have a positive impact on readmission rates among older medical patients, although the most significant impact was seen within 30 days after hospital discharge. Therefore, we believe that the current evidence supports recommending transitional care that includes both pre-and post-discharge components. However, no evidence for recommending a specific intervention exists. The key finding shows an apparent pronounced positive impact among patients categorised as "patients at risk" in "high-intensity" interventions and in interventions with duration of 1 month or more. This, however, should be seen in the light of the fact that only 11 studies met the inclusion criteria and a low certainty of evidence according to the GRADE approach.

Figure legends/captions

Figure 1. Flow chart of the study selection.

Figure 2. Quality assessment across studies. The green colour indicates strong methodological quality, yellow indicates moderate quality and red indicates

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weak quality across studies.

Supplementary Material

Supplementary 1. Selecting Tool

Supplementary 2. Search string

Supplementary 3. Quality Assessment Tool and dictionary

Supplementary 4. Exclusion of full test articles

Supplementary 5. Intervention characteristics

Supplementary 6. Forest plot grouped by study population

Supplementary 7. Forest plot grouped by intervention intensity

Supplementary 8. Forest plot grouped by length of support

Supplementary 9. Forest plot grouped by country of origin

Supplementary 10. Forest plot grouped by outcome assessment

Supplementary 11. Forest plot grouped by quality assessment

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Author Contribution

All authors listed have contributed sufficiently to the project to be included as authors according to the Vancouver Guideline. All authors have participated in the development of the study, writing the draft and all have approved the final version.

Lisa Fønss Rasmussen: First author writing the manuscript, screening records for inclusion, data extraction of included studies, quality assessment of included studies.

Louise Grode: Initial development of the study, screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Jeppe Lange: Initial development of the study, screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Ishay Barat: Initial development of the study, screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Merete Gregersen: Initial development of the study, screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Competing interests

All authors have signed the ICMJE Disclosure Statement and have nothing to disclose and there is no competing interest for any author.

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Data sharing statement

All data relevant to the study are included in the article or uploaded as supplementary information.

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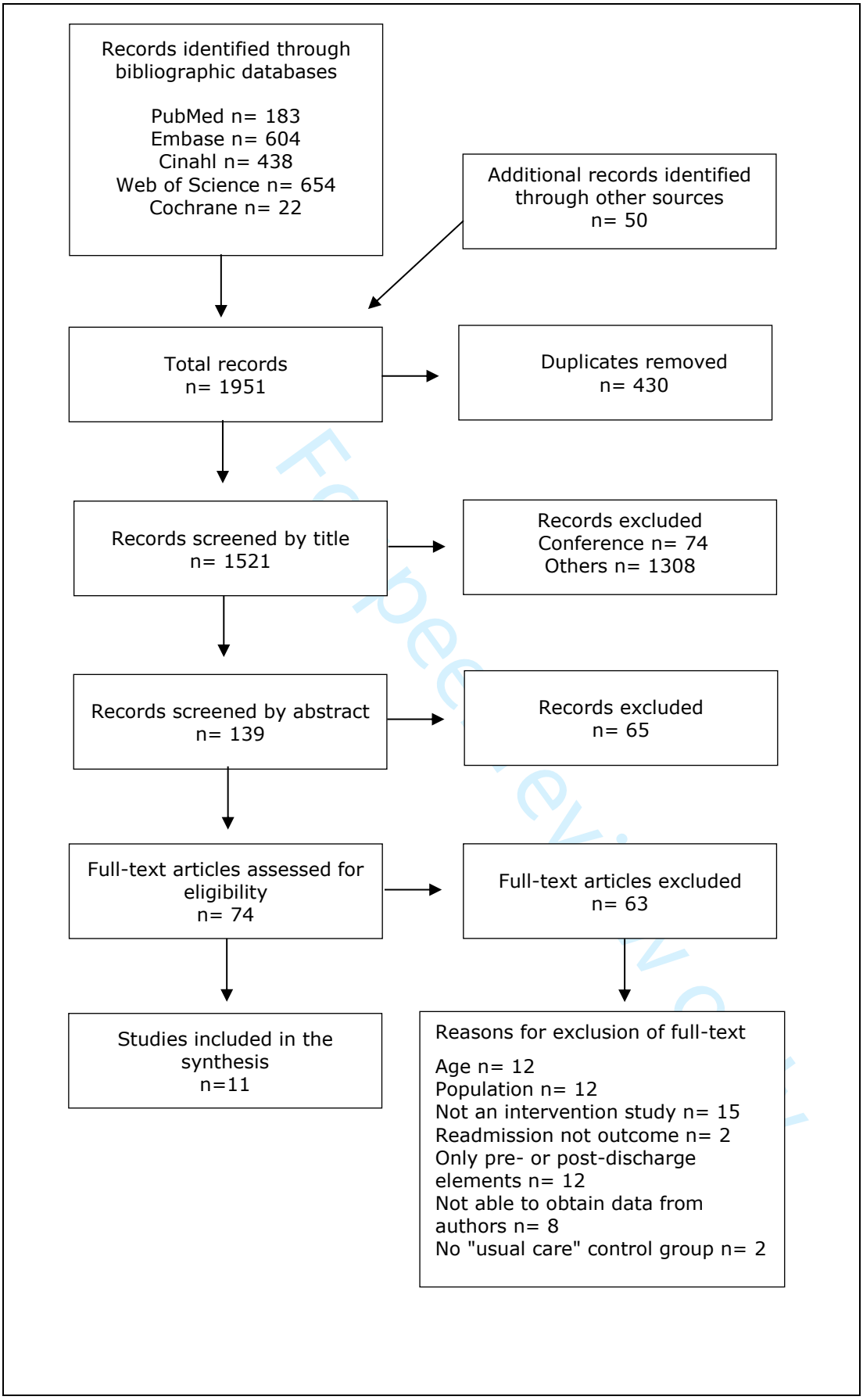
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Cochrane n= 22
Additional records identified through other sources n= 50
Total records n= 1951
Duplicates removed n= 430
Records screened by title n= 1521
Records excluded Conference n= 74 Others n= 1308
Records screened by abstract n= 139
Records excluded n= 65
Full-text articles assessed for eligibility n= 74
Full-text articles excluded n= 63
Studies included in the synthesis n=11
Reasons for exclusion of full-text
Age n= 12
Population n= 12
Not an intervention study n= 15
Readmission not outcome n= 2
Only pre- or post-discharge elements n= 12
Not able to obtain data from authors n= 8
No "usual care" control group n= 2

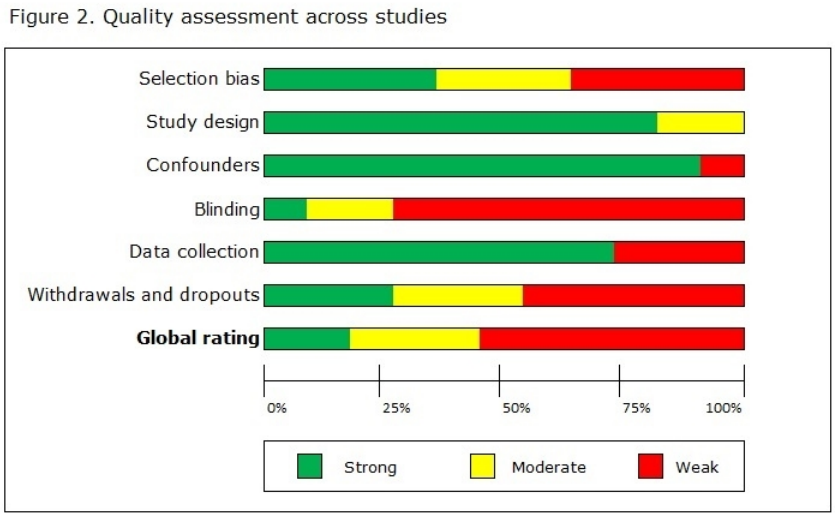


Figure 2. Quality assessment across studies. The green colour indicates strong methodological quality, yellow indicates moderate quality and red indicates weak quality across studies.

70x73mm (300 x 300 DPI)

Supplementary 1, selecting tool

Review question	"What is the impact of transitional care interventions between hospital and home on readmissions among older medical patients?"	
Reviewer Name:		Date:
Author name/study ID:		Year:
Title:		Journal:
	Include	**Exclude**
Patient population	- Elderly patients discharged from a general medical ward or emergency departments - Patients ≥ 65 years or mean age ≥ 75 years	- Patients ≤ 65 and mean age ≤ 75 - Not only medical patients - Discharging ward not specified - Only one patient group/diagnosis such as COPD, diabetes - Patients with psychiatric disorders or patients who are cognitively impaired - Other
Interventions	- Interventions in the transitional phase between hospital and home which examine the effect of the intervention on readmission rates between intervention group and control group	- No comparison/control group - Not an intervention study - Other
Comparators	- Usual or standard procedure	- Comparison between two interventions - Comparison with anything else than usual procedure - Other
Outcomes	- Readmission	- Readmission is not an outcome
Setting	- Intervention includes both pre- and post-discharge components	- Only pre-discharge components - Only post-discharge components - Other
Study design	- Quantitative studies	- Systematic reviews, meta-analysis, literature reviews - Case-reports or case-studies - Meeting abstract, study protocol - Other
Overall decision	- **Included**	- **Excluded**
Notes: Reason for exclusion:		

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Supplementary 2, search string

Search string with free-text words used in PubMed

1	elder* (title)
2	aged (title)
3	old* (title)
4	geriatric* (title)
5	1 OR 2 OR 3 OR 4
6	"discharge intervention" (title)
7	transition* (title)
8	transition* care (title)
9	6 OR 7 OR 8
10	readmission* (text word)
11	re-admission* (text word)
12	rehospitalization* (text word)
13	re-hospitalization* (text word)
14	rehospitalisation* (text word)
15	re-hospitalisation* (text word)
16	10 OR 11 OR 12 OR 13 OR 14 OR 15
17	5 AND 9 AND 16
18	Filters: published in the last 10 years; Danish; English; Norwegian;

Search string with subject headings used in PubMed

1	Patient Discharge/organization and administration (MESH)
2	Transitional Care/organization and administration (MESH)
3	Transitional care/trends (MESH)
4	1 OR 2 OR 3
5	Readmission/organization and administration (MESH)
6	Patient Readmission/trends (MESH)
7	5 OR 6
8	Aged (MESH)
9	Aged, 80 and over (MESH)
10	4 AND 7 AND 9
12	Filters: published in the last 10 years; Danish; English; Norwegian;

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Quality Assessment Tool for Quantitative Studies Dictionary



The purpose of this dictionary is to describe items in the tool thereby assisting raters to score study quality. Due to under-reporting or lack of clarity in the primary study, raters will need to make judgements about the extent that bias may be present. When making judgements about each component, raters should form their opinion based upon information contained in the study rather than making inferences about what the authors intended. Mixed methods studies can be quality assessed using this tool with the quantitative component of the study.

A) SELECTION BIAS

(Q1) Participants are more likely to be representative of the target population if they are randomly selected from a comprehensive list of individuals in the target population (score very likely). They may not be representative if they are referred from a source (e.g. clinic) in a systematic manner (score somewhat likely) or self-referred (score not likely).

(Q2) Refers to the % of subjects in the control and intervention groups that agreed to participate in the study before they were assigned to intervention or control groups.

B) STUDY DESIGN

In this section, raters assess the likelihood of bias due to the allocation process in an experimental study. For observational studies, raters assess the extent that assessments of exposure and outcome are likely to be independent. Generally, the type of design is a good indicator of the extent of bias. In stronger designs, an equivalent control group is present and the allocation process is such that the investigators are unable to predict the sequence.

Randomized Controlled Trial (RCT)

An experimental design where investigators randomly allocate eligible people to an intervention or control group. A rater should describe a study as an RCT if the randomization sequence allows each study participant to have the same chance of receiving each intervention and the investigators could not predict which intervention was next. If the investigators do not describe the allocation process and only use the words 'random' or 'randomly', the study is described as a controlled clinical trial.

See below for more details.

Was the study described as randomized?

Score YES, if the authors used words such as random allocation, randomly assigned, and random assignment.

Score NO, if no mention of randomization is made.

Was the method of randomization described?

Score YES, if the authors describe any method used to generate a random allocation sequence.

Score NO, if the authors do not describe the allocation method or describe methods of allocation such as alternation, case record numbers, dates of birth, day of the week, and any allocation procedure that is entirely transparent before assignment, such as an open list of random numbers of assignments.
If NO is scored, then the study is a controlled clinical trial.

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Was the method appropriate?

Score YES, if the randomization sequence allowed each study participant to have the same chance of receiving each intervention and the investigators could not predict which intervention was next. Examples of appropriate approaches include assignment of subjects by a central office unaware of subject characteristics, or sequentially numbered, sealed, opaque envelopes.

Score NO, if the randomization sequence is open to the individuals responsible for recruiting and allocating participants or providing the intervention, since those individuals can influence the allocation process, either knowingly or unknowingly.

If NO is scored, then the study is a controlled clinical trial.

Controlled Clinical Trial (CCT)
An experimental study design where the method of allocating study subjects to intervention or control groups is open to individuals responsible for recruiting subjects or providing the intervention. The method of allocation is transparent before assignment, e.g. an open list of random numbers or allocation by date of birth, etc.

Cohort analytic (two group pre and post)
An observational study design where groups are assembled according to whether or not exposure to the intervention has occurred. Exposure to the intervention is not under the control of the investigators. Study groups might be non-equivalent or not comparable on some feature that affects outcome.

Case control study
A retrospective study design where the investigators gather 'cases' of people who already have the outcome of interest and 'controls' who do not. Both groups are then questioned or their records examined about whether they received the intervention exposure of interest.

Cohort (one group pre + post (before and after)
The same group is pretested, given an intervention, and tested immediately after the intervention. The intervention group, by means of the pretest, act as their own control group.

Interrupted time series
A study that uses observations at multiple time points before and after an intervention (the 'interruption'). The design attempts to detect whether the intervention has had an effect significantly greater than any underlying trend over time. Exclusion: Studies that do not have a clearly defined point in time when the intervention occurred and at least three data points before and three after the intervention

Other:
One time surveys or interviews

C) CONFOUNDERS

By definition, a confounder is a variable that is associated with the intervention or exposure and causally related to the outcome of interest. Even in a robust study design, groups may not be balanced with respect to important variables prior to the intervention. The authors should indicate if confounders were controlled in the design (by stratification or matching) or in the analysis. If the allocation to intervention and control groups is randomized, the authors must report that the groups were balanced at baseline with respect to confounders (either in the text or a table).

D) BLINDING

(Q1) Assessors should be described as blinded to which participants were in the control and intervention groups. The purpose of blinding the outcome assessors (who might also be the care providers) is to protect against detection bias.

(Q2) Study participants should not be aware of (i.e. blinded to) the research question. The purpose of blinding the participants is to protect against reporting bias.

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E) DATA COLLECTION METHODS

Tools for primary outcome measures must be described as reliable and valid. If 'face' validity or 'content' validity has been demonstrated, this is acceptable. Some sources from which data may be collected are described below:

Self reported data includes data that is collected from participants in the study (e.g. completing a questionnaire, survey, answering questions during an interview, etc.).

Assessment/Screening includes objective data that is retrieved by the researchers. (e.g. observations by investigators).

Medical Records/Vital Statistics refers to the types of formal records used for the extraction of the data.

Reliability and validity can be reported in the study or in a separate study. For example, some standard assessment tools have known reliability and validity.

F) WITHDRAWALS AND DROP-OUTS

Score YES if the authors describe BOTH the numbers and reasons for withdrawals and drop-outs.

Score NO if either the numbers or reasons for withdrawals and drop-outs are not reported.

Score NOT APPLICABLE if the study was a one-time interview or survey where there was not follow-up data reported.

The percentage of participants completing the study refers to the % of subjects remaining in the study at the final data collection period in all groups (i.e. control and intervention groups).

G) INTERVENTION INTEGRITY

The number of participants receiving the intended intervention should be noted (consider both frequency and intensity). For example, the authors may have reported that at least 80 percent of the participants received the complete intervention. The authors should describe a method of measuring if the intervention was provided to all participants the same way. As well, the authors should indicate if subjects received an unintended intervention that may have influenced the outcomes. For example, co-intervention occurs when the study group receives an additional intervention (other than that intended). In this case, it is possible that the effect of the intervention may be over-estimated. Contamination refers to situations where the control group accidentally receives the study intervention. This could result in an under-estimation of the impact of the intervention.

H) ANALYSIS APPROPRIATE TO QUESTION

Was the quantitative analysis appropriate to the research question being asked?

An intention-to-treat analysis is one in which all the participants in a trial are analyzed according to the intervention to which they were allocated, whether they received it or not. Intention-to-treat analyses are favoured in assessments of effectiveness as they mirror the noncompliance and treatment changes that are likely to occur when the intervention is used in practice, and because of the risk of attrition bias when participants are excluded from the analysis.

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Component Ratings of Study:

For each of the six components A – F, use the following descriptions as a roadmap.

A) SELECTION BIAS

Good: The selected individuals are very likely to be representative of the target population (Q1 is 1) and there is greater than 80% participation (Q2 is 1).

Fair: The selected individuals are at least somewhat likely to be representative of the target population (Q1 is 1 or 2); and there is 60 - 79% participation (Q2 is 2). 'Moderate' may also be assigned if Q1 is 1 or 2 and Q2 is 5 (can't tell).

Poor: The selected individuals are not likely to be representative of the target population (Q1 is 3); or there is less than 60% participation (Q2 is 3) or selection is not described (Q1 is 4); and the level of participation is not described (Q2 is 5).

B) DESIGN

Good: will be assigned to those articles that described RCTs and CCTs.

Fair: will be assigned to those that described a cohort analytic study, a case control study, a cohort design, or an interrupted time series.

Weak: will be assigned to those that used any other method or did not state the method used.

C) CONFOUNDERS

Good: will be assigned to those articles that controlled for at least 80% of relevant confounders (Q1 is 2); or (Q2 is 1).

Fair: will be given to those studies that controlled for 60 – 79% of relevant confounders (Q1 is 1) and (Q2 is 2).

Poor: will be assigned when less than 60% of relevant confounders were controlled (Q1 is 1) and (Q2 is 3) or control of confounders was not described (Q1 is 3) and (Q2 is 4).

D) BLINDING

Good: The outcome assessor is not aware of the intervention status of participants (Q1 is 2); and the study participants are not aware of the research question (Q2 is 2).

Fair: The outcome assessor is not aware of the intervention status of participants (Q1 is 2); or the study participants are not aware of the research question (Q2 is 2).

Poor: The outcome assessor is aware of the intervention status of participants (Q1 is 1); and the study participants are aware of the research question (Q2 is 1); or blinding is not described (Q1 is 3 and Q2 is 3).

E) DATA COLLECTION METHODS

Good: The data collection tools have been shown to be valid (Q1 is 1); and the data collection tools have been shown to be reliable (Q2 is 1).

Fair: The data collection tools have been shown to be valid (Q1 is 1); and the data collection tools have not been shown to be reliable (Q2 is 2) or reliability is not described (Q2 is 3).

Poor: The data collection tools have not been shown to be valid (Q1 is 2) or both reliability and validity are not described (Q1 is 3 and Q2 is 3).

F) WITHDRAWALS AND DROP-OUTS - a rating of:

Good: will be assigned when the follow-up rate is 80% or greater (Q1 is 1 and Q2 is 1).

Fair: will be assigned when the follow-up rate is 60 – 79% (Q2 is 2) OR Q1 is 4 or Q2 is 5.

Poor: will be assigned when a follow-up rate is less than 60% (Q2 is 3) or if the withdrawals and drop-outs were not described (Q1 is No or Q2 is 4).

Not Applicable: if Q1 is 4 or Q2 is 5.

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QUALITY ASSESSMENT TOOL FOR QUANTITATIVE STUDIES



COMPONENT RATINGS

A) SELECTION BIAS

(Q1) Are the individuals selected to participate in the study likely to be representative of the target population?

1 Very likely
2 Somewhat likely
3 Not likely
4 Can't tell

(Q2) What percentage of selected individuals agreed to participate?

1 80 - 100% agreement
2 60 – 79% agreement
3 less than 60% agreement
4 Not applicable
5 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

B) STUDY DESIGN

Indicate the study design

1 Randomized controlled trial
2 Controlled clinical trial
3 Cohort analytic (two group pre + post)
4 Case-control
5 Cohort (one group pre + post (before and after))
6 Interrupted time series
7 Other specify _____
8 Can't tell

Was the study described as randomized? If NO, go to Component C.

No Yes

If Yes, was the method of randomization described? (See dictionary)

No Yes

If Yes, was the method appropriate? (See dictionary)

No Yes

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

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C) CONFOUNDERS

(Q1) Were there important differences between groups prior to the intervention?

1 Yes
2 No
3 Can't tell

The following are examples of confounders:

1 Race
2 Sex
3 Marital status/family
4 Age
5 SES (income or class)
6 Education
7 Health status
8 Pre-intervention score on outcome measure

(Q2) If yes, indicate the percentage of relevant confounders that were controlled (either in the design (e.g. stratification, matching) or analysis)?

1 80 – 100% (most)
2 60 – 79% (some)
3 Less than 60% (few or none)
4 Can't Tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

D) BLINDING

(Q1) Was (were) the outcome assessor(s) aware of the intervention or exposure status of participants?

1 Yes
2 No
3 Can't tell

(Q2) Were the study participants aware of the research question?

1 Yes
2 No
3 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

E) DATA COLLECTION METHODS

(Q1) Were data collection tools shown to be valid?

1 Yes
2 No
3 Can't tell

(Q2) Were data collection tools shown to be reliable?

1 Yes
2 No
3 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

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F) WITHDRAWALS AND DROP-OUTS

(Q1) Were withdrawals and drop-outs reported in terms of numbers and/or reasons per group?

1 Yes
2 No
3 Can't tell
4 Not Applicable (i.e. one time surveys or interviews)

(Q2) Indicate the percentage of participants completing the study. (If the percentage differs by groups, record the lowest).

1 80 -100%
2 60 - 79%
3 less than 60%
4 Can't tell
5 Not Applicable (i.e. Retrospective case-control)

RATE THIS SECTION	STRONG	MODERATE	WEAK	
See dictionary	1	2	3	Not Applicable

G) INTERVENTION INTEGRITY

(Q1) What percentage of participants received the allocated intervention or exposure of interest?

1 80 -100%
2 60 - 79%
3 less than 60%
4 Can't tell

(Q2) Was the consistency of the intervention measured?

1 Yes
2 No
3 Can't tell

(Q3) Is it likely that subjects received an unintended intervention (contamination or co-intervention) that may influence the results?

4 Yes
5 No
6 Can't tell

H) ANALYSES

(Q1) Indicate the unit of allocation (circle one)

community organization/institution practice/office individual

(Q2) Indicate the unit of analysis (circle one)

community organization/institution practice/office individual

(Q3) Are the statistical methods appropriate for the study design?

1 Yes
2 No
3 Can't tell

(Q4) Is the analysis performed by intervention allocation status (i.e. intention to treat) rather than the actual intervention received?

1 Yes
2 No
3 Can't tell

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GLOBAL RATING

COMPONENT RATINGS

Please transcribe the information from the gray boxes on pages 1-4 onto this page. See dictionary on how to rate this section.

A	**SELECTION BIAS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
B	**STUDY DESIGN**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
C	**CONFOUNDERS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
D	**BLINDING**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
E	**DATA COLLECTION METHOD**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
F	**WITHDRAWALS AND DROPOUTS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	Not Applicable

GLOBAL RATING FOR THIS PAPER (circle one):

1	STRONG	(no WEAK ratings)
2	MODERATE	(one WEAK rating)
3	WEAK	(two or more WEAK ratings)

With both reviewers discussing the ratings:

Is there a discrepancy between the two reviewers with respect to the component (A-F) ratings?

No Yes

If yes, indicate the reason for the discrepancy

1	Oversight
2	Differences in interpretation of criteria
3	Differences in interpretation of study

Final decision of both reviewers (circle one):

1	**STRONG**
2	**MODERATE**
3	**WEAK**

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Supplementary 4, exclusion of full text

Altfeld S.J., Shier G.E., Rooney M., Johnson T.J., Golden R.L., Karavolos K., Avery E., Nandi V. & Perry, A.J. 2013, "Effects of an enhanced discharge planning intervention for hospitalized older adults: a randomized trial.", The Gerontologist, vol. 53, no. 3, pp. 430-440. **Reason for exclusion:** not only medical patients.

Alvarez R, Ginsburg J, Grabowski J, Post S, Rosenberg W. The Social Work Role in Reducing 30-Day Readmissions: The Effectiveness of the Bridge Model of Transitional Care. J Gerontol Soc Work 2016 Apr;59(3):222-227. **Reason for exclusion:** not an intervention study.

Arbaje A.I., Maron D.D., Yu Q., Wendel V.I., Tanner E., Boult C., Eubank K.J. & Durso, S.C. 2010, "The geriatric floating interdisciplinary transition team.", Journal of the American Geriatrics Society, vol. 58, no. 2, pp. 364-370. **Reason for exclusion:** readmission is not an outcome.

Arnold ME, Buys L, Fullas F. Impact of pharmacist intervention in conjunction with outpatient physician follow-up visits after hospital discharge on readmission rate (Reprinted from Journal of Health-System Pharmacy Residents JHPR, vol 2, issue 4, 2014). American Journal of Health-System Pharmacy 2015 JUN 1;72(11):S36-S42. **Reason for exclusion:** only post-discharge elements, not only medical patients.

Arevalo-Flechas, L.C., Noel, P., Rottman-Sagebiel, R., Cupples, N., Conde, A., Sanchez-Reilly, S., Powers, B. & Espinoza, S. 2016, "Presence of a Caregiver Associated with Increased Odds of Readmission in a Cohort of Older Adults Included in a Transitional Care Intervention", Journal of the American Geriatrics Society, vol. 64, pp. S141-S141. **Reason for exclusion:** meeting abstract.

Bellon, J.E., Bilderback, A., Ahuja-Yende, N.S., Wilson, C., Altieri Dunn, S.C., Brodine, D. & Boninger, M.L. 2018, "University of Pittsburgh Medical Center Home Transitions Multidisciplinary Care Coordination Reduces Readmissions for Older Adults", Journal of the American Geriatrics Society. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Burt S, Berry D, Quackenbush P. Implementation of Transition in Care and Relationship Based Care to Reduce Preventable Rehospitalizations. Home Healthc Now 2015 Jul-Aug;33(7):390-393. **Reason for exclusion:** not an intervention study.

Cheen MHH, Goon CP, Ong WC, Lim PS, Wan CN, Leong MY, et al. Evaluation of a care transition program with pharmacist-provided home-based medication review for elderly Singaporeans at high risk of readmissions. Int J Qual Health Care 2017 Apr 1;29(2):200-205. **Reason for exclusion:** not only medical patients, only post-discharge elements, mean age < 75 year or age < 65 years.

Damiani G, Federico B, Venditti A, Sicuro L, Rinaldi S, Cirio F, et al. Hospital discharge planning and continuity of care for aged people in an Italian local health unit: does the care-home model reduce hospital readmission and mortality rates? BMC Health Serv Res 2009 Feb 4;9:22-6963-9-22. **Reason for exclusion:** not able to obtain data from the authors.

Dedhia P., Kravet S., Bulger J., Hinson T., Sridharan A., Kolodner K., Wright S. & Howell, E. 2009, "A quality improvement intervention to facilitate the transition of older adults from three hospitals back to their homes.", Journal of the American Geriatrics Society, vol. 57, no. 9, pp. 1540-1546. **Reason for exclusion:** only pre-discharge elements.

Dizon, M.L. & Reinking, C. 2017, "Reducing Readmissions: Nurse-Driven Interventions in the Transition of Care From the Hospital", Worldviews on evidence-based nursing, vol. 14, no. 6, pp. 432-439. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Edmans J, Bradshaw L, Franklin M, Gladman J, Conroy S. Specialist geriatric medical assessment for patients discharged from hospital acute assessment units: randomised controlled trial. BMJ 2013 Oct 8;347:f5874. **Reason for exclusion:** not able to obtain data from the authors.

Farrell T.W., TomoaiaCotisel A., Scammon D.L., Brunisholz K., Kim J., Day J., Gren L.H., Wallace S., Gunning K., Tabler J. & Magill, M.K. 2015, "Impact of an integrated transition management program

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in primary care on hospital readmissions.", Journal for healthcare quality : official publication of the National Association for Healthcare Quality, vol. 37, no. 1, pp. 81-92. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Field, T.S., Ogarek, J., Garber, L., Reed, G. & Gurwitz, J.H. 2015, "Association of early post-discharge follow-up by a primary care physician and 30-day rehospitalization among older adults", Journal of general internal medicine, vol. 30, no. 5, pp. 565-571. **Reason for exclusion:** not an intervention study.

Gardner R, Li Q, Baier RR, Butterfield K, Coleman EA, Gravenstein S. Is implementation of the care transitions intervention associated with cost avoidance after hospital discharge? J Gen Intern Med 2014 Jun;29(6):878-884. **Reason for exclusion:** readmission is not an outcome.

Gurwitz, J.H., Field, T.S., Ogarek, J., Tjia, J., Cutrona, S.L., Harrold, L.R., Gagne, S.J., Preusse, P., Donovan, J.L., Kanaan, A.O., Reed, G. & Garber, L. 2014, "An Electronic Health Record-Based Intervention to Increase Follow-Up Office Visits and Decrease Rehospitalization in Older Adults", Journal of the American Geriatrics Society, vol. 62, no. 5, pp. 865-871. **Reason for exclusion:** not only medical patients.

Hamar, B., Rula, E.Y., Wells, A.R., Coberley, C., Pope, J.E. & Varga, D. 2016, "Impact of a Scalable Care Transitions Program for Readmission Avoidance", American Journal of Managed Care, vol. 22, no. 1, pp. 28-34. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Harrison, J.D., Auerbach, A.D., Quinn, K., Kynoch, E. & Mourad, M. 2014, "Assessing the impact of nurse post-discharge telephone calls on 30-day hospital readmission rates", JGIM: Journal of General Internal Medicine, vol. 29, no. 11, pp. 1519-1525. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Hendrix C., Tepfer S., Forest S., Ziegler K., Fox V., Stein J., McConnell E.S., Hastings S.N., Schmader K. & ColonEmeric, C. 2013, "Transitional Care Partners: a hospital-to-home support for older adults and their caregivers.", Journal of the American Association of Nurse Practitioners, vol. 25, no. 8, pp. 407-414. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Heppenstall C.P., Hanger H.C., Wilkinson T.J. & Dhanak, M. 2018, "Telephone discharge support for frail, vulnerable older people discharged from hospital: Impact on readmission rates - Participant and general practitioner feedback.", Australasian journal on ageing, vol. 37, no. 2, pp. 107-112. **Reason for exclusion:** only post-discharge elements.

Hitch B, Parlier AB, Reed L, Galvin SL, Fagan EB, Wilson CG. Evaluation of a Team-Based, Transition-of-Care Management Service on 30-Day Readmission Rates. N C Med J 2016 Mar-Apr;77(2):87-92. **Reason for exclusion:** not only medical patients, mean age < 75 year or age < 65 years.

Hung D. & Leidig, R.C. 2015, "Implementing a transitional care program to reduce hospital readmissions among older adults.", Journal of nursing care quality, vol. 30, no. 2, pp. 121-129. **Reason for exclusion:** not an intervention study.

Hung D., Truong Q., Yakir M. & Nicosia, F. 2018, "Hospital-Community Partnerships to Aid Transitions for Older Adults: Applying the Care Transitions Framework.", Journal of nursing care quality, vol. 33, no. 3, pp. 221-228. **Reason for exclusion:** not an intervention study.

Hwang, U., Dresden, S.M., Rosenberg, M.S., Garrido, M.M., Loo, G., Sze, J., Gravenor, S., Courtney, D.M., Kang, R., Zhu, C.W., Vargas-Torres, C., Grudzen, C.R., Richardson, L.D. & The GEDI, W.I. 2018, "Geriatric Emergency Department Innovations: Transitional Care Nurses and Hospital Use", Journal of the American Geriatrics Society, vol. 66, no. 3, pp. 459-466. **Reason for exclusion:** only pre-discharge elements.

Jeangsawang, N., Malathum, P., Panpakdee, O., Brooten, D. & Nityasuddhi, D. 2012, "Comparison of Outcomes of Discharge Planning and Post-Discharge Follow-up Care, provided by Advanced Practice,

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Expert-by-experience, and Novice Nurses, to Hospitalized Elders with Chronic Healthcare Conditions", Pacific Rim International Journal of Nursing Research, vol. 16, no. 4, pp. 343-360. **Reason for exclusion:** not an intervention study.

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Kane R.L., Huckfeldt P., Tappen R., Engstrom G., Rojido C., Newman D., Yang Z. & Ouslander, J.G. 2017, "Effects of an intervention to reduce hospitalizations from nursing homes: A randomized implementation trial of the INTERACT program.", JAMA Internal Medicine, vol. 177, no. 9, pp. 1257-1264. **Reason for exclusion:** only post-discharge elements.

Kirkham, H., Clark, B., Paynter, J., Lewis, G. & Duncan, I. 2014, "The effect of a collaborative pharmacist-hospital care transition program on the likelihood of 30-day readmission", American Journal of Health-System Pharmacy, vol. 71, no. 9, pp. 739-745. **Reason for exclusion:** not only medical patients, mean age < 75 year or age < 65 years.

Latour, C.H., Bosmans, J.E., van Tulder, M.W., de Vos, R., Huyse, F.J., de Jonge, P., van Gemert, L.A. & Stalman, W.A. 2007, "Cost-effectiveness of a nurse-led case management intervention in general medical outpatients compared with usual care: an economic evaluation alongside a randomized controlled trial", Journal of psychosomatic research, vol. 62, no. 3, pp. 363-370. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Legrain, S., Tubach, F., Bonnet-Zamponi, D., Lemaire, A., Aquino, J., Paillaud, E., Taillandier-Heriche, E., Thomas, C., Verny, M., Pasquet, B., Moutet, A.L., Lieberherr, D. & Lacaille, S. 2011, "A New Multimodal Geriatric Discharge-Planning Intervention to Prevent Emergency Visits and Rehospitalizations of Older Adults: The Optimization of Medication in AGEd Multicenter Randomized Controlled Trial", Journal of the American Geriatrics Society, vol. 59, no. 11, pp. 2017-2028. **Reason for exclusion:** only pre-discharge elements.

Leung D.Y.P., Lee D.T.F., Lee I.F.K., Lam L.W., Lee S.W.Y., Chan M.W.M., Lam Y.M., Leung S.H., Chiu P.C., Ho N.K.F., Ip M.F. & Hui, M.M.Y. 2015, "The effect of a virtual ward program on emergency services utilization and quality of life in frail elderly patients after discharge: A pilot study.", Clinical Interventions in Aging, vol. 10, pp. 413-420. Participant age unknown, outcome readmission **Reason for exclusion:** not able to obtain data from the authors.

Lin, C.Y., Barnato, A.E. & Degenholtz, H.B. 2011, "Physician Follow-Up Visits After Acute Care Hospitalization for Elderly Medicare Beneficiaries Discharged to Noninstitutional Settings", Journal of the American Geriatrics Society, vol. 59, no. 10, pp. 1947-1954. **Reason for exclusion:** not an intervention study.

Mabire C., Bula C., Morin D. & Goulet, C. 2015, "Nursing discharge planning for older medical inpatients in Switzerland: A cross-sectional study"., Geriatric nursing (New York, N.Y.), vol. 36, no. 6, pp. 451-457. **Reason for exclusion:** not an intervention study.

Moore A.B., Krupp J.E., Dufour A.B., Sircar M., Travison T.G., Abrams A., Farris G., Mattison M.L.P. & Lipsitz, L.A. 2017, "Improving Transitions to Postacute Care for Elderly Patients Using a Novel Video-Conferencing Program: ECHO-Care Transitions.", American Journal of Medicine, vol. 130, no. 10, pp. 1199-1204. **Reason for exclusion:** not only medical patients.

Morrison, J., Palumbo, M.V. & Rambur, B. 2016, "Reducing Preventable Hospitalizations With Two Models of Transitional Care", Journal of Nursing Scholarship, vol. 48, no. 3, pp. 322-329. **Reason for exclusion:** no "usual care" control group.

Naylor MD, Bowles KH, McCauley KM, Maccoy MC, Maislin G, Pauly MV, et al. High-value transitional care: translation of research into practice. J Eval Clin Pract 2013 Oct;19(5):727-733. **Reason for exclusion:** not able to obtain data from the authors.

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Naylor M.D., Hirschman K.B., Hanlon A.L., Bowles K.H., Bradway C., Mccauley K.M. & Pauly, M.V. 2014, "Comparison of evidence-based interventions on outcomes of hospitalized, cognitively impaired older adults.", Journal of Comparative Effectiveness Research, vol. 3, no. 3, pp. 245-257. **Reason for exclusion:** only includes cognitively impaired patients.

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Supplementary 5, intervention characteristics

Study, year	Title	Description of intervention components	Providers	Context	Length of support	Treatment of comparison group
Buurman et al.[24] **2016**	Transitional Care Bridge Program Intervention	Pre-discharge: *≤ 48 hours after hospital admission*: - participants received a systematic CGA. - Care and treatment plan with prioritazion by patient and geriatric team -Multidisciplinary care provided by geriatric team Bridging: - In-person handover of CGA and care treatment plan by geriatric team to CCRN - Visit of CCRN to participant Post-discharge: *≤ 2 days after hospital discharge:* - Home visit or visit to the NH, with medication reconciliation; initiate home care; check for activities of daily living and availability of adequate help; target geriatric conditions; contact GP or NH physician *2 weeks after hospital discharge*: - Home visit or visit to the NH, evaluation of hospital-based care and treatment plan, goal setting by participant, protocol adherence for incident and prevalent geriatric conditions *6 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on the proxy and caregiver burden; further actions on geriatric conditions *12 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on care needs for incident and prevalent geriatric conditions *24 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on care needs; evaluation of geriatric conditions; handover to GP	Geriatric team A community care registered nurse (CCRN)	Three hospitals with affiliated home care organizations in the Netherlands	From ≤ 48 hours after hospital admission to 24 weeks after hospital discharge	CGA-Only Control Arm (usual care): - a systematic CGA - care and treatment plan with prioritazion by patient and geriatric team -multidisciplinary care provided by geriatric team
Chow et al.[22]	The nurse case management intervention	Two intervention arms: 1. "Home Visit" 2. "Call"	Senior year nursing students	Patients were recruited from the medical	From enrolment until 4 weeks after discharge	Social calls twice within 4 weeks.

2014		Pre-discharge in both "Home Visit" and "Call" arms: - Pre-discharge assessment: the Problem Classification Scheme, Intervention Scheme and Problem Rating Scale for Outcomes (e.g. analysis of the barriers and developed mutual goals with the patients to ensure that they were able to perform health maintenance activities, such as nutrition, monitoring of symptoms and medication adherence) Bridging: None Post-discharge - Home Visit arm: - First visit within 72 hours of discharge. - Week 2: a telephone call to evaluate the interventions and advice given during the home visit - Week 3: home visit to examine whether patients were able to continue managing their health needs after hospital discharge. - Week 4: final telephone call in the fourth week post discharge to remind patients about adherence behaviors and to motivate and support them before concluding the interventions Post-discharge – Call arm: - Week 1: first telephone call based on the patients' needs, which had been identified prior to discharge. - Week 2. call the patients: referred to the mutual goals developed by the NCM and the patients to continue with the interventions. - Week 3. call the patients referred to the mutual goals developed by the NCM and the patients to continue with the interventions. - Week 4: closing call to motivate and support the patients in maintaining self-management behaviors	Nurse Case Managers (NCM)	department of a 1700-bedded acute, general regional hospital in Hong Kong		No assessments and related interventions were conducted to the control group.
Courtney et al.[25] **2009**	Older Hospitalised Patients' Discharge Planning and In-home	Pre-discharge: *Exercise components*: - Muscle stretching, balance training, walking for endurance, and muscle strengthening using resistance exercises	A registered nurse (RN) A physio-therapist	A tertiary referral hospital in Brisbane, Australia	From enrolment until 6 months after discharge	Control group received the routine care, discharge planning, and rehabilitation advice normally provided. If

	Follow-up Protocol (OHP-DP)	*Nurse components:* - The nurse visited daily during participants' hospital stays to address concerns, facilitate the exercise program, and oversee discharge planning - Developed a transitional care plan covering the areas of functional ability and need for assistance with activities of daily living, post-discharge treatments and follow-up care, social support, chronic disease management plans and information, medication information, community services, and assistance with the exercise program. <u>Bridging</u>: - Written guidelines were provided on post-discharge management, including diagrams and specific instructions for their exercise program <u>Post-discharge</u>: - Within 48 hours of discharge, the nurse undertook a home visit to assess availability of support, address transitional concerns, provide advice and support, and ensure that the exercise program could be safely undertaken at home. - Extra home visits were provided if required - Weekly follow-up telephone calls were provided for 4 weeks, followed by monthly follow-up for a further 5 months				in-home follow-up was necessary, it was organized in the routine manner (e.g., referral to community health services)
Finlayson et al.[23] **2018**		Three intervention arms: 1. "Exercise" 2. "Nurse Home visit and Telephone follow-up" (N-HaT) 3. "Exercise and Nurse Home Visit and Telephone follow up" (ExN-HaT) <u>Exercise arm:</u> *Pre-discharge*: - Within 72 hours of admission, a physiotherapist will visit measure ADL and conduct performance tests of balance and gait. This information will be used to plan individualised exercise programs designed to improve strength, stability, coordination, endurance, mobility, and improve self confidence with respect to ADL. The exercise prescription will be developed by the patient, care givers, doctors, and ward nurses. -Physiotherapist will provide a pedometer	Advanced Practice Gerontic Nurse (APGN) A physiotherapist or exercise physiologist	Two tertiary metropolitan hospitals in Australia	From enrolment until 24 weeks after discharge	Routine hospital and follow-up care as provided by the health service. This involved a needs assessment by the hospital health staff, discharge planning, and referrals for follow up services as appropriate.

		together with simple verbal and written instructions regards usage and journaling of activity levels - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required *Bridging:* None *Post-discharge:* - An exercise physiologist conducts 6 weekly in home visits to reassess the patient's physical measures and functional capacity, evaluate progress with the exercise program, and reset program goals accordingly <u>N-HaT arm:</u> *Pre-discharge*: - Within 72 hours of admission, an APGN will undertake a health assessment, and prepare a comprehensive transitional care plan. This process includes discussing the planned discharge plan with patient's caregivers, doctor, and ward nurses to individualise the plan - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required. *Bridging:* None *Post-discharge:* - Within 48 hours after discharge, an APGN conducts a home visit. This home visit is undertaken to 1) identify sufficient caregiver support is available at home, 2) assess the home environment is safe, 3) determine the patient has the required medications and dressings (if required), 4) ensure the patient (and caregiver) fully understand their medication and treatment regimes; 5) reinforce and further explain exercise program and use of pedometer and journal in the home, and 6) provide advice and support to the caregiver				

		-Additional home visits are available by APGN if required - over the initial 4 week period after discharge, a APGN provides weekly telephone follow-up calls -APGN is available to be contacted via telephone 7 days per week - For 6 months, monthly telephone follow-up calls are conducted by the APGN. <u>ExN-HaT arm</u>: *Pre-discharge*: - Within 72 hours of admission, an APGN will undertake a health assessment, and prepare a comprehensive transitional care plan. This process includes discussing the planned discharge plan with patient's caregivers, doctor, and ward nurses to individualise the plan - Within 72 hours of admission, a physiotherapist will visit the intervention patients to determine the functional capacity of the patient using measures of ADL and performance tests of balance and gait. This information will be used to plan individualised exercise programs designed to improve strength, stability, coordination, endurance, mobility, and improve self confidence with respect to ADL. The exercise prescription will be developed by the patient, caregivers, doctors, and ward nurses. - Physiotherapist will provide a pedometer together with simple verbal and written instructions regards usage and journaling of activity levels - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required. *Bridging:* None *Post-discharge:* - Within 48 hours post discharge, an APGN provides one home visit. This home visit is undertaken to 1) identify sufficient caregiver support is available at home, 2) assess the home environment is safe, 3) determine the patient has the required medications and dressings (if				

		required), 4) ensure the patient (and caregiver) fully understand their medication and treatment regimes; 5) reinforce and further explain exercise program and use of pedometer and journal in the home, and 6) provide advice and support to the caregiver - Additional home visits are available by APGN if required - An exercise physiologist conducts 6 weekly visits to reassess the patient's physical measures and functional capacity, evaluate progress with the exercise program, and reset program goals accordingly - Over the initial 4 week period after discharge, a APGN provides weekly telephone follow-up calls - APGN is available to be contacted via telephone 7 days per week - For 6 months, monthly telephone follow-up calls are conducted by the APGN				
Koehler et al.[26] **2009**	Early Care Bundle or Supplemental Care Bundle	<u>Pre-discharge</u>: By Study Care Coordinator - Daily condition specific education - Additional time to identify and address discharge barriers - Extra discharge teaching with focus on self-management By Study Pharmacist: - Medication reconciliation at admit - Daily medication review and education - Additional medication change recommendations as felt to be indicated - Medication reconciliation at discharge - Counselling on discharge Structured documents: - personal health record (completed by the patient and family for future use) <u>Bridging</u>: Structured documents: - Supplemental discharge form given to patient and faxed to PCP <u>Post-discharge</u>: By Study Care Coordinator:	Care coordinators (CC) Clinical pharmacists (CP)	1 of 2 hospital-medicine groups at the 900-bed Baylor University Medical Center, Texas	Max 24 hours after enrolment to 1 week after discharge	Usual care: - Needs assessment - discharge planning - Medication reconciliation at admit - Daily direct patient care - Discharge medication education - Medication reconciliation at discharge - Discharge education - Medication review with new orders and as needed - Medication change recommendations - Standard BUMC discharge form - New discharge prescriptions

		- Follow up call at 5-7 days post-discharge By Study Pharmacist: - Follow up call at 5-7 days post-discharge				
Lin et al.[31] **2015**	Integrated Care and Discharge Support for elderly patients (ICDS)	Pre-discharge: - Risk stratification - Comprehensive geriatric assessment - Discharge planning are performed Bridging: None Post-discharge: Patients receive either: - ICM Case Management or - HST service *ICM Case Management* : - Each high-risk older patient is followed up by a case manager for a period of around 3 months following hospital discharge - Case managers provide home visits and telephone support and they are responsible for community service coordination and ensuring patient compliance with planned services and management. *HST service*: - Rapid and intensive community support for discharged patients, offering services such as meal delivery, household cleaning, respite care, and home assessment and modification Link nurses, ICM case managers, and the HST hold weekly multidisciplinary case conferences chaired by an ICM geriatrician. If needed, referral for rehabilitation in a geriatric day hospital, fast track clinic, or early specialist clinic follow-up can be offered to patients.	At hospital: - Link nurses - Geriatricians Community ICM Case Management: - Social workers - Physiotherapist (PT) - Occupational therapist (OT) - Advanced practice nurse (APN) Community HST service: - Nurses - PT - OT - Other allied health members	Four hospitals of Hong Kong West Cluster.	ICM case management: - From recruitment until 3 months after discharge HST service: - NR	NR
Nielsen et al.[28] **2018**	Elderly Activity Performance Intervention	Pre-discharge: *Component 1:* -Assessment of the patients' performance of daily activities - Assessment of Motor and Process Skills Based on the results in component 1, patients with limitations in performing daily activities	Occupational therapist (OT) Physiotherapist	Short-stay unit (ED) at a 1,150-bed university hospital in Denmark	From enrolment until the day after discharge	Usual practice: relevant medical treatment and care. Referral to occupational therapy and physiotherapy took place only if the medical or nursing

		received components 2 and 3 *Component 2:* - A rehabilitation plan was prescribed. The rehabilitation plan included a description of the patient's previous and current performance of daily activities and specified the need for further rehabilitation. Bridging: - Primary care was informed about the discharge, and visitation of the patient to further rehabilitation Post-discharge: *Component 3:* - Home visit by an OT the day after discharge for patients with a prescribed rehabilitation plan. The home visit aimed to enhance the patient's performance of daily activities and to start rehabilitation. - OT screened the home for safety risks and factors that potently could limit the performance of daily activities.				staff considered it necessary. If the occupational therapist was summoned to asses a patient, the occupational therapist performed a short interview and a non-standardized observation of the patient's performance of basic daily activities and the results were communicated to the municipality homecare staff. If necessary, nurses from the ED organized referral to nursing home care after discharge
Robinson et al.[32] **2015**	Integrated Transition of Care (ITC)	Pre-discharge: - Nutrition screening, and if necessary referral to a dietitian - Allied health review - Discharge medicines reconciliation and patient education by a pharmacist. Bridging: - Records were communicated to primary care providers electronically and via mail, and when required, by telephone. Post-discharge: Provided only to patients who were returning to independent community living - 1 and 3 days after discharge: a telephone assessment, education, and support. The aim was to identify problems not dealt with during the discharge process, assist with self-management, and ensure appropriate health and social supports.	Community nurses A geriatrician A pharmacist Cultural support workers	Two acute general hospitals with 536 and 250 beds in Auckland, New Zealand.	From recruitment until 3 days after discharge	NR

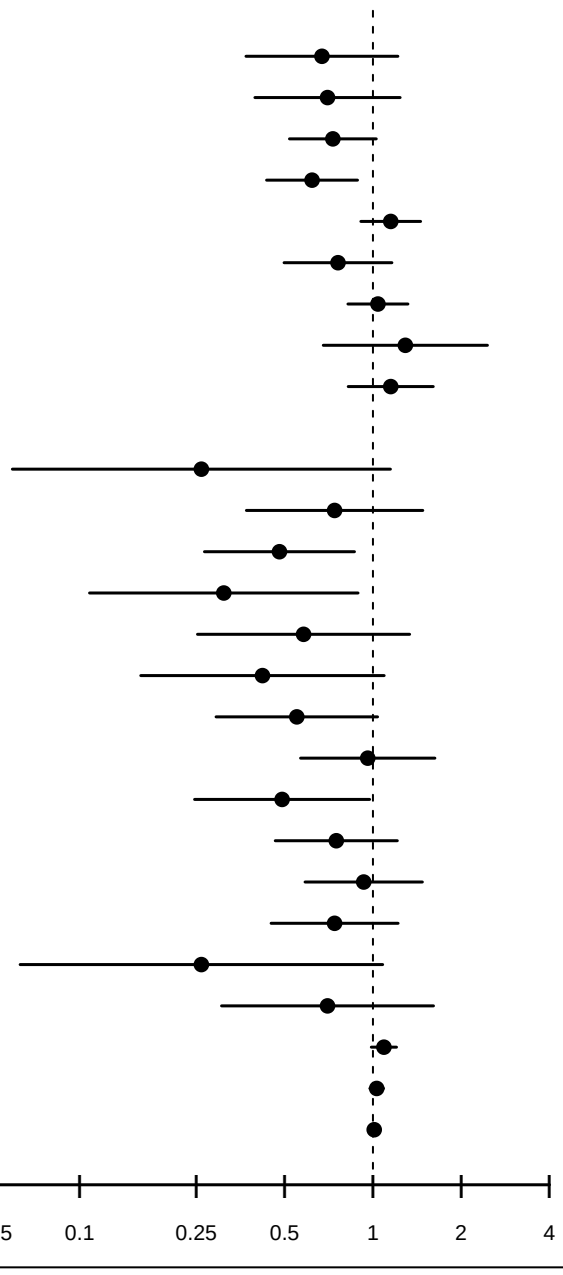
Rottman-Sagebiel et al.[29] **2018**	The Geriatrics Medication Education at Discharge project (GMED)	Pre-discharge: - An individual, face-to-face meeting between the CPS and the patient. - Medication reconciliation - Medication education regarding administration and usage of the patient's medications. - The caregiver, if any, was included in the discussion either at the bedside or by telephone following the face-to-face visit with the patient. - Discuss findings with inpatient team in person or by telephone and through documentation in the CPRS. Bridging: - All information the telephone encounter was communicated to the PCP through CPRS documentation and by telephone as needed. Post-discharge: - A telephone visit within 2 to 3 days of discharge. The purpose of the telephone visit was to perform medication reconciliation, identify and rectify medication errors, provide further patient education, and assist in facilitating appropriate follow-up by the patient's PCP, if required.	Primary care provider (PCP) A clinical pharmacist specialist (CPS) with oversight from a senior CPS with geriatric pharmacology expertise An internist or geriatrician	Veterans Memorial Hospital of the South Texas Veterans Health Care System (STVHCS)	From recruitment until 2 or 3 days after discharge	NR
Sahota et al.[27] **2017**	CIRACT	Pre-discharge: - Comprehensive assessment - Formed a rehabilitation plan - Treated daily (7 days a week, if appropriate) and the duration of rehabilitation they received depended on their needs. - The team liaised with the participant and their carer(s) to visit the participant's home to assess and provide recommendations for equipment and make adaptations and/or modifications as required Bridging: None Post-discharge: - Team visited the participant at home to assess the level of rehabilitation required	Senior occupational therapist (transition coach) Senior physiotherapist Assistant practitioner A social services practitioner	Queen's Medical Centre: a 1,800-bed hospital in Nottingham, UK.	NR	The standard THB-Rehab service was provided on weekdays only. An assessment of the participant's ability to perform certain tasks and provided recommendations for rehabilitation. Referral to community based services for provision of equipment at home, personal care and ongoing rehabilitation in community services where appropriate at

		- Further follow-up visits as deemed necessary - Appropriate referral to additional community services				discharge.
Voss et al.[30] **2011**	Care Transitions Intervention (CTI)	Pre-discharge: - Meet with the patient to establish initial rapport - Introduce the personal health record, including their main health problems, their medications, and questions for their health care providers - Arrange a home visit, ideally within 48 to 72 hours after hospital discharge Bridging: - Discuss topics related to participants with their outpatient providers Post-discharge: *Home visit within 3 days after discharge*: - Reconcile all of the patient's medication regimens - TC imparted skills for effectively communicating care needs during subsequent encounters with health care professionals. - TC also reviewed with the patient any red flags that indicated a condition was worsening and provided education about the initial steps to take to manage the red flags and when to contact the appropriate health care professional. *Phone call*: - Telephoning 3 times during a 28-day post-hospitalization discharge period. The first telephone call within 7 to 10 days generally focused on determining whether the patient had received appropriate services. In the 2 subsequent telephone calls, TC reviewed the patient's progress toward goals established during the home visit, discussed any encounters that took place with other health care professionals, reinforced the importance of maintaining and sharing the personal health record, and supported the patient's role in chronic illness self-management. The final telephone call was conducted by day 30	Transition coach (TC) who had a background in nursing or social work	6 Rhode Island acute care hospitals, including 2 community hospitals, 3 teaching hospitals, and a tertiary care center and teaching hospital, ranging from 129 beds to 719 beds	From enrolment until 30 days after discharge	From randomization to the day of discharge

NR: not reported, CGA: comprehensive geriatric assessment, NH: nursing home, GP: general practitioner, ADL: activity of daily living, PCP: primary care provider, ICM: integrated care model, HST: home support team, ED: emergency department.

Supplementary 6, forest plot grouped by study population

Author	Intervention	OA	RR (95% CI)
Unselected patients			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Patients at risk			
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)

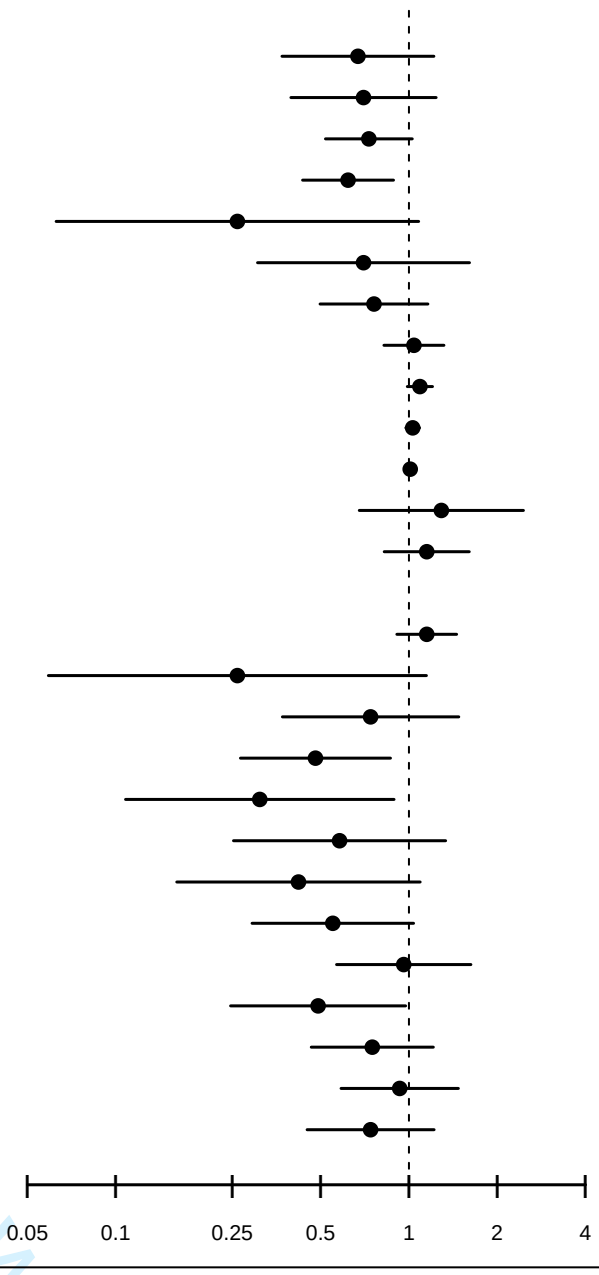


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 7, forest plot grouped by intervention intensity

Author	Intervention	OA	RR (95% CI)
Low intensity			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
High intensity			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)

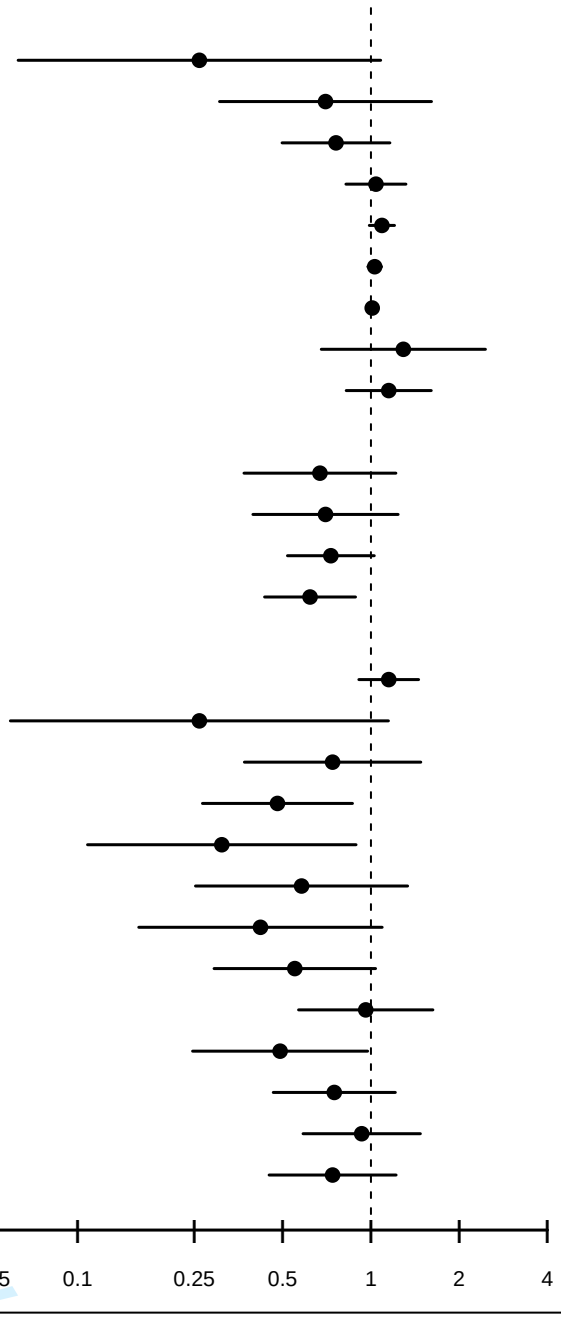


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 8, forest plot grouped by length of support

Author	Intervention	OA	RR (95% CI)
≤ 7 days			
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
8−30 days			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
≥ 31 days			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)

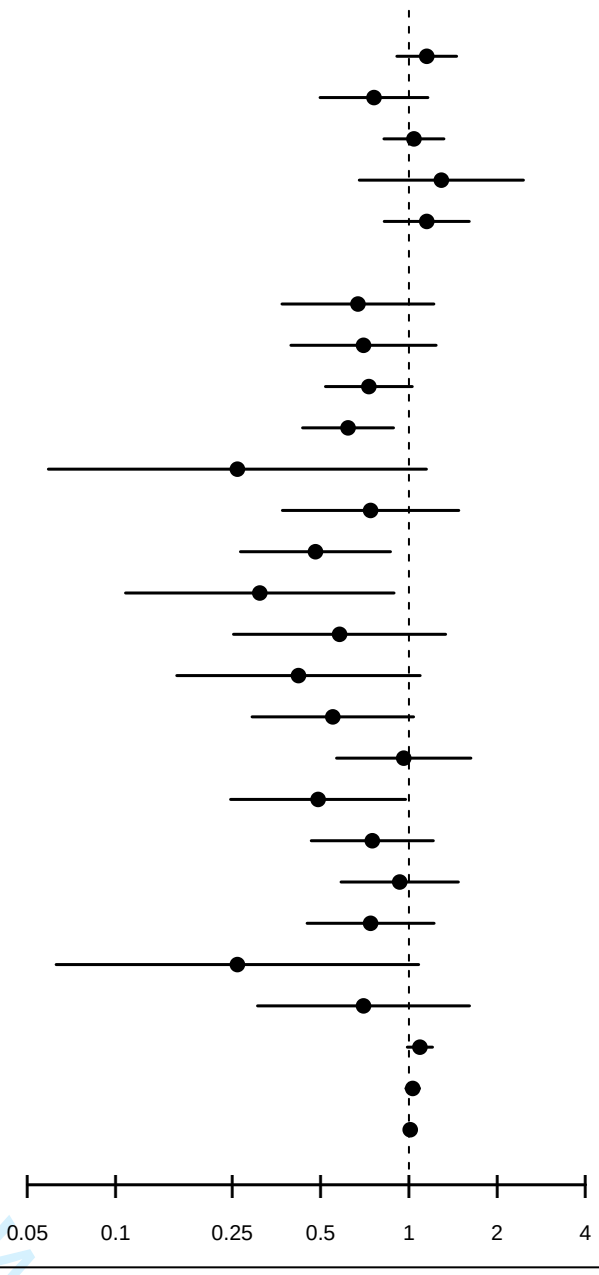


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 9, forest plot grouped by country of origin

Author	Intervention	OA	RR (95% CI)
European studies			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Non−European studies			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)

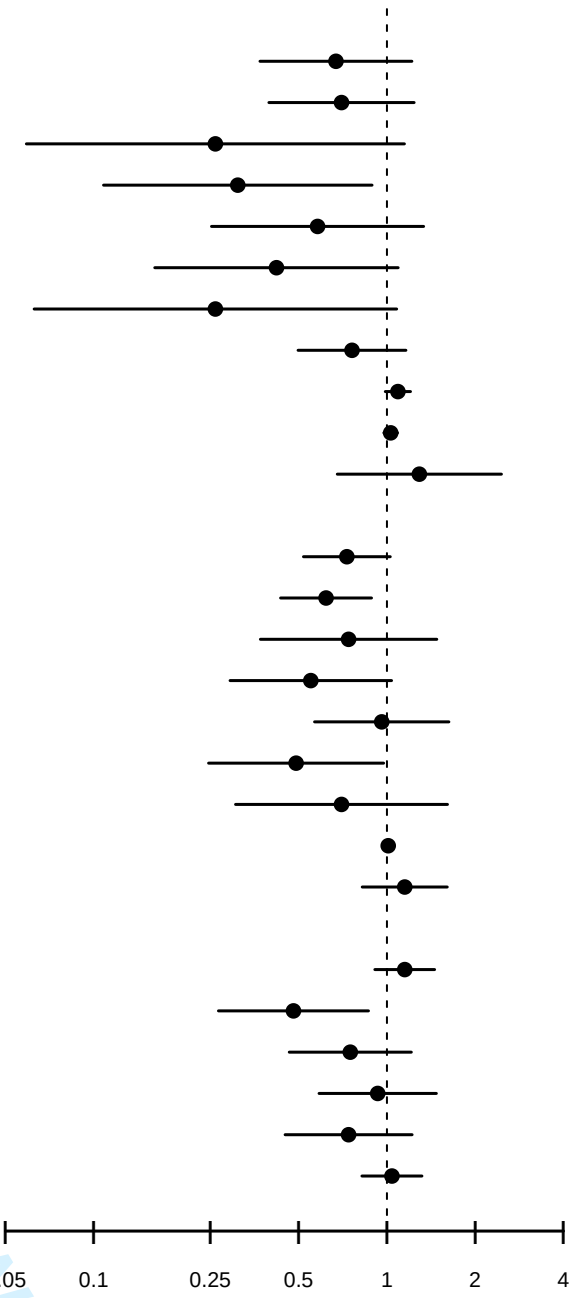


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 10, forest plot grouped by outcome assessment

Author	Intervention	RR (95% CI)
≤ **30 days**		
Chow et al.	Home visit	0.67 (0.37−1.22)
Chow et al.	Call	0.70 (0.40−1.24)
Courtney et al.		0.26 (0.06−1.15)
Finlayson et al.	EN−Hat	0.31 (0.11−0.89)
Finlayson et al.	Exercise	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	0.42 (0.16−1.09)
Koehler et al.		0.26 (0.06−1.08)
Nielsen et al.		0.76 (0.50−1.16)
Robinson et al.		1.09 (0.99−1.20)
Robinson et al.		1.03 (0.98−1.09)
Sahota et al.		1.29 (0.68−2.46)
31−91 days		
Chow et al.	Home visit	0.73 (0.52−1.03)
Chow et al.	Call	0.62 (0.43−0.89)
Courtney et al.		0.74 (0.37−1.48)
Finlayson et al.	EN−Hat	0.55 (0.29−1.04)
Finlayson et al.	Exercise	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	0.49 (0.25−0.97)
Koehler et al.		0.70 (0.30−1.61)
Robinson et al.		1.01 (0.97−1.05)
Sahota et al.		1.15 (0.82−1.61)
≥ **92 days**		
Buurman et al.		1.15 (0.91−1.45)
Courtney et al.		0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	0.75 (0.46−1.21)
Finlayson et al.	Exercise	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	0.74 (0.45−1.22)
Nielsen et al.		1.04 (0.82−1.32)

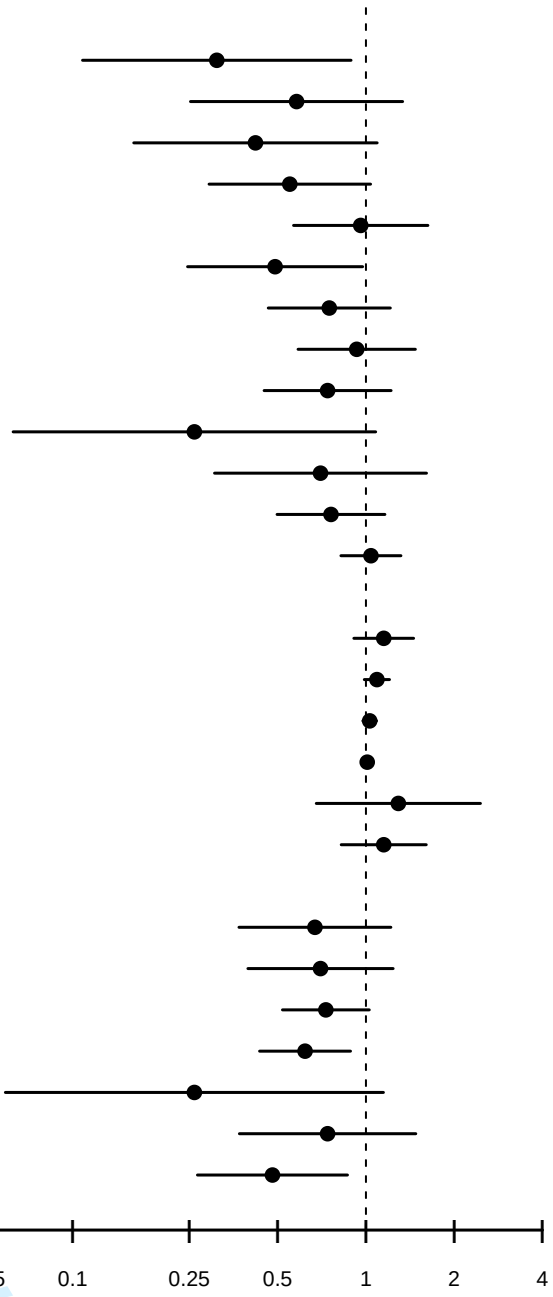


Abbreviations: RR, relative risk; CI, confidence interval

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Supplementary 11, forest plot grouped by quality assessment

Author	Intervention	OA	RR (95% CI)
Weak			
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Moderate			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Strong			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)



Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	none
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	17 Table 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Only 1 outcome Table 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	13
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	19 Fig. 2
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	None
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	19-22, 23
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	22
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

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From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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PRISMA 2009 Checklist

For more information, visit: **www.prisma-statement.org**.

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Synthesis Without Meta-analysis (SWiM) reporting items

The citation for the Synthesis Without Meta-analysis explanation and elaboration article is: Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, Hartmann-Boyce J, Ryan R, Shepperd S, Thomas J, Welch V, Thomson H. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline BMJ 2020;368:l6890 http://dx.doi.org/10.1136/bmj.l6890

SWiM is intended to complement and be used as an extension to PRISMA			
SWiM reporting item	**Item description**	**Page in manuscript where item is reported**	**Other***
Methods			
1 Grouping studies for synthesis	1a) Provide a description of, and rationale for, the groups used in the synthesis (e.g., groupings of populations, interventions, outcomes, study design)	Page 6 (data synthesis) Page 15, 17 (Sub-group analysis)	
	1b) Detail and provide rationale for any changes made subsequent to the protocol in the groups used in the synthesis	No changes made	
2 Describe the standardised metric and transformation methods used	Describe the standardised metric for each outcome. Explain why the metric(s) was chosen, and describe any methods used to transform the intervention effects, as reported in the study, to the standardised metric, citing any methodological guidance consulted	Page 6 (data synthesis)	
3 Describe the synthesis methods	Describe and justify the methods used to synthesise the effects for each outcome when it was not possible to undertake a meta-analysis of effect estimates	Page 6 (data synthesis) Page 13 (synthesis of results)	

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Synthesis Without Meta-analysis (SWiM) reporting items

4 Criteria used to prioritise results for summary and synthesis	Where applicable, provide the criteria used, with supporting justification, to select the particular studies, or a particular study, for the main synthesis or to draw conclusions from the synthesis (e.g., based on study design, risk of bias assessments, directness in relation to the review question)	Page 6 (quality assessment) Page 6 (data synthesis)	
SWiM reporting item	**Item description**	**Page in manuscript where item is reported**	**Other***
5 Investigation of heterogeneity in reported effects	State the method(s) used to examine heterogeneity in reported effects when it was not possible to undertake a meta-analysis of effect estimates and its extensions to investigate heterogeneity	Page 6 (data synthesis)	
6 Certainty of evidence	Describe the methods used to assess certainty of the synthesis findings	Page 6 (data synthesis)	
7 Data presentation methods	Describe the graphical and tabular methods used to present the effects (e.g., tables, forest plots, harvest plots). Specify key study characteristics (e.g., study design, risk of bias) used to order the studies, in the text and any tables or graphs, clearly referencing the studies included	Page 6 (data synthesis)	
Results			
8 Reporting results	For each comparison and outcome, provide a description of the synthesised findings, and the certainty of the findings. Describe the result in language that is consistent with the question the synthesis addresses, and indicate which studies contribute to the synthesis	Table 1 study characteristics Page 8 (study characteristics)	

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Synthesis Without Meta-analysis (SWiM) reporting items

		Page 12 (synthesis of results)	
Discussion			
9 Limitations of the synthesis	Report the limitations of the synthesis methods used and/or the groupings used in the synthesis, and how these affect the conclusions that can be drawn in relation to the original review question	Page 21 (strength and limitation of this review)	

PRISMA=Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

*If the information is not provided in the systematic review, give details of where this information is available (e.g., protocol, other published papers (provide citation details), or website (provide the URL)).

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BMJ Open

The impact of transitional care interventions on hospital readmissions in older medical patients: a systematic review

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Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Health services research, Evidence based practice
Keywords:	GERIATRIC MEDICINE, PREVENTIVE MEDICINE, INTERNAL MEDICINE

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TITLE PAGE

The impact of transitional care interventions on hospital readmissions in older medical patients: a systematic review

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ABSTRACT

Objectives

To identify and synthesise available evidence on the impact of transitional care interventions with both pre-and post-discharge elements on readmission rates in older medical patients.

Design

A systematic review.

Method

Inclusion criteria were: medical patients ≥65 years or mean age in study population of ≥75 years; interventions were transitional care interventions between hospital and home with both pre- and post-discharge components; outcome was hospital readmissions. Studies were excluded if they: included other patient groups than medical patients, included patients with only one diagnosis or patients with only psychiatric disorders. PubMed, The Cochrane Library, Embase, Cinahl, and Web of Science were searched from January 2008 to August 2019. Study selection at title level was undertaken by one author; the remaining selection process, data extraction and methodological quality assessment were undertaken by two authors independently. A narrative synthesis was performed, and effect sizes were estimated.

Result

We identified 1951 records and included 11 studies: five randomised trials, four non-randomised controlled trials and two pre-post cohort studies. The 11 studies represent 15 different interventions and 29 outcome results measuring readmission rates within 7-182 days after discharge. Twenty-two of the 29 outcome results showed a drop in readmission rates in the intervention groups compared with the control groups. The most significant impact was seen when interventions were of high intensity, lasted at least one month and targeted patients at risk. The methodological quality of the included studies was generally poor.

Conclusion

Transitional care interventions reduce readmission rates among older medical patients although the impact varied at different times of outcome assessment. High-quality studies examining the impact of interventions are needed, preferably complimented by a process evaluation to refine and improve future interventions.

PROSPERO registration number CRD42019121795.

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STRENGTHS AND LIMITATIONS

- Focus on discharge interventions where intervention elements are provided both at the hospital and at home
- High internal validity as PRISMA guideline has been applied
- Results may underrepresent negative study findings as negative impact often remains unpublished
- Unplanned readmissions were the only outcome assessed in this review

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INTRODUCTION

Transitional care interventions (TCIs) may be essential in older medical patients' transition from hospital to home as they may prevent adverse events and unplanned hospital readmissions. These events can have detrimental consequences for the individual patient.

The global demography is radically changing. In the European Union, elderly above 65 years are estimated to account for 29.1% of the total population in 2080 compared with 19.2% in 2016. Additionally, the fraction of the population above 80 years is expected to double between 2016 and 2080[1]. We may therefore expect a dramatic increase in healthcare service demands and costs[2]. As these changes will bring substantial challenges to healthcare systems[3 4], the potential need for TCIs will also increase. Older people needing healthcare are often medical patients with several concurrent diseases, reduced physical or mental functionalities, limited ability to provide self-care, and they are often living alone and need care from primary or secondary healthcare services[5].

Older people with complex co-morbid conditions are at high risk of adverse events and safety incidents immediately after their discharge from hospital[6]. Unplanned readmission seems to be related to insufficient discharge planning, and unintended events during discharge and transition such as medication errors and inadequate communication between hospital- and primary care professionals[7-10]. By contrast, optimised, customised and patient-centred discharge planning and transitions may reduce length of hospital stay (LOS), risk of readmission, medication discrepancies and mortality; and may as well improve the patients' activity of daily living (ADL) and reduce healthcare costs[11].

One approach to addressing these challenges is to examine the impact of interventions aimed at reducing readmissions.

Previous systematic reviews have mainly evaluated the impact of hospital- and/or home-based interventions on readmissions[12 13] and included populations with specific conditions or both medical and surgical patients. To our knowledge, no systematic review has been conducted examining the impact of TCIs that take place in both hospital and home on older medical patients' readmission rates based on recent data.

The purpose of this systematic review is to evaluate the impact of TCIs with both pre- and -post-discharge elements on readmission among older medical patients.

METHODS

This review was registered in the PROSPERO database prior to data collection (CRD42019121795)[14]. It is presented in accordance with the Preferred Reporting Items for Systematic Reviews (PRISMA)[15].

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Eligibility criteria

We identified studies that aimed to reduce readmission rates through TCIs among older medical patients. The PICO process for framing the research question was applied[16] and defined as:

i. ***P****opulation*: Older medical patients discharged from a general medical ward or emergency department (ED);
ii. ***I****ntervention*: In the transitional phase between hospital and home which examined the impact of the intervention on readmission rates. The interventions had to include both pre- and post-discharge components;
iii. ***C****omparison*: Usual care defined as standard care and treatment;
iv. ***O****utcome*: Unplanned readmission to hospital.

Selecting tool is displayed in Supplementary 1.

Studies were excluded if:

i. The population was aged under 65 years or the mean age was below 75 years
ii. They included other patient groups than medical patients (e.g. surgical patients)
iii. They included participants with only one medical diagnosis (International Classification of Disease -10th edition)
iv. Studies included participants with psychiatric disorders only
v. They compared interventions with anything other than usual care
vi. Readmission was not an outcome
vii. Intervention only included either pre- or post-discharge components; they were reviews, case reports or case studies without comparison groups.

Information sources

To identify eligible studies, we searched the following bibliographic databases: PubMed, The Cochrane Library, Embase, Cinahl and Web of Science from January 2008 to August 2019. An extensive snowball search was performed where the reference lists from relevant studies, systematic reviews and included studies were examined. Additionally, publication lists from prominent researchers within the field were examined. Grey literature was searched in all relevant resources listed by Paez[17] from January 2008 to August 2019. Authors from relevant study protocols and grey literature were contacted in order to examine whether the studies were published or study results were available. The Negative Results Scientific Journal was searched.

Search strategy

The search string (Supplementary 2) was developed in collaboration between the authors and a university research librarian. We used key terms, free text-words, subject headings, index terms and appertaining synonyms, which were

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identified through relevant theory and research. The searches were limited by only including studies published in English or Scandinavian languages. The bibliographic searches were conducted on 13 and 14 December 2018 and the searches were regularly updated until 31 August 2019.

Study selection
First, titles were screened for their potential relevance according to population and outcome by the first author (LFR). Second, two authors independently screened titles and abstracts for intervention eligibility (LFR and MG). Third, an assessment of the full text was performed by two authors independently (always including LFR). In case of disagreement, a third author, who was chosen *a priori*, was consulted.

Data extraction
The Cochrane Data Extraction Form was modified to fit the present patient group and intervention type[18]. Data from the included studies were extracted by two researchers independently (always including LFR). Extracted data included study characteristics and results such as author, year of publication, country, study design, setting, participants, study size, outcomes, follow-up time and impact of intervention in numbers and/or per cent. Only data on the outcome "readmission" were extracted and analysed.

Quality assessment
"The Quality Assessment Tool for Quantitative Studies" (EPHPP) (Supplementary 3) was applied to assess bias in each study included in the review. The validated tool is recommended by The Cochrane Collaboration[16 19] and provides a standardised means to assess study quality. The EPHPP assesses six methodological dimensions: selection bias, study design, confounders, blinding, data collection as well as withdrawals and dropouts. It distributes the overall methodological rating into a strong, moderate or weak measure of internal validity[20]. Quality assessment will be conducted at study level. All studies meeting the inclusion criteria are included in the synthesis regardless of the results of the quality assessments. Quality assessment across studies will be analysed and the impact hereof will be discussed.

Data synthesis
Inspired by Pigott and Shepperd[21], the following aspects that can entail heterogeneity will be assessed: 1) context, 2) target population, 3) intervention, 4) methodological features and 5) researcher characteristics and reporting context. This assessment will result in a descriptive synthesis or a meta-analysis of data. Risk ratios (RR) and their 95% confidence intervals (CIs) are calculated[16(section 6.4.1)] when possible and presented in forest plots stratified by subgroups. Only main results will be presented and discussed according to sub-groups based on study- and intervention characteristics and review findings. The certainty of the synthesised results will be assessed using GRADE[16(section 14-2-1)].

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Patient and public involvement

Patients or the public were not involved in this study.

RESULTS

Study selection

In total, 1,951 records were identified. Of those, 1,901 records were identified through bibliographic databases and 50 records were found through other sources. After removing duplicates, 1,521 records were screened by title and abstract. Seventy-four records were considered for full-text review of which 11 met the eligibility criteria (Figure 1). The 63 remaining records were excluded due to participants' age, population, study design, readmission not listed as an outcome, only pre- or post-discharge elements in the intervention, not able to obtain data from authors and no "usual care" control group (Supplementary 4).

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Study characteristics

The 11 included studies represented 15 different interventions and 29 outcome assessments measuring readmission rates at different time points. Two studies were multi-arm studies[22 23]. Study characteristics are shown in Table 1. Five of the included studies were randomised trials (RCTs)[23-27], four non-randomised controlled trials (NRCTs)[22 28-30] and two pre-post cohort studies[31 32].
Sample sizes of individual studies ranged from 41 to 19,157. Allocation to intervention or control group was performed on an individual level. The majority of studies reported that readmissions were unplanned or acute[23-28 32]. Outcome assessments were conducted 7-182 days after discharge. In addition to readmission, the 11 studies assessed outcomes according to physical functioning[24 27]; cognitive functioning[24]; quality of life[22 25 27]; time to either hospital readmission or discharge from nursing homes[24 28]; self-efficacy[22]; self-rated health[22]; visits to emergency departments, general practitioners or allied health professional[25 26 28 31 32]; length of stay[27 31]; co-morbidity[27]; cost-effectiveness[27] and mortality[24 28].Three studies took place in USA[26 29 30] , two in Australia[23 25], two in Hong Kong[22 31], one in the Netherlands [24], one in the United Kingdom[27], one in Denmark[28] and one in New Zealand[32]. The studies were published between 2009 and 2018.

Study population

In total, approximately 24,500 patients were evaluated in our review; 8,800 in the intervention groups and 15,700 in the control groups. The mean age of the participants in the intervention groups was approximately 78 years (range 74.9–83.6), whilst the mean age among the control groups was approximately 79 years (range 75.2-84.5). Approximately 59% (range 3.6-85) of the participants in the intervention groups were females and 52% (range 2.2-67) of the participants in the control group were females.

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Table 1 Study characteristic

Author Year	Country Setting	Design	Intervention	Study population	Mean age, years (SD) or median (range)	Female n, (%)	Outcomes	OA	Readmission (event/total)	RR (CI)
Buurman[24] 2016	Netherland Multi-centre study	RCT	Transitional Care Bridge Program Intervention	T: 674 I: 337 C:337 Unselected patients	I: 79.7 (7.3) C: 80.0 (7.8)	I: 195 (57.9) C: 195 (57.9)	-Readmission - Katz Index of ADL - Mortality - Cognitive functioning - Time to hospital readmission - Time to discharge from a nursing home	182	I: 106/316 C: 88/303	1.15 (0.91−1.45)
Chow[22] 2014	Hong Kong Single-centre study	NRCT	The nurse case management intervention	T: 312 Ia: 96 Ib: 108 C: 108 Unselected patients	Ia: 75.00 (60–92) Ib: 75.50 (60–89) C: 77.00 (60–89)	Ia: 46 (52.9) Ib: 52 (54.2) C: 49 (50.0)	- Readmission - Quality of life - Self-efficacy - Self-rated health	28 84	Ia:14 /91 Ib: 16/100 C: 24/105 Ia: 32/97 Ib: 30/106 C: 59/130	Ia: 0.67 (0.37−1.22) Ib: 0.70 (0.40−1.24) Ia: 0.73 (0.52−1.03) Ib: 0.62 (0.43−0.89)
Courtney[25] 2009	Australia Single-centre study	RCT	Older Hospitalised Patients' Discharge Planning and In-home Follow-up Protocol (OHP-DP)	T: 128 I: 64 C: 64 Patients at risk	I: 78.1 (6.3) C: 79.4 (7.3)	I: 36 (62.1) C: 40 (62.5)	- Readmission - Visits to emergency department - Visits to general practitioner - Visits to allied health professional - Health-related quality of life	28 84 168	I: 2/49 C: 9/58 I: 10/49 C: 16/58 I: 11/49 C: 27/58	0.26 (0.06−1.15) 0.74 (0.37−1.48) 0.48 (0.27−0.87)
Finlayson[23] 2018	Australia Multi-centre study	RCT	NR	T: 222 Ic: 57 Id:56 Ie:54 C:55 Patients at risk	Ic: 77.1 (7.64) Id: 77.6 (6.50) Ie: 77.8 (6.23) C: 77.9 (6.20)	Ic: 46 (80.7) Id: 42 (75.0) Ie: 37 (68.5) C: 37 (67.3)	- Readmission	28 84 168	Ic: 4/53 Id: 7/49 Ie: 5/49 C: 13/53 Ic: 11/53 Id: 16/44 Ie: 9/48 C: 19/50	Ic: 0.31 (0.11−0.89) Id: 0.58 (0.25−1.33) Ie: 0.42 0.42 (0.16−1.09) Ic: 0.55 (0.29−1.04) Id: 0.96 (0.57−1.63)

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									Ic: 18/52 Id: 18/42 Ie: 16/47 C: 23/50	Ie: 0.49 (0.25−0.97) Ic: 0.75 (0.46−1.21) Id: 0.93 (0.59−1.47) Ie: 0.74 (0.45−1.22)
Koehler[26] 2009	USA Single-centre study	RCT (pilot)	Elderly Care Bundle	T: 41 I: 20 C:21 Patients at risk	I: 77.2 (5.3) C: 79.8 (5.6)	I: 17 (85) C: 13 (62)	- Readmission - Emergency department visits	30 60	I: 2/20 C: 8/21 I: 6/20 C: 9/21	0.26 (0.06−1.08) 0.70 (0.30−1.61)
Lin[31] 2015	Hong Kong, Multi-centre study	Cohort analytic (two groups pre - post study)	Integrated Care and Discharge Support for elderly patients (ICDS)	T: NR I: 1,090 C: NR Patients at risk	Total: 80.4 ± 7.6 I: NR C: NR	T: 557 (51.1)	- Readmission -AED attendance - LOS	182 pre 182 post	NR Reports a statistically significant reduction on 47% in readmissions	NR
Nielsen[28] 2018	Denmark Single-centre study	NRCT	Elderly Activity Performance Intervention	T: 375 I:144 C: 231 Unselected patients	I: 81 (7.9) C: 78 (8.6)	I: 79 (55) C: 122 (53)	- Readmission - All-cause mortality - Number of contacts to general practitioners and emergency departments without admission - Time to first readmission	30 182	I: 25/139 C: 55/231 I: 64/144 C: 99/231	0.76 (0.50−1.16) 1.04 (0.82−1.32)
Robinson[32] 2015	New Zealand Multi-centre study	Cohort analytic (two groups pre - post study)	Integrated Transition of Care (ITC)	T: 19157 I: 5172 C:13,985 Patients at risk	I: 78.2 (9.2) C: 77.6 (9.1)	I: 2486 (48.1) C: 6765 (48.4)	- Readmission - Emergency attendances	7 28 90	I: 500/5,172 C: 1,239/13,985 I: 1370/5,172 C: 3,588/13,985 I: 2,281/5,172 C: 6,079/13,985	1.09 (0.99−1.20) 1.03 (0.98−1.09) 1.01 (0.97−1.05)
Rottman-Sagebiel[29] 2018	USA Single-centre study	NRCT	The Geriatrics Medication Education at Discharge project (GMED)	T: 1,624 I: 435 C: 1,189 Patients at risk	I: 74.9 (7.6) C: 75.2 (8.35)	I: 14 (3.6) C: 26 (2.2)	- Readmission	30	NR	NR

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Sahota[27] 2017	UK Single-centre study	RCT	The Community In-reach Rehabilitation and Care Transition (CIRACT)	T: 250 I: 125 C: 125 Unselected patients	I: 83.6 (6.6) C: 84.5 (5.9)	I: 82 (66) C: 79 (63)	- Readmission - LOS - Day 91-super spell bed days -Functional ability - Co-morbidity - Health-related quality of life - Cost-effectiveness analysis	28 91	I: 18/106 C: 14/106 I: 45/106 C: 39/106	1.29 (0.68–2.46) 1.15 (0.82–1.61)
Voss[30] 2011	USA Multi-centre study	NRCT	Care Transitions Intervention (CTI)	T: 1888 I: 1042 C: 846 Unselected patients	I: NR C: NR	I: NR C: NR	- Readmission	30	NR	NR

T: total, OA: Readmission outcome assessment in days after hospital discharge, RR: relative risk, NR: not reported or author could not access the raw data.
CGA: comprehensive geriatric assessment, LOS: length of hospital stay
I: intervention, Ia: intervention "Home visit", Ib: intervention "Call", C: control, Ic: EN-HaT (Exercise and Nurse Home Visit and Telephone follow up), Id: Exercise, Ie: N-HaT (Nurse Home visit and Telephone follow-up)

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Transitional care interventions

All studies included pre- and post-discharge components and some included bridging components. The most pronounced difference between the interventions was in the pre-discharge phase, whereas elements in the bridging and post-discharge phase were somewhat similar. Table 2 outlines the intervention components, and Supplementary 5 contains detailed descriptions of the interventions.

Pre-discharge components

Across studies, pre-discharge components consisted of health record or plan, discharge planning, involvement of caregivers, concerns or barriers, patient education, physical exercise (physical exercise (muscle strength, stretch, balance, and walking), medication reconciliation, counselling, nutritional screening, pre-discharge home visit or patient assessments (Comprehensive Geriatric Assessment, Problem Classification Scheme, Assessment of Patients' Performance of Daily Activities and Assessment of Motor and Process Skills).

Bridging components

Bridging components consisted of written hand-over between hospital and primary sector, telephone hand-over or in-person hand-over.

Post-discharge components

Post-discharge components comprised of home visits, telephone follow-up, referral to additional healthcare services (rehabilitation, early specialist follow-up, general practitioner or other community services), nurse availability from hospital discharge until seven days after discharge or rapid and intensive community support.

Synthesis of results

All five aspects that may introduce heterogeneity were analysed and a pronounced diversity between the included studies was found. Additionally, some studies had incomplete reporting of study effects. Therefore, because of the heterogeneity, a meta-analysis could not be conducted.

Considering the impact on outcome level, 22 of the 29 outcome results (76%) showed a positive impact on readmission in the intervention groups compared with the control groups. Three studies did not report readmission rates but reported a positive impact on readmission rates, and it was not possible to obtain these numbers from the authors[29-31]. It was not possible to calculate RRs and their 95% CIs for those studies, and they are therefore not presented in the forest plots.

The impact size (RR) from individual studies was calculated using the reported readmission rates from eight studies. RRs ranged from 0.26 to 1.29. Nineteen outcome results were <1 of which one was close to 1 (RR=0.96). Seven estimates were >1 of which three were close to 1 (RR= 1.01, 1.03, 1.04). Of the 29 outcome results, five were statistically significant[22 23 25 26 31].

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Considering the impact on study level, seven of the 11 studies showed an entirely positive impact on readmissions[22 23 25 26 29-31], whilst one study presented both a positive impact and no impact[28], and three studies showed no impact at all[24 27].

Subgroup analysis

Study population

Supplementary 6 exhibits impact according to the included study population. There is a clear difference in impacts between the groups. In total, 14 of 17 (82%) interventions including "patients at risk" reported positive impacts on readmission rates. In contrast, only five out of nine (56%) interventions including "unselected patients" reported positive impacts. In addition, the positive impacts were larger among interventions including "patients at risk" than among interventions with "unselected patients". Three of the four statistically significant results are found among the "patients at risk" group. However, the 95% Cis are wider among "patients at risk" compared with the "unselected patients" indicating less precise effect estimates.

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Table 2 Intervention components

Study	Pre-discharge components												Bridging components					Post-discharge components					
	Patient assessment	Personal health record and/or plan	Education	Medication reconciliation	Physical exercise	Address concerns and/or barriers	Counselling	Discharge planning	Nutrition screening	Caregiver involvement	Home visit to assess future needs	Multidisciplinary care	In-person hand-over	Written hand-over	Telephone hand-over	Community-based nurse visits hospital	Communications path unknown	Number of home visits	Number of telephone follow-up	Nurse availability	Referral to additional services	Intensive community support	Intervention intensity
Buurman et al.[24]	+	+										+	+			+		5[g]					9
Chow et al.[22] (1)	+	+				+												2[g]	2				6
Chow et al.[22] (2)	+	+				+													4				6
Courtney et al.[25]	+	+			+	+		+				+		+				1[a,g]	9				12
Finlayson et al.[23] (3)	+	+			+					+		+						6[b]					10
Finlayson et al.[23] (4)	+	+						+		+		+						1[a,g]	10	+			12
Finlayson et al.[23] (5)	+	+			+			+		+		+						1[a,g] 6[b]	10	+			12
Koehler et al.[26]		+	+	+		+	+	+		+		+		+				2[c]					7
Lin et al.[31] (6)	+							+				+						?[h]	?				6
Lin et al.[31] (7)	+							+				+									+[d]	+	5
Nielsen et al.[28]	+	+[d]			+												+	1[g]					4
Robinson et al.[32]	+		+[f]	+[f]					+			+		+	+[d]				2[e]				7
Rottman-Sagebiel et al.[29]			+[f]	+[f]						+		+		+	+[d]				1		+[d]		6

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Sahota et al.[27]	+	+								+	+	+						1[a,h]			+[d]		3
Voss et al.[30]		+													+			1[g]	3				7

1; post-discharge home visit group, 2; post-discharge call group , 3; Exercise intervention, 4; N-Hat (Nurse Home visit and Telephone follow-up) intervention, 5; Ex-Hat (Exercise and Nurse Home Visit and Telephone follow up) intervention, 6; ICM case management group, 7; HST service group a; additional home visits available if needed, b; by physiotherapist, c; one call by nurse and one call by pharmacist, d; if deemed necessary, e; to participants who return to independent living, f; by pharmacist, g; first visit within 3 days after discharge, h; time for first home visit unknown, ?; number not reported

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Intervention intensity
Calculation of intervention intensity was inspired by Verhaegh et al.[33] Ten interventions were categorised as low-intensity[22 26-32] and five were categorised as high-intensity[23-25].
Supplementary 7 illustrates a clear difference. The vast majority of interventions with a high intensity reported a positive impact on readmission rates whereas only half of the interventions with a low intensity showed a positive impact. The impacts were larger and statistically significant among high-intensive interventions.

Length of support
The interventions lasted from enrolment until 1 day to 6 months after hospital discharge. Four studies lasted between 1 and 7 days[26 28 29 32], two studies lasted 28 and 30 days[22 30], one study 84 days[31], two 168 days[23 24], one 182 days[25] and one study did not report the duration of the intervention[27].
Supplementary 8 displays the impact on readmission rates stratified according to the duration of the interventions. A short length of support is associated with less or no impact on readmission rates. A length of support of one month or more is associated with positive, larger and statistically significant impacts.

Country of origin
Studies conducted outside the European countries seemed to have a greater impact on readmissions than studies conducted within the European Union and all statistically significant results are found in studies conducted in Non-European countries (Supplementary 9).

Outcome assessment
The impact on readmissions was largest within 30 days after hospital discharge. The impact decreased hereafter and was similar between 1 month and 6 months after discharge. Most statistically significant results are found when outcomes are assessed between 1 and 3 months (Supplementary 10).

Quality assessment within studies
Of the 11 studies, two studies were assessed to have a strong methodological quality[22 25], three had a moderate[24 27 32] and six had a weak methodological quality[23 26 28-31]. Of the RCTs, one had a strong[25], two had a moderate[24 27] and two had a weak quality[23 26]. Of the NRCTs, one study had a strong[22] and three had a weak quality[28-30]. One of the pre-post cohort studies had a moderate[32] and one had a weak quality[31] (Table 3).

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Table 3 Quality assessment

Author	Selection bias	Study design	Confounders	Blinding	Data collection methods	Withdrawals and drop-outs	Global rating
Buurman et al.[24]	Moderate	Strong	Strong	Strong	Strong	Weak	**Moderate**
Chow et al.[22]	Moderate	Strong	Strong	Moderate	Strong	Strong	**Strong**
Courtney et al.[25]	Moderate	Strong	Strong	Moderate	Strong	Moderate	**Strong**
Finlayson et al.[23]	Weak	Strong	Strong	Weak	Weak	Moderate	**Weak**
Koehler et al.[26]	Weak	Strong	Strong	Weak	Strong	Weak	**Weak**
Lin et al.[31]	Strong	Moderate	Weak	Weak	Weak	Strong	**Weak**
Nielsen et al.[28]	Strong	Strong	Strong	Weak	Strong	Weak	**Weak**
Robinson et al.[32]	Strong	Moderate	Strong	Weak	Strong	Strong	**Moderate**
Rottman-Sagebiel et al.[29]	Weak	Strong	Strong	Weak	Weak	Weak	**Weak**
Sahota et al.[27]	Strong	Strong	Strong	Weak	Strong	Moderate	**Moderate**
Voss et al.[30]	Weak	Strong	Strong	Weak	Strong	Weak	**Weak**

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Quality assessment across studies
The majority of the studies did not meet the criteria in the components *selection bias* and *blinding* and were thus rated moderate or weak. The vast majority of studies met the criteria of *study design*, *confounders* and *data collection* and were therefore rated strong (Figure 2).

DISCUSSION

We found that the majority of interventions in the transitional phase between hospital and home appear to reduce readmission rates among older patients discharged from a medical ward.
However, some studies reported both a positive impact and no impact on the readmission rate following similar care interventions. These divergent results may have several plausible explanations as discussed below.

Explanations related to study characteristics

Country of origin
Studies conducted in European countries have less impact than studies conducted in non-European countries. The impact of complex interventions is, among others, altered by the context of the implementation[34 35], and differences in impact between countries may therefore be explained by diversity in the social, political, economic, clinical and geographical setting. The accessibility, type, character, quality and overall comprehensiveness of healthcare services provided may also play a role[36]. The USA and Australia have a long history of discharge planning and transitional care, and these countries therefore have a high quantity of research as well as refined strategies and guidelines[37-39].

Study population
The impact on readmissions is greater among "patients at risk" than among "unselected older patients". This is to be expected as readmission rates are higher among "patients at risk" than among "unselected patients". Furthermore, "patients at risk" are more frail and may have a higher degree of morbidity, which could have affected the risk of readmission[40].

Readmission rates prior to the study
Pre-intervention readmission rates may mirror differences in impact between studies. It may be assumed that hospitals with low pre-intervention readmission rates may experience no reduction of post-intervention readmission rates as the remaining readmissions may not be preventable. In contrast, it may also be assumed that hospitals with high pre-intervention readmission rates will achieve a reduction in post-intervention readmission rates.

Sample size

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Several studies had small sample sizes. As seen in Table 1, studies with small sample sizes reported a stronger impact than studies with larger sample sizes. This is in line with Dechartres et al. who also reported larger effect sizes in small-to-moderate-sized trials than in larger trials[41] One reason may be that small studies are more prone to publication bias than larger studies[42].

Explanations related to review findings

Interventions

Our findings suggest that the intensity of interventions influences the impact on readmission rates as high-intensity interventions generally have a stronger impact than low-intensity interventions. This is in line with previous findings presented by Verhaegh et al.[33]. The higher quantity of elements means that more aspects in a complex patient cohort and complex settings can be addressed.

Intervention components that at first glance appear to be similar across studies may comprise different features. The diverging impact on readmissions may be affected by differences in intervention contents across studies. It is not possible to make an intervention component analysis across studies that evaluates which components positively affect readmissions. However, some trends are seen in Table 2. Interventions with a positive impact on readmission comprise the following components; patient assessment, personal health record or plan, concerns and barriers, discharge planning, caregiver involvement, home visits and telephone follow-up.

Intervention fidelity

The impact on readmission rates may be affected by the fidelity of the interventions[43]. However, several studies do not describe intervention fidelity, which may be due to inconsistency in monitoring of the implementation or lack of transparency in reporting the study. These shortcomings thus hinder assessment of whether lack of positive impact is caused by poor implementation of the intervention or if the intervention did not work.

Outcome assessment

The timing of the outcome assessment has an important bearing on the possible preventive readmission rate[44]. The findings of this review suggest that interventions have the largest impact within the first 30 days after discharge. A potential impact is likely to have obliterated if outcome is assessed later than 30 days after discharge.

Residual confounding

Several of the included studies did not adjust for potential confounders such as length of stay, fall within the past 12 months, living conditions, prior admissions, poor overall health condition and functional disability[45-47]. This could have affected the internal validity and the impact on readmission rates.

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The lack of analysis adjusting for potential confounders is highlighted in previous studies[48], underlining the problematic trend in this research area.

Risk of bias
Differences in the quality assessment and thus risk of bias may explain the variation in the impact of the included studies[16]. Supplementary 11 shows that studies assessed to have weak and strong methodological quality have a positive impact on readmission rates whereas studies with moderate quality have no impact.

Statistical analysis
Several authors fail to report whether the statistical analysis is performed based on the first readmission. One patient may therefore represent several readmissions within a specific follow-up period. If this is the case, results from those studies may report a higher readmission rate and thus greater impact than studies using unique observations in their statistical analysis.

Quality assessment
The methodological quality of the majority of the included studies was low, thus indicating a high risk of bias. Low global rating can reflect both the methodological quality and insufficient reporting of the methodology. The latter makes it difficult to accurately assess the true quality and thus the risk of bias. If the low rating is caused by insufficient methodological reporting, it may have no effect on the impact. However, if the low rating reflects methodological problems, it probably underestimates the true impact of the interventions.
The component "blinding" represents the weakest rating. Blinding of participants was not always possible due to the nature of the interventions. Blinding is crucial in pharmaceutical trials and in other clinical studies. However, little is known about the benefits or disadvantages of blinding participants in complex interventions and how bias due to blinding may affect the results. Blinding outcome assessors may not be crucial when outcome is collected through digital records.

COMPARISON WITH PREVIOUS RESEARCH
Prior systematic reviews included other study populations such as surgical patients or patients with only one medical condition. The present review includes a broader population, namely older medical patients. Direct comparison of the present findings with previous findings is therefore difficult. The present review suggests that transitional care interventions reduce the risk of readmission among older medical patients.
These findings are in line with findings in similar reviews that found positive effects of TCIs on hospital readmission[49-51]. These reviews also found that the included studies had a very-low-to-moderate methodological quality.

LIMITATION OF INCLUDED EVIDENCE BASE

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Transparency across studies is lacking even if the CONSORT statement recommends reporting a sufficient description of interventions[52]. A call for more transparency in clinical trials and adherence to appropriate guidelines is also reported elsewhere[53]. If the template for the Intervention Description and Replication (TIDieR) is followed, the replicability may improve, thus making it possible to build on prior research findings[54].
Evaluating TCIs provides insight into whether interventions reduce readmissions among intervention groups compared with control groups. However, we gain no knowledge about causality between exposure and outcome. In complex interventions, such as TCIs, it is difficult to evaluate which mechanisms or components result in change[55]. A process evaluation may have captured the fidelity of the interventions and may thus have provided insight into which mechanisms and components actually work[56]. Process evaluation has previously been requested[57], stressing the unmet need to identify essential components in TCIs. We have requested missing intervention details and other relevant data from authors; some requests were met while others were not. Lastly, all GRADE domains except one (indirectness of evidence) were assessed and found to downgrade the evidence and hence, lower the certainty of the evidence of this review.

STRENGTHS AND LIMITATIONS OF THIS REVIEW

The study has several strengths. This review adhered to PRISMA[15] and SWiM[58] which ensures that all important methodological considerations were made. By reducing the risk of bias in this review, systematic errors were minimised during all stages of the process.
Despite these strengths, we note some limitations. The eligibility criteria were somewhat narrow, which limited the number of included studies. This, however, was deemed necessary to meet the need for evidence focusing on clinical settings where a large number of hospital wards are general medical wards.
Only studies published in English, Danish, Norwegian or Swedish were eligible for inclusion in this review. Therefore, studies published in other languages were omitted, potentially excluding useful evidence.
This review only focuses on hospital readmission as an outcome although the included studies evaluated the intervention impact on multiple outcomes. It is therefore possible that some studies report a positive impact in terms of other outcomes than readmission.
Positive findings are more likely to be published in English language journals, whereas negative findings are more often published in local-language journals[59]. Publication bias is therefore likely present. During this review process, we have contacted several researchers who had registered TCIs on ClinicalTrials.gov. All researchers who found no or negative impact either struggled to publish or decided not to publish. The above-mentioned issues may result in an underrepresentation of negative results in the evidence base and thus in this present review.

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IMPLICATIONS AND FUTURE RESEARCH

Future research projects may benefit from the knowledge gained from the present review when designing and developing new studies. Knowing that interventions with a minimum duration of 1 month that have a high intensity and target high-risk patients may result in more effective interventions.
This review highlights an unmet need for studies of high methodological quality that evaluate the impact of TCIs among older medical patients. Therefore, future research can benefit from a higher level of adherence to relevant guidelines and more detailed descriptions of 1) interventions and comparison groups, 2) the implementation process and 3) actions taken to minimise bias and confounding. Further research should be undertaken to investigate the process evaluations of complex interventions to identify how and why interventions either work or do not work. To develop a broader picture of TCIs, additional studies need to focus on psychological outcomes. The societal cost of such interventions versus individual and societal benefits needs to be further evaluated. High-risk patients such as physically disabled, chronically ill patients may benefit from these kinds of interventions regardless of age and co-morbidity [60].

CONCLUSION

The majority of TCIs have a positive impact on readmission rates among older medical patients, although the most significant impact was seen within 30 days after hospital discharge. Therefore, we believe that the current evidence supports recommending transitional care that includes both pre-and post-discharge components. However, no evidence for recommending a specific intervention exists. The key finding shows an apparent pronounced positive impact among patients categorised as "patients at risk" in "high-intensity" interventions and in interventions with duration of 1 month or more. This, however, should be seen in the light of the fact that only 11 studies met the inclusion criteria and a low certainty of evidence according to the GRADE approach.

Figure legends/captions

Figure 1. Flow chart of the study selection.

Figure 2. Quality assessment across studies. The green colour indicates strong methodological quality, yellow indicates moderate quality and red indicates

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weak quality across studies.

Supplementary Material

Supplementary 1. Selecting Tool

Supplementary 2. Search string

Supplementary 3. Quality Assessment Tool and dictionary

Supplementary 4. Exclusion of full test articles

Supplementary 5. Intervention characteristics

Supplementary 6. Forest plot grouped by study population

Supplementary 7. Forest plot grouped by intervention intensity

Supplementary 8. Forest plot grouped by length of support

Supplementary 9. Forest plot grouped by country of origin

Supplementary 10. Forest plot grouped by outcome assessment

Supplementary 11. Forest plot grouped by quality assessment

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Author Contribution

All authors listed have contributed sufficiently to the project to be included as authors according to the Vancouver Guideline. All authors have participated in the development of the study, writing the draft and all have approved the final version.

Lisa Fønss Rasmussen: First author writing the manuscript, screening records for inclusion, data extraction of included studies, quality assessment of included studies.

Louise Grode: Initial development of the study, screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Jeppe Lange: Initial development of the study, screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Ishay Barat: Initial development of the study, screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Merete Gregersen: Initial development of the study, screening records for inclusion, data extraction of included studies, quality assessment of included studies, reading and giving feedback on the manuscript and approving the final manuscript.

Competing interests

All authors have signed the ICMJE Disclosure Statement and have nothing to disclose and there is no competing interest for any author.

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Data sharing statement

All data relevant to the study are included in the article or uploaded as supplementary information.

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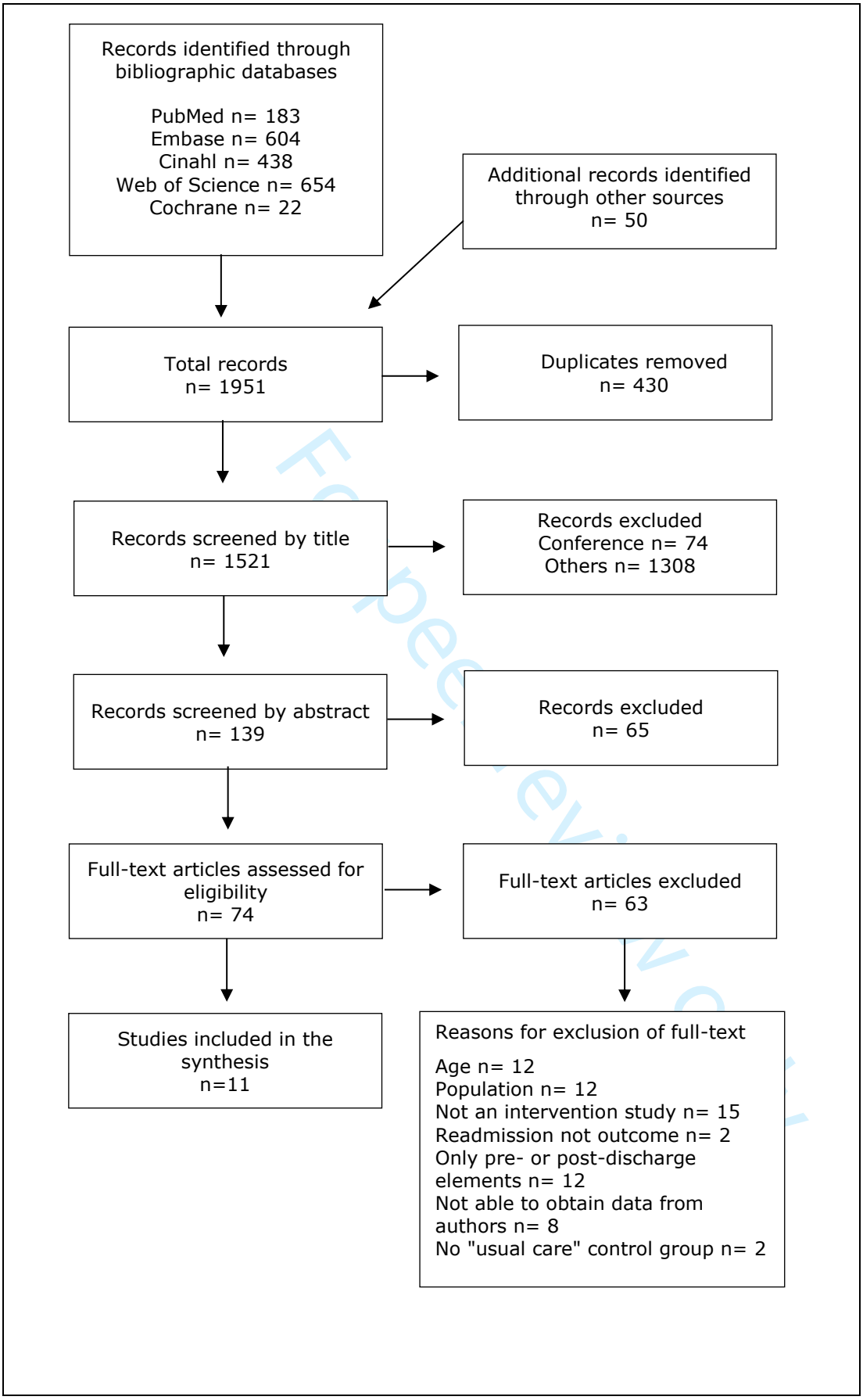
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Records identified through bibliographic databases
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Web of Science n= 654
Cochrane n= 22
Additional records identified through other sources n= 50
Total records n= 1951
Duplicates removed n= 430
Records screened by title n= 1521
Records excluded Conference n= 74 Others n= 1308
Records screened by abstract n= 139
Records excluded n= 65
Full-text articles assessed for eligibility n= 74
Full-text articles excluded n= 63
Studies included in the synthesis n=11
Reasons for exclusion of full-text
Age n= 12
Population n= 12
Not an intervention study n= 15
Readmission not outcome n= 2
Only pre- or post-discharge elements n= 12
Not able to obtain data from authors n= 8
No "usual care" control group n= 2

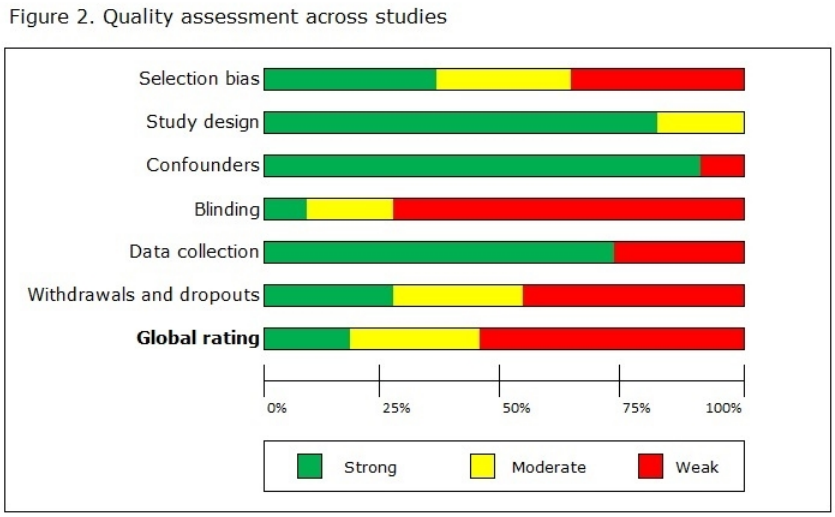


Figure 2. Quality assessment across studies. The green colour indicates strong methodological quality, yellow indicates moderate quality and red indicates weak quality across studies.

70x73mm (300 x 300 DPI)

Supplementary 1, selecting tool

Review question	"What is the impact of transitional care interventions between hospital and home on readmissions among older medical patients?"	
Reviewer Name:		Date:
Author name/study ID:		Year:
Title:		Journal:
	Include	**Exclude**
Patient population	- Elderly patients discharged from a general medical ward or emergency departments - Patients ≥ 65 years or mean age ≥ 75 years	- Patients ≤ 65 and mean age ≤ 75 - Not only medical patients - Discharging ward not specified - Only one patient group/diagnosis such as COPD, diabetes - Patients with psychiatric disorders or patients who are cognitively impaired - Other
Interventions	- Interventions in the transitional phase between hospital and home which examine the effect of the intervention on readmission rates between intervention group and control group	- No comparison/control group - Not an intervention study - Other
Comparators	- Usual or standard procedure	- Comparison between two interventions - Comparison with anything else than usual procedure - Other
Outcomes	- Readmission	- Readmission is not an outcome
Setting	- Intervention includes both pre- and post-discharge components	- Only pre-discharge components - Only post-discharge components - Other
Study design	- Quantitative studies	- Systematic reviews, meta-analysis, literature reviews - Case-reports or case-studies - Meeting abstract, study protocol - Other
Overall decision	- **Included**	- **Excluded**
Notes: Reason for exclusion:		

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Supplementary 2, search string

Search string with free-text words used in PubMed

1	elder* (title)
2	aged (title)
3	old* (title)
4	geriatric* (title)
5	1 OR 2 OR 3 OR 4
6	"discharge intervention" (title)
7	transition* (title)
8	transition* care (title)
9	6 OR 7 OR 8
10	readmission* (text word)
11	re-admission* (text word)
12	rehospitalization* (text word)
13	re-hospitalization* (text word)
14	rehospitalisation* (text word)
15	re-hospitalisation* (text word)
16	10 OR 11 OR 12 OR 13 OR 14 OR 15
17	5 AND 9 AND 16
18	Filters: published in the last 10 years; Danish; English; Norwegian;

Search string with subject headings used in PubMed

1	Patient Discharge/organization and administration (MESH)
2	Transitional Care/organization and administration (MESH)
3	Transitional care/trends (MESH)
4	1 OR 2 OR 3
5	Readmission/organization and administration (MESH)
6	Patient Readmission/trends (MESH)
7	5 OR 6
8	Aged (MESH)
9	Aged, 80 and over (MESH)
10	4 AND 7 AND 9
12	Filters: published in the last 10 years; Danish; English; Norwegian;

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Quality Assessment Tool for Quantitative Studies Dictionary



The purpose of this dictionary is to describe items in the tool thereby assisting raters to score study quality. Due to under-reporting or lack of clarity in the primary study, raters will need to make judgements about the extent that bias may be present. When making judgements about each component, raters should form their opinion based upon information contained in the study rather than making inferences about what the authors intended. Mixed methods studies can be quality assessed using this tool with the quantitative component of the study.

A) SELECTION BIAS

(Q1) Participants are more likely to be representative of the target population if they are randomly selected from a comprehensive list of individuals in the target population (score very likely). They may not be representative if they are referred from a source (e.g. clinic) in a systematic manner (score somewhat likely) or self-referred (score not likely).

(Q2) Refers to the % of subjects in the control and intervention groups that agreed to participate in the study before they were assigned to intervention or control groups.

B) STUDY DESIGN

In this section, raters assess the likelihood of bias due to the allocation process in an experimental study. For observational studies, raters assess the extent that assessments of exposure and outcome are likely to be independent. Generally, the type of design is a good indicator of the extent of bias. In stronger designs, an equivalent control group is present and the allocation process is such that the investigators are unable to predict the sequence.

Randomized Controlled Trial (RCT)

An experimental design where investigators randomly allocate eligible people to an intervention or control group. A rater should describe a study as an RCT if the randomization sequence allows each study participant to have the same chance of receiving each intervention and the investigators could not predict which intervention was next. If the investigators do not describe the allocation process and only use the words 'random' or 'randomly', the study is described as a controlled clinical trial.

See below for more details.

Was the study described as randomized?

Score YES, if the authors used words such as random allocation, randomly assigned, and random assignment.

Score NO, if no mention of randomization is made.

Was the method of randomization described?

Score YES, if the authors describe any method used to generate a random allocation sequence.

Score NO, if the authors do not describe the allocation method or describe methods of allocation such as alternation, case record numbers, dates of birth, day of the week, and any allocation procedure that is entirely transparent before assignment, such as an open list of random numbers of assignments.
If NO is scored, then the study is a controlled clinical trial.

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Was the method appropriate?

Score YES, if the randomization sequence allowed each study participant to have the same chance of receiving each intervention and the investigators could not predict which intervention was next. Examples of appropriate approaches include assignment of subjects by a central office unaware of subject characteristics, or sequentially numbered, sealed, opaque envelopes.

Score NO, if the randomization sequence is open to the individuals responsible for recruiting and allocating participants or providing the intervention, since those individuals can influence the allocation process, either knowingly or unknowingly.

If NO is scored, then the study is a controlled clinical trial.

Controlled Clinical Trial (CCT)
An experimental study design where the method of allocating study subjects to intervention or control groups is open to individuals responsible for recruiting subjects or providing the intervention. The method of allocation is transparent before assignment, e.g. an open list of random numbers or allocation by date of birth, etc.

Cohort analytic (two group pre and post)
An observational study design where groups are assembled according to whether or not exposure to the intervention has occurred. Exposure to the intervention is not under the control of the investigators. Study groups might be non-equivalent or not comparable on some feature that affects outcome.

Case control study
A retrospective study design where the investigators gather 'cases' of people who already have the outcome of interest and 'controls' who do not. Both groups are then questioned or their records examined about whether they received the intervention exposure of interest.

Cohort (one group pre + post (before and after)
The same group is pretested, given an intervention, and tested immediately after the intervention. The intervention group, by means of the pretest, act as their own control group.

Interrupted time series
A study that uses observations at multiple time points before and after an intervention (the 'interruption'). The design attempts to detect whether the intervention has had an effect significantly greater than any underlying trend over time. Exclusion: Studies that do not have a clearly defined point in time when the intervention occurred and at least three data points before and three after the intervention

Other:
One time surveys or interviews

C) CONFOUNDERS

By definition, a confounder is a variable that is associated with the intervention or exposure and causally related to the outcome of interest. Even in a robust study design, groups may not be balanced with respect to important variables prior to the intervention. The authors should indicate if confounders were controlled in the design (by stratification or matching) or in the analysis. If the allocation to intervention and control groups is randomized, the authors must report that the groups were balanced at baseline with respect to confounders (either in the text or a table).

D) BLINDING

(Q1) Assessors should be described as blinded to which participants were in the control and intervention groups. The purpose of blinding the outcome assessors (who might also be the care providers) is to protect against detection bias.

(Q2) Study participants should not be aware of (i.e. blinded to) the research question. The purpose of blinding the participants is to protect against reporting bias.

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E) DATA COLLECTION METHODS

Tools for primary outcome measures must be described as reliable and valid. If 'face' validity or 'content' validity has been demonstrated, this is acceptable. Some sources from which data may be collected are described below:

Self reported data includes data that is collected from participants in the study (e.g. completing a questionnaire, survey, answering questions during an interview, etc.).

Assessment/Screening includes objective data that is retrieved by the researchers. (e.g. observations by investigators).

Medical Records/Vital Statistics refers to the types of formal records used for the extraction of the data.

Reliability and validity can be reported in the study or in a separate study. For example, some standard assessment tools have known reliability and validity.

F) WITHDRAWALS AND DROP-OUTS

Score YES if the authors describe BOTH the numbers and reasons for withdrawals and drop-outs.

Score NO if either the numbers or reasons for withdrawals and drop-outs are not reported.

Score NOT APPLICABLE if the study was a one-time interview or survey where there was not follow-up data reported.

The percentage of participants completing the study refers to the % of subjects remaining in the study at the final data collection period in all groups (i.e. control and intervention groups).

G) INTERVENTION INTEGRITY

The number of participants receiving the intended intervention should be noted (consider both frequency and intensity). For example, the authors may have reported that at least 80 percent of the participants received the complete intervention. The authors should describe a method of measuring if the intervention was provided to all participants the same way. As well, the authors should indicate if subjects received an unintended intervention that may have influenced the outcomes. For example, co-intervention occurs when the study group receives an additional intervention (other than that intended). In this case, it is possible that the effect of the intervention may be over-estimated. Contamination refers to situations where the control group accidentally receives the study intervention. This could result in an under-estimation of the impact of the intervention.

H) ANALYSIS APPROPRIATE TO QUESTION

Was the quantitative analysis appropriate to the research question being asked?

An intention-to-treat analysis is one in which all the participants in a trial are analyzed according to the intervention to which they were allocated, whether they received it or not. Intention-to-treat analyses are favoured in assessments of effectiveness as they mirror the noncompliance and treatment changes that are likely to occur when the intervention is used in practice, and because of the risk of attrition bias when participants are excluded from the analysis.

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Component Ratings of Study:

For each of the six components A – F, use the following descriptions as a roadmap.

A) SELECTION BIAS

Good: The selected individuals are very likely to be representative of the target population (Q1 is 1) and there is greater than 80% participation (Q2 is 1).

Fair: The selected individuals are at least somewhat likely to be representative of the target population (Q1 is 1 or 2); and there is 60 - 79% participation (Q2 is 2). 'Moderate' may also be assigned if Q1 is 1 or 2 and Q2 is 5 (can't tell).

Poor: The selected individuals are not likely to be representative of the target population (Q1 is 3); or there is less than 60% participation (Q2 is 3) or selection is not described (Q1 is 4); and the level of participation is not described (Q2 is 5).

B) DESIGN

Good: will be assigned to those articles that described RCTs and CCTs.

Fair: will be assigned to those that described a cohort analytic study, a case control study, a cohort design, or an interrupted time series.

Weak: will be assigned to those that used any other method or did not state the method used.

C) CONFOUNDERS

Good: will be assigned to those articles that controlled for at least 80% of relevant confounders (Q1 is 2); or (Q2 is 1).

Fair: will be given to those studies that controlled for 60 – 79% of relevant confounders (Q1 is 1) and (Q2 is 2).

Poor: will be assigned when less than 60% of relevant confounders were controlled (Q1 is 1) and (Q2 is 3) or control of confounders was not described (Q1 is 3) and (Q2 is 4).

D) BLINDING

Good: The outcome assessor is not aware of the intervention status of participants (Q1 is 2); and the study participants are not aware of the research question (Q2 is 2).

Fair: The outcome assessor is not aware of the intervention status of participants (Q1 is 2); or the study participants are not aware of the research question (Q2 is 2).

Poor: The outcome assessor is aware of the intervention status of participants (Q1 is 1); and the study participants are aware of the research question (Q2 is 1); or blinding is not described (Q1 is 3 and Q2 is 3).

E) DATA COLLECTION METHODS

Good: The data collection tools have been shown to be valid (Q1 is 1); and the data collection tools have been shown to be reliable (Q2 is 1).

Fair: The data collection tools have been shown to be valid (Q1 is 1); and the data collection tools have not been shown to be reliable (Q2 is 2) or reliability is not described (Q2 is 3).

Poor: The data collection tools have not been shown to be valid (Q1 is 2) or both reliability and validity are not described (Q1 is 3 and Q2 is 3).

F) WITHDRAWALS AND DROP-OUTS - a rating of:

Good: will be assigned when the follow-up rate is 80% or greater (Q1 is 1 and Q2 is 1).

Fair: will be assigned when the follow-up rate is 60 – 79% (Q2 is 2) OR Q1 is 4 or Q2 is 5.

Poor: will be assigned when a follow-up rate is less than 60% (Q2 is 3) or if the withdrawals and drop-outs were not described (Q1 is No or Q2 is 4).

Not Applicable: if Q1 is 4 or Q2 is 5.

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QUALITY ASSESSMENT TOOL FOR QUANTITATIVE STUDIES



COMPONENT RATINGS

A) SELECTION BIAS

(Q1) Are the individuals selected to participate in the study likely to be representative of the target population?

1 Very likely
2 Somewhat likely
3 Not likely
4 Can't tell

(Q2) What percentage of selected individuals agreed to participate?

1 80 - 100% agreement
2 60 – 79% agreement
3 less than 60% agreement
4 Not applicable
5 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

B) STUDY DESIGN

Indicate the study design

1 Randomized controlled trial
2 Controlled clinical trial
3 Cohort analytic (two group pre + post)
4 Case-control
5 Cohort (one group pre + post (before and after))
6 Interrupted time series
7 Other specify _____
8 Can't tell

Was the study described as randomized? If NO, go to Component C.

No Yes

If Yes, was the method of randomization described? (See dictionary)

No Yes

If Yes, was the method appropriate? (See dictionary)

No Yes

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

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C) CONFOUNDERS

(Q1) Were there important differences between groups prior to the intervention?

1 Yes
2 No
3 Can't tell

The following are examples of confounders:

1 Race
2 Sex
3 Marital status/family
4 Age
5 SES (income or class)
6 Education
7 Health status
8 Pre-intervention score on outcome measure

(Q2) If yes, indicate the percentage of relevant confounders that were controlled (either in the design (e.g. stratification, matching) or analysis)?

1 80 – 100% (most)
2 60 – 79% (some)
3 Less than 60% (few or none)
4 Can't Tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

D) BLINDING

(Q1) Was (were) the outcome assessor(s) aware of the intervention or exposure status of participants?

1 Yes
2 No
3 Can't tell

(Q2) Were the study participants aware of the research question?

1 Yes
2 No
3 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

E) DATA COLLECTION METHODS

(Q1) Were data collection tools shown to be valid?

1 Yes
2 No
3 Can't tell

(Q2) Were data collection tools shown to be reliable?

1 Yes
2 No
3 Can't tell

RATE THIS SECTION	STRONG	MODERATE	WEAK
See dictionary	1	2	3

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F) WITHDRAWALS AND DROP-OUTS

(Q1) Were withdrawals and drop-outs reported in terms of numbers and/or reasons per group?

1 Yes
2 No
3 Can't tell
4 Not Applicable (i.e. one time surveys or interviews)

(Q2) Indicate the percentage of participants completing the study. (If the percentage differs by groups, record the lowest).

1 80 -100%
2 60 - 79%
3 less than 60%
4 Can't tell
5 Not Applicable (i.e. Retrospective case-control)

RATE THIS SECTION	STRONG	MODERATE	WEAK	
See dictionary	1	2	3	Not Applicable

G) INTERVENTION INTEGRITY

(Q1) What percentage of participants received the allocated intervention or exposure of interest?

1 80 -100%
2 60 - 79%
3 less than 60%
4 Can't tell

(Q2) Was the consistency of the intervention measured?

1 Yes
2 No
3 Can't tell

(Q3) Is it likely that subjects received an unintended intervention (contamination or co-intervention) that may influence the results?

4 Yes
5 No
6 Can't tell

H) ANALYSES

(Q1) Indicate the unit of allocation (circle one)

community organization/institution practice/office individual

(Q2) Indicate the unit of analysis (circle one)

community organization/institution practice/office individual

(Q3) Are the statistical methods appropriate for the study design?

1 Yes
2 No
3 Can't tell

(Q4) Is the analysis performed by intervention allocation status (i.e. intention to treat) rather than the actual intervention received?

1 Yes
2 No
3 Can't tell

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GLOBAL RATING

COMPONENT RATINGS

Please transcribe the information from the gray boxes on pages 1-4 onto this page. See dictionary on how to rate this section.

A	**SELECTION BIAS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
B	**STUDY DESIGN**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
C	**CONFOUNDERS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
D	**BLINDING**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
E	**DATA COLLECTION METHOD**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	
F	**WITHDRAWALS AND DROPOUTS**	**STRONG**	**MODERATE**	**WEAK**	
		1	2	3	Not Applicable

GLOBAL RATING FOR THIS PAPER (circle one):

1	STRONG	(no WEAK ratings)
2	MODERATE	(one WEAK rating)
3	WEAK	(two or more WEAK ratings)

With both reviewers discussing the ratings:

Is there a discrepancy between the two reviewers with respect to the component (A-F) ratings?

No Yes

If yes, indicate the reason for the discrepancy

1	Oversight
2	Differences in interpretation of criteria
3	Differences in interpretation of study

Final decision of both reviewers (circle one):

1	**STRONG**
2	**MODERATE**
3	**WEAK**

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Supplementary 4, exclusion of full text

Altfeld S.J., Shier G.E., Rooney M., Johnson T.J., Golden R.L., Karavolos K., Avery E., Nandi V. & Perry, A.J. 2013, "Effects of an enhanced discharge planning intervention for hospitalized older adults: a randomized trial.", The Gerontologist, vol. 53, no. 3, pp. 430-440. **Reason for exclusion:** not only medical patients.

Alvarez R, Ginsburg J, Grabowski J, Post S, Rosenberg W. The Social Work Role in Reducing 30-Day Readmissions: The Effectiveness of the Bridge Model of Transitional Care. J Gerontol Soc Work 2016 Apr;59(3):222-227. **Reason for exclusion:** not an intervention study.

Arbaje A.I., Maron D.D., Yu Q., Wendel V.I., Tanner E., Boult C., Eubank K.J. & Durso, S.C. 2010, "The geriatric floating interdisciplinary transition team.", Journal of the American Geriatrics Society, vol. 58, no. 2, pp. 364-370. **Reason for exclusion:** readmission is not an outcome.

Arnold ME, Buys L, Fullas F. Impact of pharmacist intervention in conjunction with outpatient physician follow-up visits after hospital discharge on readmission rate (Reprinted from Journal of Health-System Pharmacy Residents JHPR, vol 2, issue 4, 2014). American Journal of Health-System Pharmacy 2015 JUN 1;72(11):S36-S42. **Reason for exclusion:** only post-discharge elements, not only medical patients.

Arevalo-Flechas, L.C., Noel, P., Rottman-Sagebiel, R., Cupples, N., Conde, A., Sanchez-Reilly, S., Powers, B. & Espinoza, S. 2016, "Presence of a Caregiver Associated with Increased Odds of Readmission in a Cohort of Older Adults Included in a Transitional Care Intervention", Journal of the American Geriatrics Society, vol. 64, pp. S141-S141. **Reason for exclusion:** meeting abstract.

Bellon, J.E., Bilderback, A., Ahuja-Yende, N.S., Wilson, C., Altieri Dunn, S.C., Brodine, D. & Boninger, M.L. 2018, "University of Pittsburgh Medical Center Home Transitions Multidisciplinary Care Coordination Reduces Readmissions for Older Adults", Journal of the American Geriatrics Society. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Burt S, Berry D, Quackenbush P. Implementation of Transition in Care and Relationship Based Care to Reduce Preventable Rehospitalizations. Home Healthc Now 2015 Jul-Aug;33(7):390-393. **Reason for exclusion:** not an intervention study.

Cheen MHH, Goon CP, Ong WC, Lim PS, Wan CN, Leong MY, et al. Evaluation of a care transition program with pharmacist-provided home-based medication review for elderly Singaporeans at high risk of readmissions. Int J Qual Health Care 2017 Apr 1;29(2):200-205. **Reason for exclusion:** not only medical patients, only post-discharge elements, mean age < 75 year or age < 65 years.

Damiani G, Federico B, Venditti A, Sicuro L, Rinaldi S, Cirio F, et al. Hospital discharge planning and continuity of care for aged people in an Italian local health unit: does the care-home model reduce hospital readmission and mortality rates? BMC Health Serv Res 2009 Feb 4;9:22-6963-9-22. **Reason for exclusion:** not able to obtain data from the authors.

Dedhia P., Kravet S., Bulger J., Hinson T., Sridharan A., Kolodner K., Wright S. & Howell, E. 2009, "A quality improvement intervention to facilitate the transition of older adults from three hospitals back to their homes.", Journal of the American Geriatrics Society, vol. 57, no. 9, pp. 1540-1546. **Reason for exclusion:** only pre-discharge elements.

Dizon, M.L. & Reinking, C. 2017, "Reducing Readmissions: Nurse-Driven Interventions in the Transition of Care From the Hospital", Worldviews on evidence-based nursing, vol. 14, no. 6, pp. 432-439. **Reason for exclusion:** mean age < 75 year or age < 65 years.

Edmans J, Bradshaw L, Franklin M, Gladman J, Conroy S. Specialist geriatric medical assessment for patients discharged from hospital acute assessment units: randomised controlled trial. BMJ 2013 Oct 8;347:f5874. **Reason for exclusion:** not able to obtain data from the authors.

Farrell T.W., TomoaiaCotisel A., Scammon D.L., Brunisholz K., Kim J., Day J., Gren L.H., Wallace S., Gunning K., Tabler J. & Magill, M.K. 2015, "Impact of an integrated transition management program

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in primary care on hospital readmissions.", Journal for healthcare quality : official publication of the National Association for Healthcare Quality, vol. 37, no. 1, pp. 81-92. **Reason for exclusion:** mean age < 75 year or age < 65 years.

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Gardner R, Li Q, Baier RR, Butterfield K, Coleman EA, Gravenstein S. Is implementation of the care transitions intervention associated with cost avoidance after hospital discharge? J Gen Intern Med 2014 Jun;29(6):878-884. **Reason for exclusion:** readmission is not an outcome.

Gurwitz, J.H., Field, T.S., Ogarek, J., Tjia, J., Cutrona, S.L., Harrold, L.R., Gagne, S.J., Preusse, P., Donovan, J.L., Kanaan, A.O., Reed, G. & Garber, L. 2014, "An Electronic Health Record-Based Intervention to Increase Follow-Up Office Visits and Decrease Rehospitalization in Older Adults", Journal of the American Geriatrics Society, vol. 62, no. 5, pp. 865-871. **Reason for exclusion:** not only medical patients.

Hamar, B., Rula, E.Y., Wells, A.R., Coberley, C., Pope, J.E. & Varga, D. 2016, "Impact of a Scalable Care Transitions Program for Readmission Avoidance", American Journal of Managed Care, vol. 22, no. 1, pp. 28-34. **Reason for exclusion:** mean age < 75 year or age < 65 years.

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Hitch B, Parlier AB, Reed L, Galvin SL, Fagan EB, Wilson CG. Evaluation of a Team-Based, Transition-of-Care Management Service on 30-Day Readmission Rates. N C Med J 2016 Mar-Apr;77(2):87-92. **Reason for exclusion:** not only medical patients, mean age < 75 year or age < 65 years.

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Hwang, U., Dresden, S.M., Rosenberg, M.S., Garrido, M.M., Loo, G., Sze, J., Gravenor, S., Courtney, D.M., Kang, R., Zhu, C.W., Vargas-Torres, C., Grudzen, C.R., Richardson, L.D. & The GEDI, W.I. 2018, "Geriatric Emergency Department Innovations: Transitional Care Nurses and Hospital Use", Journal of the American Geriatrics Society, vol. 66, no. 3, pp. 459-466. **Reason for exclusion:** only pre-discharge elements.

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Expert-by-experience, and Novice Nurses, to Hospitalized Elders with Chronic Healthcare Conditions", Pacific Rim International Journal of Nursing Research, vol. 16, no. 4, pp. 343-360. **Reason for exclusion:** not an intervention study.

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Leung D.Y.P., Lee D.T.F., Lee I.F.K., Lam L.W., Lee S.W.Y., Chan M.W.M., Lam Y.M., Leung S.H., Chiu P.C., Ho N.K.F., Ip M.F. & Hui, M.M.Y. 2015, "The effect of a virtual ward program on emergency services utilization and quality of life in frail elderly patients after discharge: A pilot study.", Clinical Interventions in Aging, vol. 10, pp. 413-420. Participant age unknown, outcome readmission **Reason for exclusion:** not able to obtain data from the authors.

Lin, C.Y., Barnato, A.E. & Degenholtz, H.B. 2011, "Physician Follow-Up Visits After Acute Care Hospitalization for Elderly Medicare Beneficiaries Discharged to Noninstitutional Settings", Journal of the American Geriatrics Society, vol. 59, no. 10, pp. 1947-1954. **Reason for exclusion:** not an intervention study.

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Morrison, J., Palumbo, M.V. & Rambur, B. 2016, "Reducing Preventable Hospitalizations With Two Models of Transitional Care", Journal of Nursing Scholarship, vol. 48, no. 3, pp. 322-329. **Reason for exclusion:** no "usual care" control group.

Naylor MD, Bowles KH, McCauley KM, Maccoy MC, Maislin G, Pauly MV, et al. High-value transitional care: translation of research into practice. J Eval Clin Pract 2013 Oct;19(5):727-733. **Reason for exclusion:** not able to obtain data from the authors.

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Nazir A., Khan B., Counsell S., Henderson M., Gao S. & Boustani, M. 2015, "Impact of an inpatient geriatric consultative service on outcomes for cognitively impaired patients.", Journal of Hospital Medicine, vol. 10, no. 5, pp. 275-280. **Reason for exclusion:** only pre-discharge elements.

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Parsons, M., Parsons, J., Rouse, P., Pillai, A., Mathieson, S., Parsons, R., Smith, C. & Kenealy, T. 2018, "Supported Discharge Teams for older people in hospital acute care: a randomised controlled trial", Age and Ageing, vol. 47, no. 2, pp. 288-294. **Reason for exclusion:** not only medical patients.

Peel N.M., Chan K.W. & Hubbard, R.E. 2015, "Outcomes of cognitively impaired older people in Transition Care.", Australasian journal on ageing, vol. 34, no. 1, pp. 53-57. Participant age unknown, outcome readmission. Outcome of intervention compared between cognitively impaired and cognitively intact patients, and not between patient groups receiving intervention and those who receive usual care?? **Reason for exclusion:** only cognitively impaired patients.

Peel, N.M., Hubbard, R.E. & Gray, L.C. 2013, "Impact of Post-Acute Transition Care for Frail Older People: A Prospective Study", The Journal of frailty & aging, vol. 2, no. 3, pp. 165-171. **Reason for exclusion:** no "usual care" control group.

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Rosted E, Wagner L, Hendriksen C, Poulsen I. Geriatric nursing assessment and intervention in an emergency department: a pilot study. Int J Older People Nurs 2012 Jun;7(2):141-151. **Reason for exclusion:** not an intervention study.

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patients", Scandinavian journal of primary health care, vol. 28, no. 3, pp. 146-153. **Reason for exclusion:** only pre-discharge elements.

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Supplementary 5, intervention characteristics

Study, year	Title	Description of intervention components	Providers	Context	Length of support	Treatment of comparison group
Buurman et al.[24] **2016**	Transitional Care Bridge Program Intervention	Pre-discharge: *≤ 48 hours after hospital admission*: - participants received a systematic CGA. - Care and treatment plan with prioritazion by patient and geriatric team -Multidisciplinary care provided by geriatric team Bridging: - In-person handover of CGA and care treatment plan by geriatric team to CCRN - Visit of CCRN to participant Post-discharge: *≤ 2 days after hospital discharge:* - Home visit or visit to the NH, with medication reconciliation; initiate home care; check for activities of daily living and availability of adequate help; target geriatric conditions; contact GP or NH physician *2 weeks after hospital discharge*: - Home visit or visit to the NH, evaluation of hospital-based care and treatment plan, goal setting by participant, protocol adherence for incident and prevalent geriatric conditions *6 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on the proxy and caregiver burden; further actions on geriatric conditions *12 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on care needs for incident and prevalent geriatric conditions *24 weeks after hospital discharge*: - Home visit or visit to the NH, focusing on care needs; evaluation of geriatric conditions; handover to GP	Geriatric team A community care registered nurse (CCRN)	Three hospitals with affiliated home care organizations in the Netherlands	From ≤ 48 hours after hospital admission to 24 weeks after hospital discharge	CGA-Only Control Arm (usual care): - a systematic CGA - care and treatment plan with prioritazion by patient and geriatric team -multidisciplinary care provided by geriatric team
Chow et al.[22]	The nurse case management intervention	Two intervention arms: 1. "Home Visit" 2. "Call"	Senior year nursing students	Patients were recruited from the medical	From enrolment until 4 weeks after discharge	Social calls twice within 4 weeks.

2014		Pre-discharge in both "Home Visit" and "Call" arms: - Pre-discharge assessment: the Problem Classification Scheme, Intervention Scheme and Problem Rating Scale for Outcomes (e.g. analysis of the barriers and developed mutual goals with the patients to ensure that they were able to perform health maintenance activities, such as nutrition, monitoring of symptoms and medication adherence) Bridging: None Post-discharge - Home Visit arm: - First visit within 72 hours of discharge. - Week 2: a telephone call to evaluate the interventions and advice given during the home visit - Week 3: home visit to examine whether patients were able to continue managing their health needs after hospital discharge. - Week 4: final telephone call in the fourth week post discharge to remind patients about adherence behaviors and to motivate and support them before concluding the interventions Post-discharge – Call arm: - Week 1: first telephone call based on the patients' needs, which had been identified prior to discharge. - Week 2. call the patients: referred to the mutual goals developed by the NCM and the patients to continue with the interventions. - Week 3. call the patients referred to the mutual goals developed by the NCM and the patients to continue with the interventions. - Week 4: closing call to motivate and support the patients in maintaining self-management behaviors	Nurse Case Managers (NCM)	department of a 1700-bedded acute, general regional hospital in Hong Kong		No assessments and related interventions were conducted to the control group.
Courtney et al.[25] **2009**	Older Hospitalised Patients' Discharge Planning and In-home	Pre-discharge: *Exercise components*: - Muscle stretching, balance training, walking for endurance, and muscle strengthening using resistance exercises	A registered nurse (RN) A physio-therapist	A tertiary referral hospital in Brisbane, Australia	From enrolment until 6 months after discharge	Control group received the routine care, discharge planning, and rehabilitation advice normally provided. If

	Follow-up Protocol (OHP-DP)	*Nurse components:* - The nurse visited daily during participants' hospital stays to address concerns, facilitate the exercise program, and oversee discharge planning - Developed a transitional care plan covering the areas of functional ability and need for assistance with activities of daily living, post-discharge treatments and follow-up care, social support, chronic disease management plans and information, medication information, community services, and assistance with the exercise program. <u>Bridging</u>: - Written guidelines were provided on post-discharge management, including diagrams and specific instructions for their exercise program <u>Post-discharge</u>: - Within 48 hours of discharge, the nurse undertook a home visit to assess availability of support, address transitional concerns, provide advice and support, and ensure that the exercise program could be safely undertaken at home. - Extra home visits were provided if required - Weekly follow-up telephone calls were provided for 4 weeks, followed by monthly follow-up for a further 5 months				in-home follow-up was necessary, it was organized in the routine manner (e.g., referral to community health services)
Finlayson et al.[23] **2018**		Three intervention arms: 1. "Exercise" 2. "Nurse Home visit and Telephone follow-up" (N-HaT) 3. "Exercise and Nurse Home Visit and Telephone follow up" (ExN-HaT) <u>Exercise arm:</u> *Pre-discharge*: - Within 72 hours of admission, a physiotherapist will visit measure ADL and conduct performance tests of balance and gait. This information will be used to plan individualised exercise programs designed to improve strength, stability, coordination, endurance, mobility, and improve self confidence with respect to ADL. The exercise prescription will be developed by the patient, care givers, doctors, and ward nurses. -Physiotherapist will provide a pedometer	Advanced Practice Gerontic Nurse (APGN) A physiotherapist or exercise physiologist	Two tertiary metropolitan hospitals in Australia	From enrolment until 24 weeks after discharge	Routine hospital and follow-up care as provided by the health service. This involved a needs assessment by the hospital health staff, discharge planning, and referrals for follow up services as appropriate.

		together with simple verbal and written instructions regards usage and journaling of activity levels - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required *Bridging:* None *Post-discharge:* - An exercise physiologist conducts 6 weekly in home visits to reassess the patient's physical measures and functional capacity, evaluate progress with the exercise program, and reset program goals accordingly <u>N-HaT arm:</u> *Pre-discharge*: - Within 72 hours of admission, an APGN will undertake a health assessment, and prepare a comprehensive transitional care plan. This process includes discussing the planned discharge plan with patient's caregivers, doctor, and ward nurses to individualise the plan - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required. *Bridging:* None *Post-discharge:* - Within 48 hours after discharge, an APGN conducts a home visit. This home visit is undertaken to 1) identify sufficient caregiver support is available at home, 2) assess the home environment is safe, 3) determine the patient has the required medications and dressings (if required), 4) ensure the patient (and caregiver) fully understand their medication and treatment regimes; 5) reinforce and further explain exercise program and use of pedometer and journal in the home, and 6) provide advice and support to the caregiver				

		-Additional home visits are available by APGN if required - over the initial 4 week period after discharge, a APGN provides weekly telephone follow-up calls -APGN is available to be contacted via telephone 7 days per week - For 6 months, monthly telephone follow-up calls are conducted by the APGN. <u>ExN-HaT arm</u>: *Pre-discharge*: - Within 72 hours of admission, an APGN will undertake a health assessment, and prepare a comprehensive transitional care plan. This process includes discussing the planned discharge plan with patient's caregivers, doctor, and ward nurses to individualise the plan - Within 72 hours of admission, a physiotherapist will visit the intervention patients to determine the functional capacity of the patient using measures of ADL and performance tests of balance and gait. This information will be used to plan individualised exercise programs designed to improve strength, stability, coordination, endurance, mobility, and improve self confidence with respect to ADL. The exercise prescription will be developed by the patient, caregivers, doctors, and ward nurses. - Physiotherapist will provide a pedometer together with simple verbal and written instructions regards usage and journaling of activity levels - APGN will visit patients 2nd daily thereafter until discharge to establish and implement the program, monitor progress, and modify transitional care plan if required. *Bridging:* None *Post-discharge:* - Within 48 hours post discharge, an APGN provides one home visit. This home visit is undertaken to 1) identify sufficient caregiver support is available at home, 2) assess the home environment is safe, 3) determine the patient has the required medications and dressings (if				

		required), 4) ensure the patient (and caregiver) fully understand their medication and treatment regimes; 5) reinforce and further explain exercise program and use of pedometer and journal in the home, and 6) provide advice and support to the caregiver - Additional home visits are available by APGN if required - An exercise physiologist conducts 6 weekly visits to reassess the patient's physical measures and functional capacity, evaluate progress with the exercise program, and reset program goals accordingly - Over the initial 4 week period after discharge, a APGN provides weekly telephone follow-up calls - APGN is available to be contacted via telephone 7 days per week - For 6 months, monthly telephone follow-up calls are conducted by the APGN				
Koehler et al.[26] **2009**	Early Care Bundle or Supplemental Care Bundle	<u>Pre-discharge</u>: By Study Care Coordinator - Daily condition specific education - Additional time to identify and address discharge barriers - Extra discharge teaching with focus on self-management By Study Pharmacist: - Medication reconciliation at admit - Daily medication review and education - Additional medication change recommendations as felt to be indicated - Medication reconciliation at discharge - Counselling on discharge Structured documents: - personal health record (completed by the patient and family for future use) <u>Bridging</u>: Structured documents: - Supplemental discharge form given to patient and faxed to PCP <u>Post-discharge</u>: By Study Care Coordinator:	Care coordinators (CC) Clinical pharmacists (CP)	1 of 2 hospital-medicine groups at the 900-bed Baylor University Medical Center, Texas	Max 24 hours after enrolment to 1 week after discharge	Usual care: - Needs assessment - discharge planning - Medication reconciliation at admit - Daily direct patient care - Discharge medication education - Medication reconciliation at discharge - Discharge education - Medication review with new orders and as needed - Medication change recommendations - Standard BUMC discharge form - New discharge prescriptions

		- Follow up call at 5-7 days post-discharge By Study Pharmacist: - Follow up call at 5-7 days post-discharge				
Lin et al.[31] **2015**	Integrated Care and Discharge Support for elderly patients (ICDS)	Pre-discharge: - Risk stratification - Comprehensive geriatric assessment - Discharge planning are performed Bridging: None Post-discharge: Patients receive either: - ICM Case Management or - HST service *ICM Case Management* : - Each high-risk older patient is followed up by a case manager for a period of around 3 months following hospital discharge - Case managers provide home visits and telephone support and they are responsible for community service coordination and ensuring patient compliance with planned services and management. *HST service*: - Rapid and intensive community support for discharged patients, offering services such as meal delivery, household cleaning, respite care, and home assessment and modification Link nurses, ICM case managers, and the HST hold weekly multidisciplinary case conferences chaired by an ICM geriatrician. If needed, referral for rehabilitation in a geriatric day hospital, fast track clinic, or early specialist clinic follow-up can be offered to patients.	At hospital: - Link nurses - Geriatricians Community ICM Case Management: - Social workers - Physiotherapist (PT) - Occupational therapist (OT) - Advanced practice nurse (APN) Community HST service: - Nurses - PT - OT - Other allied health members	Four hospitals of Hong Kong West Cluster.	ICM case management: - From recruitment until 3 months after discharge HST service: - NR	NR
Nielsen et al.[28] **2018**	Elderly Activity Performance Intervention	Pre-discharge: *Component 1:* -Assessment of the patients' performance of daily activities - Assessment of Motor and Process Skills Based on the results in component 1, patients with limitations in performing daily activities	Occupational therapist (OT) Physiotherapist	Short-stay unit (ED) at a 1,150-bed university hospital in Denmark	From enrolment until the day after discharge	Usual practice: relevant medical treatment and care. Referral to occupational therapy and physiotherapy took place only if the medical or nursing

		received components 2 and 3 *Component 2:* - A rehabilitation plan was prescribed. The rehabilitation plan included a description of the patient's previous and current performance of daily activities and specified the need for further rehabilitation. Bridging: - Primary care was informed about the discharge, and visitation of the patient to further rehabilitation Post-discharge: *Component 3:* - Home visit by an OT the day after discharge for patients with a prescribed rehabilitation plan. The home visit aimed to enhance the patient's performance of daily activities and to start rehabilitation. - OT screened the home for safety risks and factors that potently could limit the performance of daily activities.				staff considered it necessary. If the occupational therapist was summoned to asses a patient, the occupational therapist performed a short interview and a non-standardized observation of the patient's performance of basic daily activities and the results were communicated to the municipality homecare staff. If necessary, nurses from the ED organized referral to nursing home care after discharge
Robinson et al.[32] **2015**	Integrated Transition of Care (ITC)	Pre-discharge: - Nutrition screening, and if necessary referral to a dietitian - Allied health review - Discharge medicines reconciliation and patient education by a pharmacist. Bridging: - Records were communicated to primary care providers electronically and via mail, and when required, by telephone. Post-discharge: Provided only to patients who were returning to independent community living - 1 and 3 days after discharge: a telephone assessment, education, and support. The aim was to identify problems not dealt with during the discharge process, assist with self-management, and ensure appropriate health and social supports.	Community nurses A geriatrician A pharmacist Cultural support workers	Two acute general hospitals with 536 and 250 beds in Auckland, New Zealand.	From recruitment until 3 days after discharge	NR

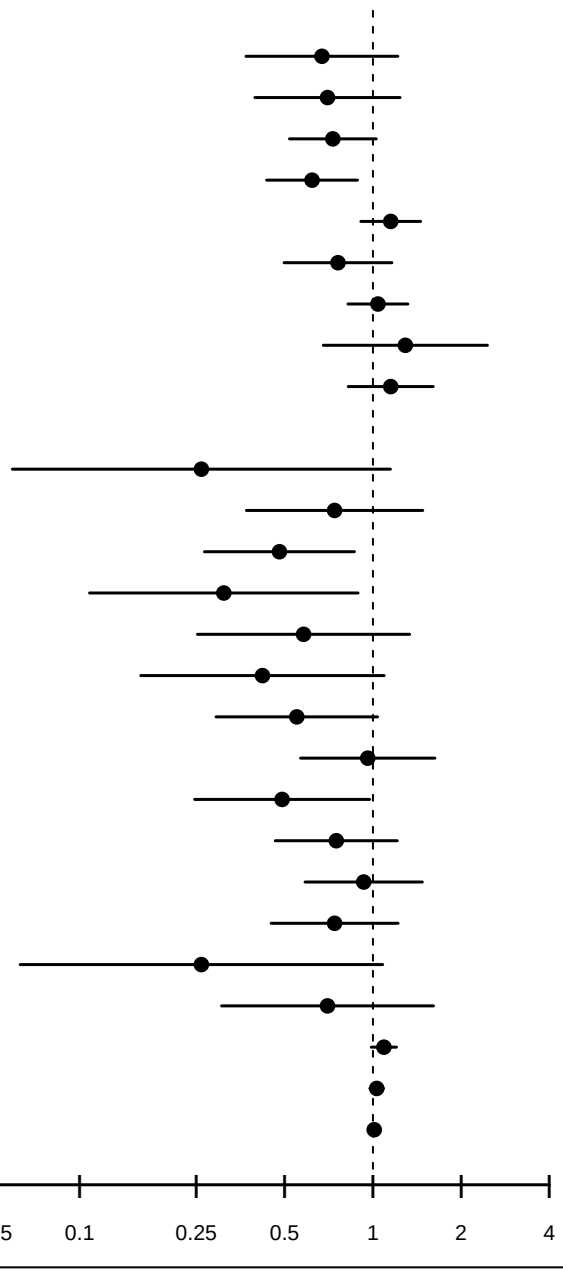
Rottman-Sagebiel et al.[29] **2018**	The Geriatrics Medication Education at Discharge project (GMED)	Pre-discharge: - An individual, face-to-face meeting between the CPS and the patient. - Medication reconciliation - Medication education regarding administration and usage of the patient's medications. - The caregiver, if any, was included in the discussion either at the bedside or by telephone following the face-to-face visit with the patient. - Discuss findings with inpatient team in person or by telephone and through documentation in the CPRS. Bridging: - All information the telephone encounter was communicated to the PCP through CPRS documentation and by telephone as needed. Post-discharge: - A telephone visit within 2 to 3 days of discharge. The purpose of the telephone visit was to perform medication reconciliation, identify and rectify medication errors, provide further patient education, and assist in facilitating appropriate follow-up by the patient's PCP, if required.	Primary care provider (PCP) A clinical pharmacist specialist (CPS) with oversight from a senior CPS with geriatric pharmacology expertise An internist or geriatrician	Veterans Memorial Hospital of the South Texas Veterans Health Care System (STVHCS)	From recruitment until 2 or 3 days after discharge	NR
Sahota et al.[27] **2017**	CIRACT	Pre-discharge: - Comprehensive assessment - Formed a rehabilitation plan - Treated daily (7 days a week, if appropriate) and the duration of rehabilitation they received depended on their needs. - The team liaised with the participant and their carer(s) to visit the participant's home to assess and provide recommendations for equipment and make adaptations and/or modifications as required Bridging: None Post-discharge: - Team visited the participant at home to assess the level of rehabilitation required	Senior occupational therapist (transition coach) Senior physiotherapist Assistant practitioner A social services practitioner	Queen's Medical Centre: a 1,800-bed hospital in Nottingham, UK.	NR	The standard THB-Rehab service was provided on weekdays only. An assessment of the participant's ability to perform certain tasks and provided recommendations for rehabilitation. Referral to community based services for provision of equipment at home, personal care and ongoing rehabilitation in community services where appropriate at

		- Further follow-up visits as deemed necessary - Appropriate referral to additional community services				discharge.
Voss et al.[30] **2011**	Care Transitions Intervention (CTI)	Pre-discharge: - Meet with the patient to establish initial rapport - Introduce the personal health record, including their main health problems, their medications, and questions for their health care providers - Arrange a home visit, ideally within 48 to 72 hours after hospital discharge Bridging: - Discuss topics related to participants with their outpatient providers Post-discharge: *Home visit within 3 days after discharge*: - Reconcile all of the patient's medication regimens - TC imparted skills for effectively communicating care needs during subsequent encounters with health care professionals. - TC also reviewed with the patient any red flags that indicated a condition was worsening and provided education about the initial steps to take to manage the red flags and when to contact the appropriate health care professional. *Phone call*: - Telephoning 3 times during a 28-day post-hospitalization discharge period. The first telephone call within 7 to 10 days generally focused on determining whether the patient had received appropriate services. In the 2 subsequent telephone calls, TC reviewed the patient's progress toward goals established during the home visit, discussed any encounters that took place with other health care professionals, reinforced the importance of maintaining and sharing the personal health record, and supported the patient's role in chronic illness self-management. The final telephone call was conducted by day 30	Transition coach (TC) who had a background in nursing or social work	6 Rhode Island acute care hospitals, including 2 community hospitals, 3 teaching hospitals, and a tertiary care center and teaching hospital, ranging from 129 beds to 719 beds	From enrolment until 30 days after discharge	From randomization to the day of discharge

NR: not reported, CGA: comprehensive geriatric assessment, NH: nursing home, GP: general practitioner, ADL: activity of daily living, PCP: primary care provider, ICM: integrated care model, HST: home support team, ED: emergency department.

Supplementary 6, forest plot grouped by study population

Author	Intervention	OA	RR (95% CI)
Unselected patients			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Patients at risk			
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)

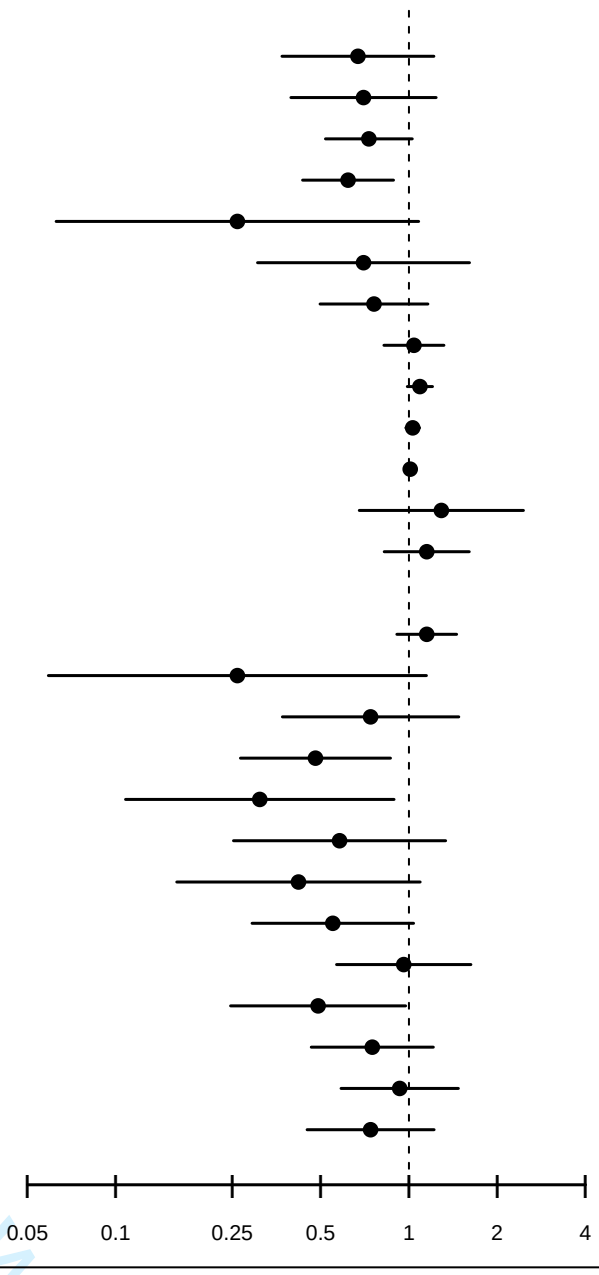


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 7, forest plot grouped by intervention intensity

Author	Intervention	OA	RR (95% CI)
Low intensity			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
High intensity			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)

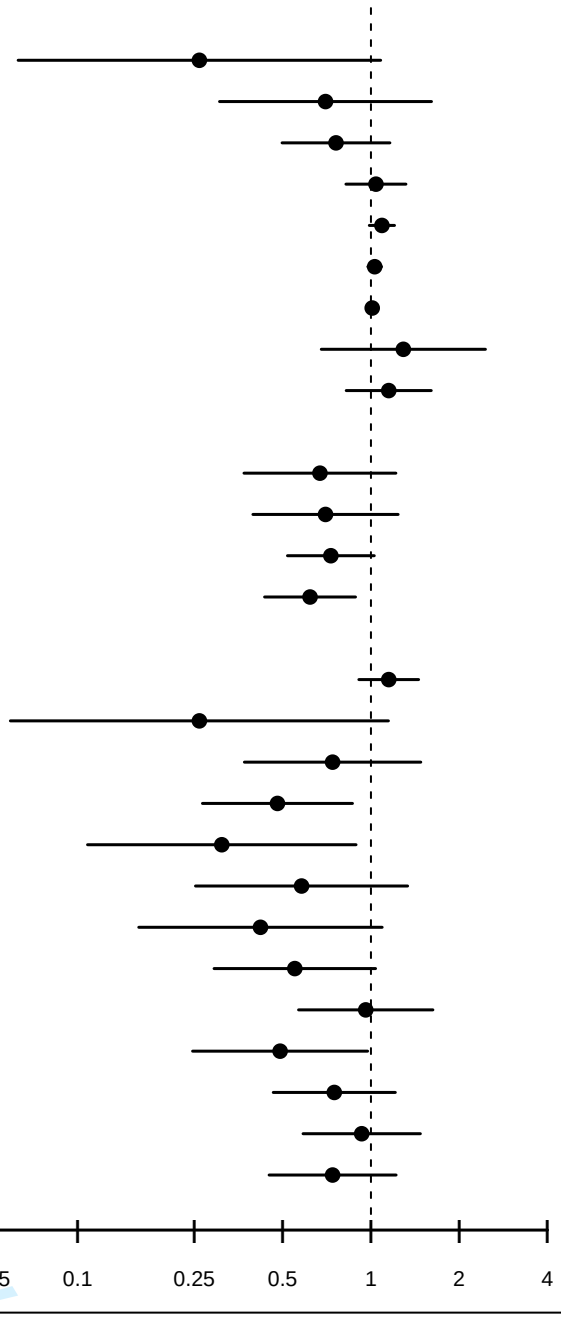


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 8, forest plot grouped by length of support

Author	Intervention	OA	RR (95% CI)
≤ 7 days			
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
8−30 days			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
≥ 31 days			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)

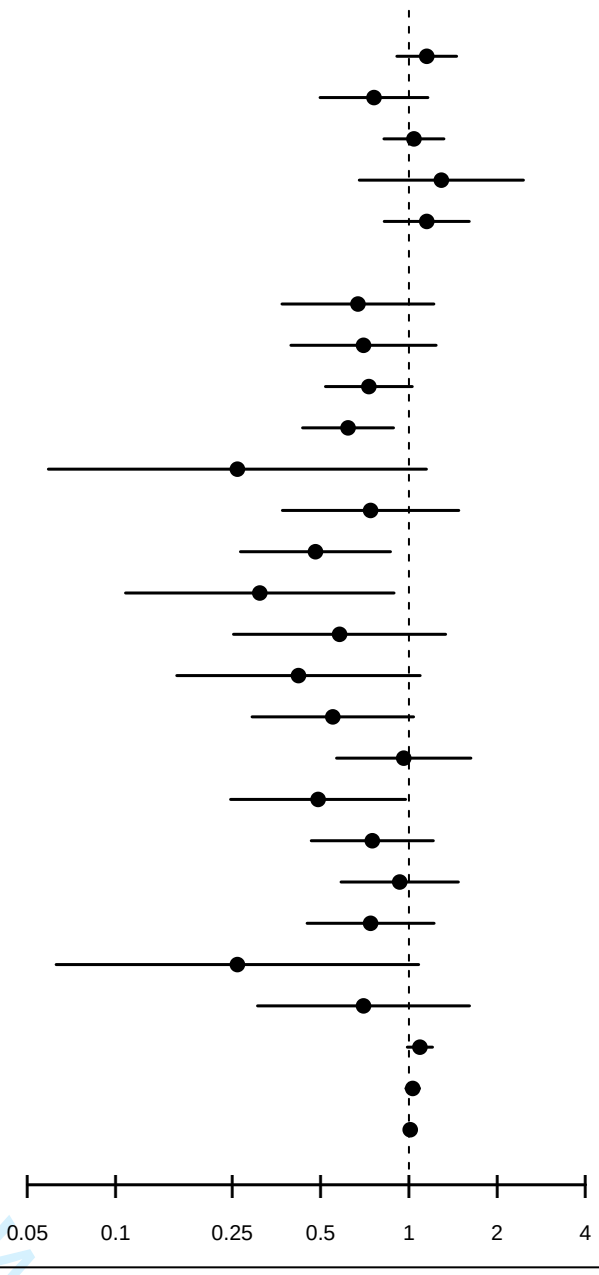


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 9, forest plot grouped by country of origin

Author	Intervention	OA	RR (95% CI)
European studies			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Non−European studies			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)

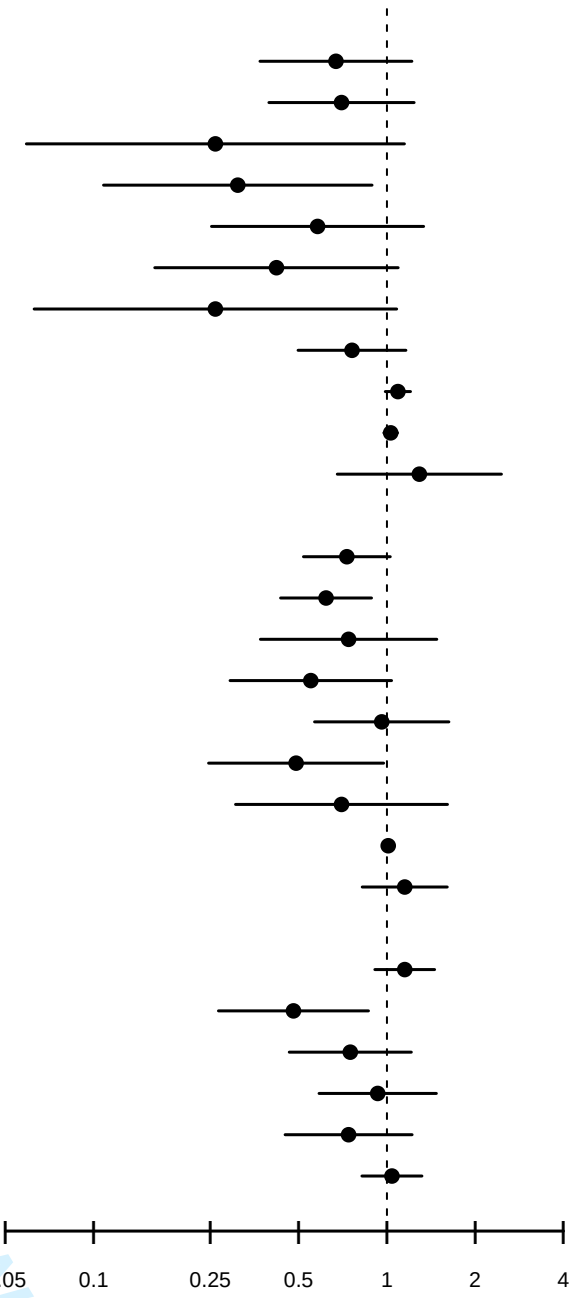


Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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Supplementary 10, forest plot grouped by outcome assessment

Author	Intervention	RR (95% CI)
≤ **30 days**		
Chow et al.	Home visit	0.67 (0.37−1.22)
Chow et al.	Call	0.70 (0.40−1.24)
Courtney et al.		0.26 (0.06−1.15)
Finlayson et al.	EN−Hat	0.31 (0.11−0.89)
Finlayson et al.	Exercise	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	0.42 (0.16−1.09)
Koehler et al.		0.26 (0.06−1.08)
Nielsen et al.		0.76 (0.50−1.16)
Robinson et al.		1.09 (0.99−1.20)
Robinson et al.		1.03 (0.98−1.09)
Sahota et al.		1.29 (0.68−2.46)
31−91 days		
Chow et al.	Home visit	0.73 (0.52−1.03)
Chow et al.	Call	0.62 (0.43−0.89)
Courtney et al.		0.74 (0.37−1.48)
Finlayson et al.	EN−Hat	0.55 (0.29−1.04)
Finlayson et al.	Exercise	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	0.49 (0.25−0.97)
Koehler et al.		0.70 (0.30−1.61)
Robinson et al.		1.01 (0.97−1.05)
Sahota et al.		1.15 (0.82−1.61)
≥ **92 days**		
Buurman et al.		1.15 (0.91−1.45)
Courtney et al.		0.48 (0.27−0.87)
Finlayson et al.	EN−Hat	0.75 (0.46−1.21)
Finlayson et al.	Exercise	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	0.74 (0.45−1.22)
Nielsen et al.		1.04 (0.82−1.32)

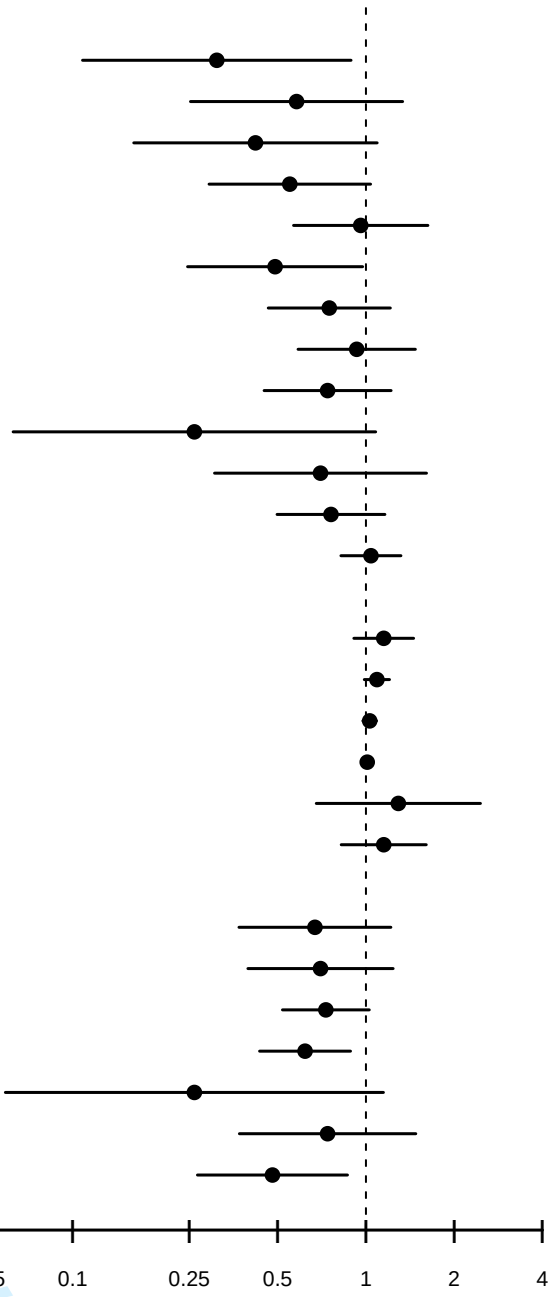


Abbreviations: RR, relative risk; CI, confidence interval

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Supplementary 11, forest plot grouped by quality assessment

Author	Intervention	OA	RR (95% CI)
Weak			
Finlayson et al.	EN−Hat	≤ 30 days	0.31 (0.11−0.89)
Finlayson et al.	Exercise	≤ 30 days	0.58 (0.25−1.33)
Finlayson et al.	N−Hat	≤ 30 days	0.42 (0.16−1.09)
Finlayson et al.	EN−Hat	31−91 days	0.55 (0.29−1.04)
Finlayson et al.	Exercise	31−91 days	0.96 (0.57−1.63)
Finlayson et al.	N−Hat	31−91 days	0.49 (0.25−0.97)
Finlayson et al.	EN−Hat	≥ 92 days	0.75 (0.46−1.21)
Finlayson et al.	Exercise	≥ 92 days	0.93 (0.59−1.47)
Finlayson et al.	N−Hat	≥ 92 days	0.74 (0.45−1.22)
Koehler et al.		≤ 30 days	0.26 (0.06−1.08)
Koehler et al.		31−91 days	0.70 (0.30−1.61)
Nielsen et al.		≤ 30 days	0.76 (0.50−1.16)
Nielsen et al.		≥ 92 days	1.04 (0.82−1.32)
Moderate			
Buurman et al.		≥ 92 days	1.15 (0.91−1.45)
Robinson et al.		≤ 30 days	1.09 (0.99−1.20)
Robinson et al.		≤ 30 days	1.03 (0.98−1.09)
Robinson et al.		31−91 days	1.01 (0.97−1.05)
Sahota et al.		≤ 30 days	1.29 (0.68−2.46)
Sahota et al.		31−91 days	1.15 (0.82−1.61)
Strong			
Chow et al.	Home visit	≤ 30 days	0.67 (0.37−1.22)
Chow et al.	Call	≤ 30 days	0.70 (0.40−1.24)
Chow et al.	Home visit	31−91 days	0.73 (0.52−1.03)
Chow et al.	Call	31−91 days	0.62 (0.43−0.89)
Courtney et al.		≤ 30 days	0.26 (0.06−1.15)
Courtney et al.		31−91 days	0.74 (0.37−1.48)
Courtney et al.		≥ 92 days	0.48 (0.27−0.87)



Abbreviations: OA, outcome assessment; RR, relative risk; CI, confidence interval

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	none
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	17 Table 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Only 1 outcome Table 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	13
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	19 Fig. 2
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	None
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	19-22, 23
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	22
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

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From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Synthesis Without Meta-analysis (SWiM) reporting items

The citation for the Synthesis Without Meta-analysis explanation and elaboration article is: Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, Hartmann-Boyce J, Ryan R, Shepperd S, Thomas J, Welch V, Thomson H. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline BMJ 2020;368:l6890 http://dx.doi.org/10.1136/bmj.l6890

SWiM is intended to complement and be used as an extension to PRISMA			
SWiM reporting item	**Item description**	**Page in manuscript where item is reported**	**Other***
Methods			
1 Grouping studies for synthesis	1a) Provide a description of, and rationale for, the groups used in the synthesis (e.g., groupings of populations, interventions, outcomes, study design)	Page 6 (data synthesis) Page 15, 17 (Sub-group analysis)	
	1b) Detail and provide rationale for any changes made subsequent to the protocol in the groups used in the synthesis	No changes made	
2 Describe the standardised metric and transformation methods used	Describe the standardised metric for each outcome. Explain why the metric(s) was chosen, and describe any methods used to transform the intervention effects, as reported in the study, to the standardised metric, citing any methodological guidance consulted	Page 6 (data synthesis)	
3 Describe the synthesis methods	Describe and justify the methods used to synthesise the effects for each outcome when it was not possible to undertake a meta-analysis of effect estimates	Page 6 (data synthesis) Page 13 (synthesis of results)	

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Synthesis Without Meta-analysis (SWiM) reporting items

4 Criteria used to prioritise results for summary and synthesis	Where applicable, provide the criteria used, with supporting justification, to select the particular studies, or a particular study, for the main synthesis or to draw conclusions from the synthesis (e.g., based on study design, risk of bias assessments, directness in relation to the review question)	Page 6 (quality assessment) Page 6 (data synthesis)	
SWiM reporting item	**Item description**	**Page in manuscript where item is reported**	**Other***
5 Investigation of heterogeneity in reported effects	State the method(s) used to examine heterogeneity in reported effects when it was not possible to undertake a meta-analysis of effect estimates and its extensions to investigate heterogeneity	Page 6 (data synthesis)	
6 Certainty of evidence	Describe the methods used to assess certainty of the synthesis findings	Page 6 (data synthesis)	
7 Data presentation methods	Describe the graphical and tabular methods used to present the effects (e.g., tables, forest plots, harvest plots). Specify key study characteristics (e.g., study design, risk of bias) used to order the studies, in the text and any tables or graphs, clearly referencing the studies included	Page 6 (data synthesis)	
Results			
8 Reporting results	For each comparison and outcome, provide a description of the synthesised findings, and the certainty of the findings. Describe the result in language that is consistent with the question the synthesis addresses, and indicate which studies contribute to the synthesis	Table 1 study characteristics Page 8 (study characteristics)	

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Synthesis Without Meta-analysis (SWiM) reporting items

		Page 12 (synthesis of results)	
Discussion			
9 Limitations of the synthesis	Report the limitations of the synthesis methods used and/or the groupings used in the synthesis, and how these affect the conclusions that can be drawn in relation to the original review question	Page 21 (strength and limitation of this review)	

PRISMA=Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

*If the information is not provided in the systematic review, give details of where this information is available (e.g., protocol, other published papers (provide citation details), or website (provide the URL)).

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